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PET REPUBLIK: A WEB-BASED INTEGRATED PLATFORM FOR ONLINE ORDERING AND SERVICE APPOINTMENT MANAGEM...

 Davao del Norte State College

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CHAPTER I

INTRODUCTION

A. Company Overview

The John & Jai Pet Republik Pet Shop is a small, yet growing pet supplies business located in Panabo City, Davao del Norte, Philippines. It was officially established and started on April 28, 2021, owned and managed by Ms. Roselle T. Janiola, who has a passion for animals and envisioned to support the needs of pet owners in the local community. Following this, the shop has earned the unwavering trust of many loyal customers in the area. The business operates mainly in two service areas: retail and grooming. The retail division includes a selection of items for various categories of pet dogs and cats, including a wide range of both dry and wet pet food with an emphasis on natural and organic choices, while also offering treats, hygiene products, shampoos, grooming tools, and accessories such as collars, leashes, and feeding bowls.

On the service side, the shop also offers pet grooming. These are carried out by Mr. Allan Alia, the Senior Groomer, and Ms. Marjlin Constancio, the Assistant Groomer. They specialize in pet bathing, hair trimming, and even nail clipping to help animals feel relaxed and well cared for. Both groomers are experienced in handling different sizes of pets and personalities, making sure the safety and comfort during the grooming process. Their dedication to quality service has helped the shop develop a positive reputation among pet owners in the city.

The shop has a welcoming and pet-friendly environment, with a team that is approachable, hardworking, and passionate about animal care. Customers frequently express their appreciation for the personalized attention given to each pet and the consistent care shown by the staff. While the business is relatively small, it plays an important role in the local community by providing accessible and reliable pet care services.

Currently, the shop operates using manual processes for grooming appointments, customer records, and inventory tracking. Customers frequently

walk in or call the shop for service inquiries and booking schedules. Though this traditional manual system works, it can sometimes lead to delays and miscommunication, especially on busy days. Despite this, the business continues to provide exceptional services to its customers.

John & Jai Pet Republik Pet Shop remains committed to giving pets the best care possible while supporting the needs of pet owners. It stands as a trusted and valued part of Panabo City's local pet care landscape, offering both compassion and convenience in every visit.

B. Organizational Structure

The organizational chart of John & Jai Pet Republik Pet shop shows a small but well-organized team at the pet shop under the direction of the owner Ms. Roselle T. Janiola. In this department, she oversees two major sections for the shop: the cashier area and the grooming section. Dymple Tacder works as a cashier of the pet shop with Danela Cabouinason as her associate who also assists her to complete the transactions and help customers when buying products. Allan Alia is the senior groomer for the pet area and is the person in charge of grooming services. With her companion, Ms. Marjlin Constancio, who helps with bathing, drying, grooming and other pet care tasks.

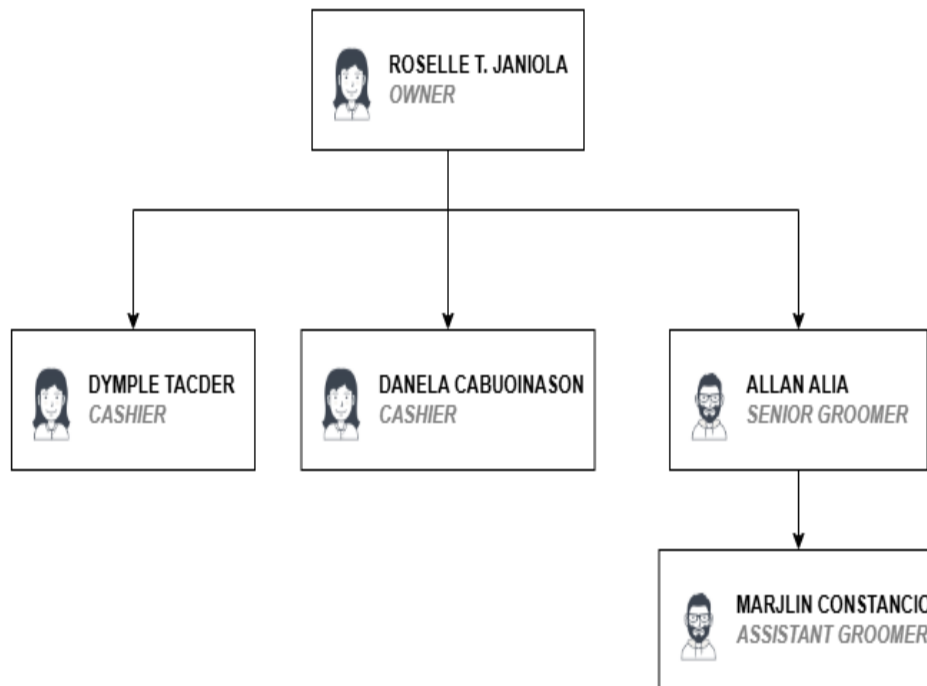


Figure 1. Organizational Structure of the John & Jai Pet Republik Pet shop.

C. Problem Encountered

John & Jai Pet Republik is a small locally owned pet supply store in Panabo City that sells pet foods and grooming for pets. The business has a loyal customer base but has its own set of challenges in terms of how the store is growing and expanding and meeting the needs of customers within the market. One of the biggest flaws of the John and Jai Pet Republik is the lack of a web-based system: there is no web-based system where people can check out available products, book grooming sessions, or even place an order. That means people need to call, message, or even physically go to the store to ask questions or schedule services – not always convenient and frequently results in confusion and delays.

Another issue is how to report on sales. Without a good system to track transactions the store does not know exactly how well they are doing, or which product is selling the most effectively. This makes making decisions and planning harder. Also, the store does not have any way to track the expiration dates of pet

food & other pet items that have a short shelf life, as the expired items can be forgotten about and negatively affect the quality of service.

Managing inventory is an even more difficult task because everything is manually tracked thus it can be easy to miss when stock levels are low, or they are piling up, so you run out of stuff you need or spend money on items that are already stocked. Last but not least the fact that appointments are recorded by hand or through messages so there have been instances of double bookings or missing an appointment which creates stress for staff and customers and may affect customers' trust.

The problems encountered by the John & Jai Pet Republik are:

- Traditional manual bookings can lead to mistakes, like double bookings.
- Struggle on monitoring their data sales.
- Lack of a system that can monitor the product stocks and expiration dates.
- Challenges in generating reports needed for decision-making and planning.
- Orders are recorded incorrectly due to manual encoding.

D. Project Overview

The Pet Republik web-based system aims to create a centralized platform for John & Jai Pet Republik Pet Shop such as their pet foods, accessories and also appointment services within the company. The proposed project is designed to address the challenges faced by most pet owners in Panabo City, such as the limited access digitally to pet products, the lack of e- marketplace in pet related products, and a poor integration of business such as of the Pet Republik company. The overview of this project includes the description of its objectives, highlighting how the proposed system or platform will offer its convenience, accessibility, and better pet care problem solutions.

The Pet Republik Web will feature an online store where users can buy pet foods and other products digitally from the store. It will also have an appointment service management, allowing customers to book a schedule from John & Jai pet

grooming services. Through this platform, the customers of the company will have an easier access in terms of buying pet related products and their other services. The system's user-friendly design will allow an easy product listing, searching, and purchasing that ensures a smooth shopping and other pet services experience to the customers. The proposed web application to Pet Republik Pet Shop enables the pet owners to have a reliable and accessible platform that simplifies pet care, promotes responsible ownership, and connects them to the new and integrated system of John & Jai Company.

E. Existing Business Process

Pet Store Process:

Figure 2 illustrates the process followed when a customer makes a purchase at a pet store. The process begins when the customer goes to the store and starts browsing products. After browsing, the customer selects their desired items and if they wish to add more items, they repeat the browsing and selection then add the item. Once the customer is finished, they proceed to the cashier where the cashier checks the selected items, receives payment, provides a receipt and then gives the purchased items to the customer thus completing the transaction.

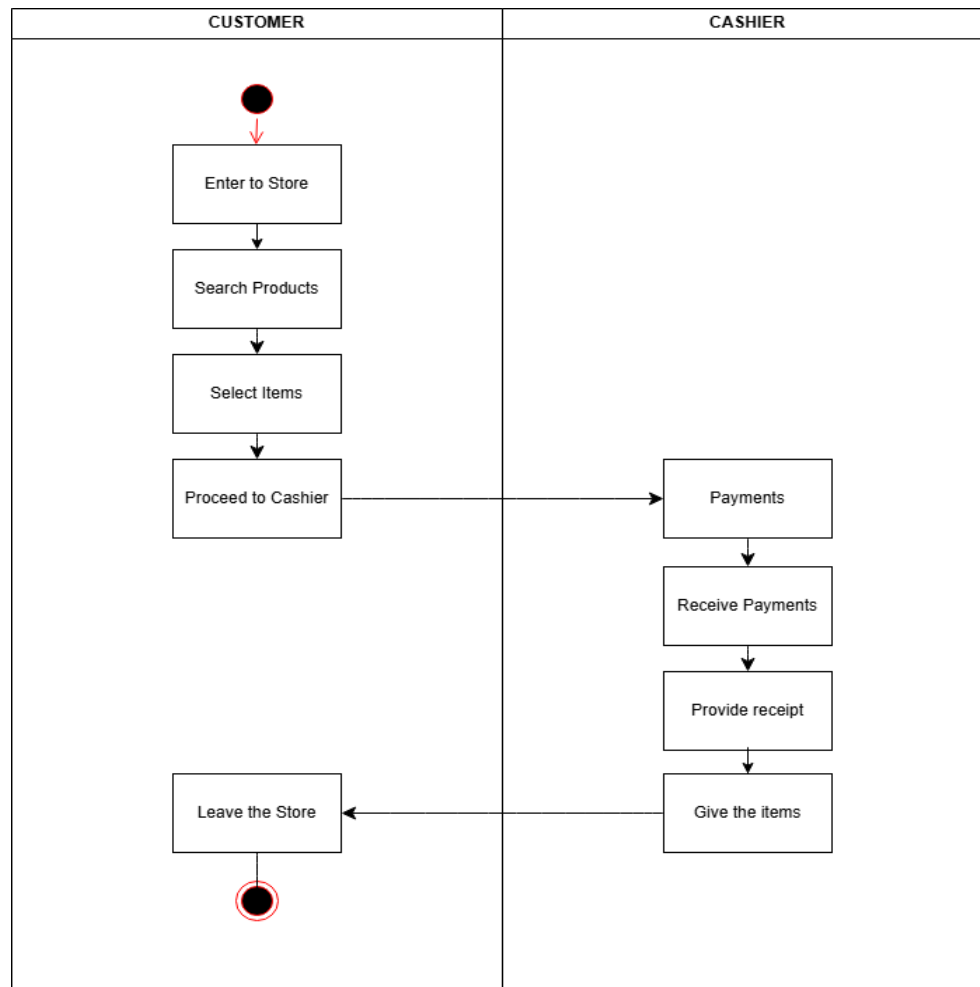


Figure 2. Pet Store Purchase Process

Figure 3 displays the flow of the John & Jai Pet Republik Pet Shop's pet grooming service process. It starts when the customer visits the pet grooming store and consults with the staff to check the availability and other offers of grooming services. If the service is not available at the moment, the customer is informed of the estimated vacant time or pet grooming schedule. After it is done the customer proceeds to issue a payment to the receptionist and then the cashier will collect the payment before the grooming service begins. Once the grooming is finished, the customer returns to the store to pick up their pet, marking the end of the service.

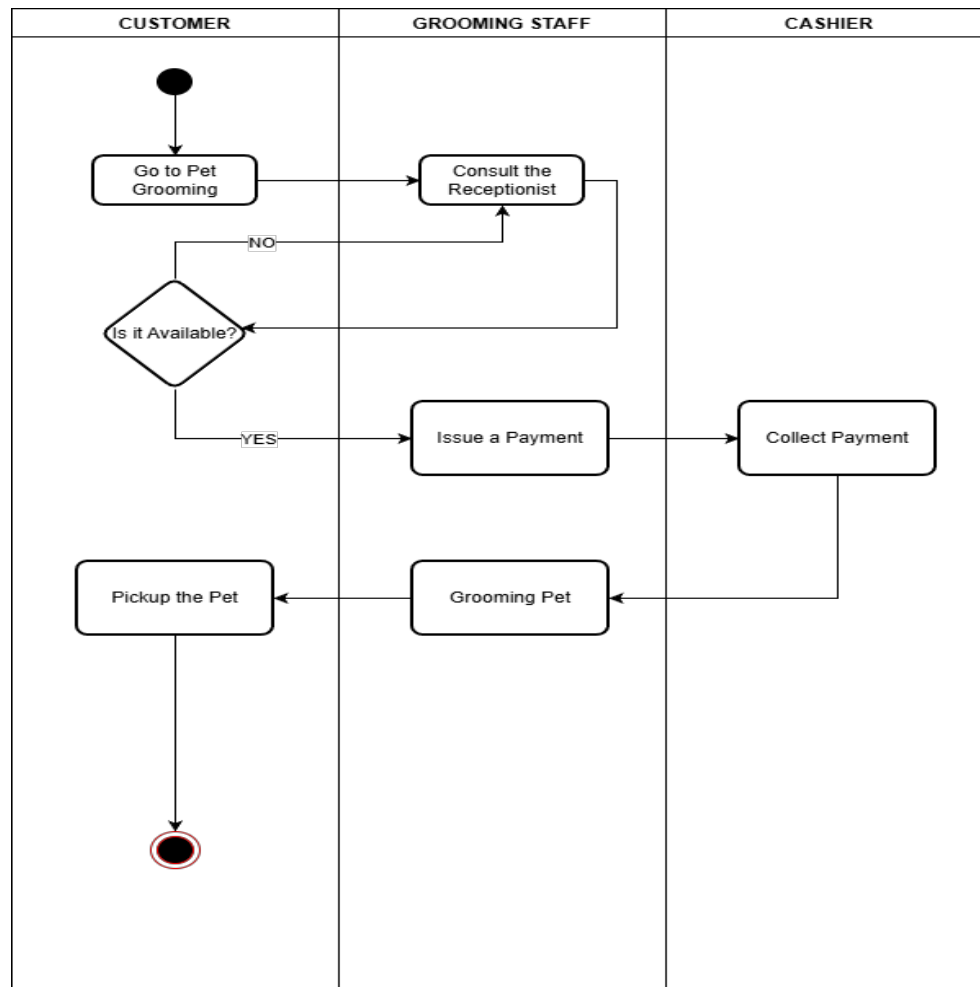


Figure 3. Pet Grooming Service Flow

F. Objectives of the Project

General Objectives:

The Pet Republik web-based system aims to digitalize the way pet owners in Panabo City access pet stores online from John & Jai Company, whether buying pet related products or stay informed about their offer pet grooming services. The main goal of this project is to create a centralized web-based platform from John & Jai Company Services that offers online ordering and appointment booking for their other offered services, replacing the manual process and unorganized methods the pet shop currently operates on. Furthermore, the system will enhance its company's ability to compete in other businesses. This system will also allow

users to easily buy pet food or items online and also appoint booking services regarding their pet grooming services.

Specific Objectives:

The system aims to accomplish the following objectives in enhancing its service accessibility and efficiency through Pet Republik:

1. The system allows admin to:
 - 1.1 Log in to their account.
 - 1.2 Manage user accounts.
 - 1.3 Manage products, categories, delivery fees and services.
 - 1.4 Communicate with the cashier.
 - 1.5 Tracks inventory, stock movements, customer orders and activity logs.
 - 1.6 Generate detailed reports.
2. The system allows cashiers to:
 - 2.1 Log in to their account.
 - 2.2 Add and update products.
 - 2.3 Manage customers' orders and bookings.
 - 2.4 Update orders and booking status.
 - 2.5 Communicate with customers and admin.
 - 2.6 Track inventory, POS and stock movements.
3. The system allows customers to:
 - 3.1 Create and log in to their account.
 - 3.2 Place orders and book pet services.
 - 3.3 Track orders and bookings status.
 - 3.4 Communicate with the cashier.
4. The system will generate reports including:
 - 4.1 Sales
 - 4.2 Bookings
 - 4.3 Customers
 - 4.4 Pet Service

G. Scope and Limitation

This proposed project aims to develop a digital marketplace specifically designed for pet- related products and services. It allows users to buy online and make booking reservations in available services of the company. It also has a messenger function to monitor the pet care services. The scope includes account management for different user roles like Admin, Customer and Staff User with each having a predefined access control and responsibilities. The system composes software technologies like PHP, JavaScript, and XAMPP that integrate and manage the front-end interactions and back-end data processing, while FIGMA supports the interface design. However, the limitation of the system is located at the local server deployment using XAMPP and does not yet support real-time communication, mobile responsiveness, or third-party integrations such as online payment gateways. Below is the table that explains what the scope and limitations of the system in detail are.

Table 1. Scope and Limitations

People	● Admin - Has full access to the application, including managing users, handling administrative tasks, and maintaining the integrity and security of the system. The admin is also responsible for registering cashier accounts to ensure accurate system operations. In addition, the admin can monitor all transactions, manage products and services, track inventory, and access the analytics and reporting functions of the system. However, the system currently lacks real-time communication features, which may cause delays in coordination between the admin and cashiers. To improve operational efficiency and meet the high usability standards expected by the users. ● Cashiers - They are the ones who operate the customers' orders and have the ability to stock and restock products. They also operate the customers booking services of the company, the pet grooming services. ● Customer - Uses the system to order pet foods or accessories online from the Pet Republik Pet Shop. They can also book a service reservation that is available in the system (which the company offers).
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Technologies	HARDWARE: ● PC/Laptop - This helps the cashiers to make an easier stocking and restocking of their products. ● Cellphone - Allows the customer to order online and book reservations to their available services. ● Printer – allows the cashier to print invoice or receipt to the customer. SOFTWARE: ● XAMPP - Serves as the local server environment that holds user data and other application-related information through its database support. ● JavaScript - Adds interactivity and dynamic behavior to enhance the user experience within the application. ● PHP - Functions as both the front-end and back-end scripting language, managing user input, database connections and overall application logic. ● COMPOSER - Tools used for managing libraries or packages and relies dependencies in PHP projects. ● NODE.JS - Helps run JavaScript code outside from a web browser, this software helps JavaScript build Backend applications. ● GIT- Helps track changes in the codes, it is a version control system that tracks, saves and manages the code changes. ● Web Sockets or Pusher - Enables real-time communication within the system by instantly sending notifications and updates between the admin, cashier, and customers, reducing delays in order processing, bookings, and transaction monitoring.
Data	● Admin Information - Includes administrator details such as name, contact number, and account credential information. ● User Information - Contains registered clients' personal details and account-related data. ● Analytics - Generate reports of the company's revenue, total sales and other information. ● Transaction History - Records details of selling and purchasing transactions, including exact date and time for tracking purposes.

Process	● Creating User Account - Customers create an account and input the necessary information, while only the admin can create a cashier account for its efficiency in Role Management. ● Online Ordering - The Customer orders online and adds to cart the available products within the system. It has two payment methods which are pickups and cash-on-delivery. ● Service Booking - The Pet Republik Pet Shop has additional services to their company, which is pet grooming. Customers can book a reservation online to reduce its human error and improve customer satisfaction.
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H. Related Systems and Works

Pet care has undergone dramatic changes in the digital age, with the exponential growth of e-commerce platforms and user-generated marketplaces, online service scheduling systems, etc. In response, small and medium-sized enterprises (SMEs) working in the pet care industry are adopting digital solutions such as a well-integrated pet supply and service management platform to improve operations, increase customer engagement, and stay competitive. A well-integrated pet supply and service management platform can help optimize operations by automatically monitoring pet inventory levels, listing product items from available vendors, and automatically scheduling grooming and spa services. According to an empirical study exploring the service blueprinting in pet e-commerce platforms, such digital frameworks enable better service delivery, enabling personalized customer experiences [1].

They still not only rely on manual recordkeeping and scheduling but also have problems with overbooking, lost customer records, as well as insufficient stock tracking. In a recent case study on inventory management based on local animal product shop that was conducted in Tanzania, moreover, it was found that unoptimized stock processes lead to financial implications and operational delays in both day and nighttime for the business, thus emphasizing the need to put digital software in place [2]. Meanwhile, local implementation of pet care management

systems has shown in some cases that the absence of central platforms leads to delayed updates of services and low customer retention [3].

Around the world, there is growing interest in deploying a complete platform for combining e-commerce and other services functions. An IoT-based pet care solution developed in India provides evidence for the potential of connected technologies to automate supply monitoring and monitoring pet health [4]. A mobile app designed for Kampar residents in Malaysia combines artificial intelligence (AI) elements with veterinary and grooming appointments in order to increase responsiveness [5]. Similar results can be observed using real time automation for developing online pet shops that meet ethical criteria in Indonesia. Indeed, real time automation is proven to reduce stock mismatches and increase delivery turnaround time [6].

User-driven marketplaces are particularly influential for the pet care sector. Studies of vertical e-commerce suggest that by allowing pet owners to post and sell their own products, they not only stimulate engagement to the marketplace but also increase customer loyalty [7]. Community-sourced product listings foster small product sellers, expand the supply of niche products, and replicate successful social commerce models such as the Facebook Marketplace and Etsy [8]. Furthermore, there is evidence that peer-to-peer selling combined with user-generated ratings and reviews can have a significantly influencing effect on the decision to purchase in the pet food and accessory market [9].

In the Philippines, small pet businesses often encounter operational challenges related to both sales and customer service due to their reliance on manual processes. These limitations frequently result in inefficient customer handling, difficulties in tracking sales and inventory, and limited scalability. A study on pet store businesses in Metro Manila emphasized that many operators underutilize app or computer-assisted tools, which could otherwise enhance financial oversight and overall business efficiency [10]. These findings highlight the overall need for a digital infrastructure to link sales, inventory management and appointment bookings for local businesses to improve efficiency, reduce human errors and be more competitive. The inefficiency of these systems in turn leads to

customer dissatisfaction and prevents local businesses from scaling up. Financial assessments on micro, small and medium enterprises (MSMEs) in Cebu also show that underdevelopment in digital infrastructure is a major impediment to growth [11]. Emerging digital platforms have shown promise in filling the operational gaps. For instance, one consumer-to-consumer (C2C) e-commerce mobile application developed for pet stores in Metro Manila integrated product catalogs with real-time booking functions, resulting in faster order fulfillment and faster coordination of services [12]. Successful platforms typically provide dashboards, stock alerts, automated reminders and data analytics in order to facilitate data-driven decision making and business scalability. Other examples, such as *VETGO* and *PetCare*, emphasize how automation supports service consistency, increases customer capacity, and reduces dependence on manual labor [13] - [14].

Developing a centralized digital platform that combines pet supply inventory management, peer-to-peer product listings, and grooming or veterinary scheduling tools presents a holistic solution to the evolving needs of the pet care sector. Such a system supports operational efficiency, improves customer satisfaction, and encourages local entrepreneurship. Furthermore, it aligns with the Philippine government's push for digital transformation among MSMEs and supports the United Nations Sustainable Development Goal 9, which promotes resilient infrastructure and inclusive innovation [15]. By automating essential functions and empowering both vendors and consumers, this platform contributes to a more competitive, responsive, and sustainable pet care ecosystem.

The proposed Pet Republik Pet Shop system aligns with the goals set forth in PDP 2028 Part III Chapter 8 of the Philippine Development Plan (PDP) 2023–2028, specially in enhancing and advancing research and development (R&D) such as technology and innovation for MSMEs. This suggested platform address and identified challenges such as limited digital automation, poor integration of market-user data and a low commercialization of technology, by offering a centralized system in market-driven and customer-centered web-based solutions for pet care services and product sales. It embodies the PDP's strategies by promoting digital transformation in a small business context into fostering

innovation and thus creating and enabling an environment for entrepreneurship. Moreover, Pet Republik's use of an analytics system that generated sales reports and customer service automation reflects the PDP's emphasis on technology utilization, IP support and strengthening its innovation ecosystems showcasing how local technology initiatives can contribute to the inclusive and competitive economy envisioned in the PDP 2028.

In addition, the proposed system also complements and aligns with the pillar "Maginhawa" of AmBisyon Natin 2040 – namely, a comfortable life for all Filipinos. Through making pet products and services more accessible and readily available, reducing the manual operation of the business, and providing more stable income opportunities for the business owners, digital systems also help to make life accessible, efficient and economically resilient. In short, it is a step forward for Filipino families to have modern, tech-driven services that are responsive to their life needs in today's rapidly evolving lifestyle.

In line with our institutional agenda "Upgrade traditional industry processes and cybersecurity", the proposed Pet Republik Pet Shop system embodies the goals of DNSC's institutional strategy of "Upgrade traditional industry processes and cybersecurity." This is through upgrading the core operations of pet care micro-enterprises with secure digital innovations. Pet Republik replaces the old handwritten appointment books, physical inventory checks, and track sales by face-to-face meetings – with an integrated and web-based platform that automates inventory management, real-time grooming appointment services, and digital sales reporting. These improvements significantly boost operational efficiency, minimize errors, and enable business owners to make better and more informed business decisions based on informed data. Pet Republik also provides some basic security features and secure data storage that ensures that customer information and records are protected from being compromised when using a manual or unsecured digital system.

CHAPTER II

METHODOLOGY

A. Startup and Feasibility

This section presents the Budget and Summary alongside its Operational and Technical Feasibility, providing a thorough evaluation of the proposed digitized pet store application with functionality and potential for success. It will evaluate whether the proposed system project can effectively deliver innovative features such as an online pet store with different kinds of products and accessories, appointment booking for pet grooming services as it will progressively ensure customer loyalty programs. By exploring the financial requirements, technical capabilities and operational needs, this section will likely ensure that the proposed web application system is not only achievable but also aligned with market demand and business goals, supporting a smooth and sustainable web application system launch in John & Jai Pet Republik Pet Shop.

Budget Summary

Table 2 provides a holistic view of the overall financial and time allocation required to complete the proposed project Pet Republik System from initiation to its closure maintenance and support. Spanning approximately 125.20 days, the project's estimated total cost is around ₱237,651.50 covering all the critical phases from its initial gathering information and analysis throughout the design, development, testing, deployment, training and documentation, and maintenance. This overall budget summary ensures that the necessary resources, timeline and costs have been carefully planned to support the successful delivery of its functionality and user-friendly system that meets the intended business goals and customer needs in the business.

Table 2. Budget Summary

	Phase	Duration (Mandays)	Start Date	End Date	Estimated Cost
1.0	Initiation & Planning	12	Thu, 03-Apr-25	Sun,04-May-25	₱ 28,968.10
1.1	Define project goals and scope	2.50	Thu, 03-Apr-25	Mon, 14-Apr-25	₱ 5,770.00
1.2	Conduct feasibility study	3.30	Tue, 15-Apr-25	Tue, 22-Apr-25	₱ 5,098.50
1.3	Identify stakeholders and form the team	1.60	Wed, 23-Apr-25	Thu, 24-Apr-25	₱ 6,164.80
1.4	Prepare project charter	1.60	Fri, 25-Apr-25	Sat, 26-Apr-25	₱ 6,164.80
1.5	Develop project plan & timeline	2.50	Sun, 27-Apr-25	Sun, 04-May-25	₱ 5,770.00
2.0	Requirements Gathering	14.70	Mon, 05-May-25	Fri, 06-Jun-25	₱ 30,764.80
2.1	Conduct stakeholder interviews	2.40	Mon, 05-May-25	Sat, 10-May-25	₱ 3,708.00
2.2	Define functional requirements	2.20	Sun, 11-May-25	Tue, 20-May-25	₱ 8,476.60
2.3	Define non-functional requirements	2.60	Thu, 22-May-25	Sat, 24-May-25	₱ 4,017.00
2.4	Prepare Software Requirements Specification (SRS)	3.60	Sun, 25-May-25	Fri, 30-May-25	₱ 5,562.00
2.5	Get requirement approval	3.90	Sat, 01-Jun-25	Fri, 06-Jun-25	₱ 9,001.20
3.0	System Design	11.30	Sat, 07-Jun-25	Sun, 06-Jul-25	₱ 26,270.40

3.1	Design system architecture	2.90	Sat, 07-Jun-25	Thu, 12-Jun-25	₱ 5,597.00
3.2	Create ERD / database schema	1.90	Fri, 13-Jun-25	Wed, 18-Jun-25	₱ 3,667.00
3.3	Design UI/UX mockups	2.00	Wed, 18-Jun-25	Wed, 25-Jun-25	₱ 3,860.00
3.4	Prepare technical design document	2.20	Thu, 26-Jun-25	Mon, 30-Jun-25	₱ 3,399.00
3.5	Review & approve designs	2.30	Tue, 01-Jul-25	Sun, 06-Jul-25	₱ 9,747.40
4.0	Development	23.50	Mon, 07-Jul-25	Mon, 03-Nov-25	₱ 34,432.00
4.1	Set up development environment	2.30	Mon, 07-Jul-25	Tue, 15-Jul-25	₱ 3,128.00
4.2	Frontend development	4.20	Wed, 16-Jul-25	Sun, 03-Aug-25	₱ 5,712.00
4.3	Backend development	5.00	Sat, 23-Aug-25	Mon, 01-Sep-25	₱ 6,800.00
4.4	Integration of frontend and backend	1.60	Sun, 05-Oct-25	Sun, 26-Oct-25	₱ 4,648.00
4.5	Unit testing by developers	2.30	Mon, 27-Oct-25	Mon, 03-Nov-25	₱ 3,128.00
5.0	Testing & QA	10.80	Tue, 04-Nov-25	Sat, 28-Feb-26	₱ 16,686.00
5.1	Prepare test cases and scenarios	1.70	Tue, 04-Nov-25	Thu, 06-Nov-25	₱2,626.50
5.2	Perform functional testing	1.60	Thu, 06-Nov-25	Mon, 10-Nov-25	₱2,472.00
5.3	Perform usability testing	1.70	Mon, 10-Nov-25	Sat, 15-Nov-25	₱2,626.50

5.4	Perform security and performance testing	1.50	Sat,15-Nov-25	Fri, 28-Nov-25	₱2,317.50
5.5	Fix bugs and retest	1.70	Wed,03-Dec-25	Sun, 28-Dec-25	₱2,626.50
5.6	Get UAT (User Acceptance Testing) approval	2.60	Mon, 12-Jan-26	Sat, 28-Feb-26	₱4,017.00
6.0	Deployment	40.40	Mon, 09-Mar-26	Fri, 17-Apr-26	₱ 80,406.00
6.1	Set up production environment	5.00	Mon,09-Mar-26	Thu, 12-Mar-26	₱6,800.00
6.2	Deploy system	5.00	Fri, 13-Mar-26	Tue, 17-Mar-26	₱6,800.00
6.3	Perform smoke testing	4.40	Wed,18-Mar-26	Fri, 20-Mar-26	₱6,798.00
6.4	Final go-live	26.00	Sat,21-Mar-26	Fri,17-Apr-26	₱60,008.00
7.0	Training & Documentation	7.30	Sat, 18-Apr-26	Thu, 30-Apr-26	₱ 11,278.50
7.1	Create user manuals	3.30	Sat, 18-Apr-26	Thu, 23-Apr-26	₱5,098.50
7.2	Train end-users	2.20	Fri, 24-Apr-26	Sun, 26-Apr-26	₱3,399.00
7.3	Prepare technical documentation	1.80	Mon, 27-Apr-26	Thu, 30-Apr-26	₱2,781.00
8.0	Maintenance & Support	5.70	Tue, 05-May-26	Sat, 30-May-26	₱ 8,845.70
8.1	Monitor system performance	1.60	Tue, 05-May-26	Sun, 10-May-26	₱2,176.00
8.2	Bug fixing and patching	1.90	Mon, 11-May-26	Wed, 20-May-26	₱2,584.00

8.3	Handle user feedback and updates	1.30	Wed,20-May-26	Tue, 26-May-26	₱2,008.50
8.4	Plan for system scaling or future enhancements	0.90	Mon, 25-May-26	Sat, 30-May-26	₱2,077.20
	TOTAL	125.20			₱237,651.50

Manpower Cost Estimates

Table 3 displays the summarized estimation of the overall manpower costs required for the successful completion of the proposed project Pet Republik System. The overall manpower cost of this project amounts to ₱237,651.50, with a total of 172.90 mandays. This will cover the distribution of each key personnel in the project including the project sponsor, project manager, system programmer, system designer and the system analyst. This summary makes sure that the human resource requirements are well-accounted for, reflecting the integration of expertise and efforts needed to deliver the proposed project efficiently and effectively within the planned time frame.

Table 3. Manpower Cost Estimates

Position	Total Man-days	Rate/Day	Cost
Project Manager	43.50	₱ 2,308.00	₱ 100,398.00
Project Sponsor	38.40		₱ -
System Programmer	37.00	₱ 1,360.00	₱ 50,320.00
System Designer	9.10	₱ 1,930.00	₱ 17,563.00
System Analyst	44.90	₱ 1,545.00	₱ 69,370.50
Total	172.90		₱ 237,651.50

Operational Feasibility

The company operates primarily through in-store sales, which they manage with manual use of inventory and spreadsheets. Customer interactions are limited to face-to-face engagement and a walk in or appointment booking for pet grooming services thus leading to occasional double-booking or missed appointments. The company already has an existing computer setup that can be utilized to deploy the proposed system, making the operational setup largely prepared. Although the availability of IT resources is limited, the company can provide a small portion of the technology needed. The company is also willing to assist the project team with anything in terms of the question needed in the system.

Technical Feasibility

The main goal of the proposed project is to develop a web-based system called Pet Republik for John & Jai Pet Republik Pet Shop company here in Panabo City. The system will have important features such as online ordering of pet foods and accessories, booking of pet grooming services and also a delivery section service that has a low percentage of fees for the users.

Through the environmental scanning on May 5, 2025, the interview in the company revealed that they are fully supporting the proposed pet store application and their company is the basis of the system. As the company John & Jai Pet Republik Pet Shop have two more services, a pet grooming services, the project team included it as a feature in the proposed application, integrating its business process into a digitized customer service. The project team thoroughly examined and mapped out the current business processes of the company to confirm that the proposed system is practical, achievable and well-suited to the business's current capabilities and resources.

B. Work Plan and Schedule Feasibility

Project Development Methodology

The Modified Waterfall Model is a software development that approaches the following specific phases such as gathering requirements, design, implementation, testing and deployment. Unlike the traditional model's, it allows limited revisions to earlier stages and offers more flexibility and reduces the risk of errors during later phases [16], [17]. This model is suitable for the development of Pet Republik Pet Shop as it provides a more structured process while still allowing adjustments based on early feedback [18].

Pet Republik Pet Shop will handle the tasks such as user registration, product listings, service booking and real-time updates, which requires careful planning and accurate execution. The Modified Waterfall Model supports these needs through clear documentation and phase-by-phase testing [19], [20], ensuring its system reliability and reducing the chances of failure upon the deployment. According to recent studies, this model is effective for systems that have clearly defined requirements but may benefit from minor changes during development [21], [22], making it ideal for business applications like Pet Republik that require stability and predictability in implementation.

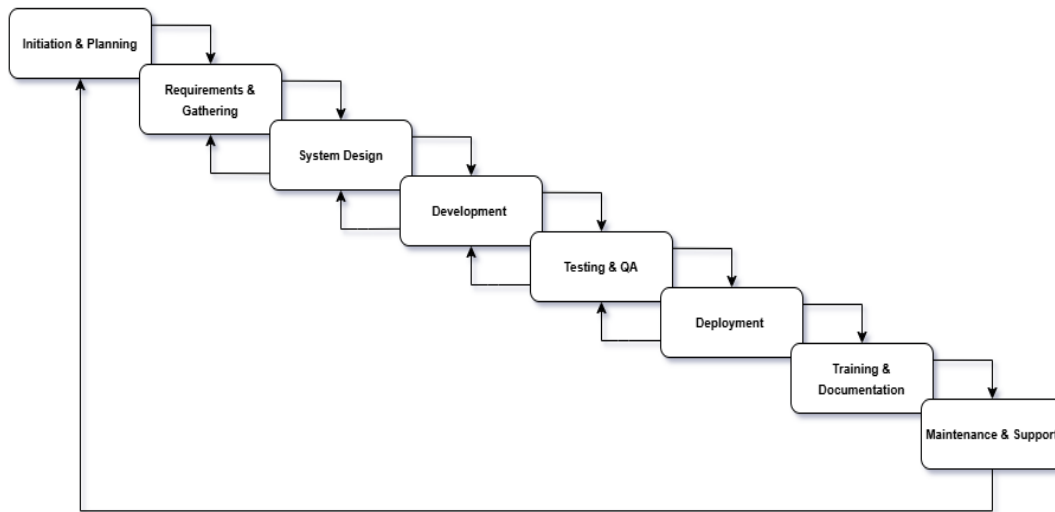


Figure 4. Modified Waterfall Model

Integrated & Proposed New Processes

The integrated and proposed new processes for John & Jai Pet Republik Pet Shop helps to enhance and improve their business sales and customer satisfaction. The figure presents a holistic approach of streamlining the customer, cashier and the admin account management. By enhancing the applications flow, verification to admin level users' oversight and from user account creation, these processes aim to foster a seamless digital experience. The system design ensures that both customers, cashiers and administrators can interact with the system efficiently and effectively. This promotes an ease of access, improves its service delivery and elevates customer satisfaction. From logical progression and verification of system response, the unified workflow supports its business growth and enhances their operational reliability.

Figure 5 shows the login process of the Pet Republik web system for Customers, Cashiers, and Administrators. The process starts when a user opens the system and enters a username and password. The system checks the details. If the information is wrong, the user is asked to enter it again. If it is correct, the system displays the dashboard. After logging in, users can browse the system until

they choose to exit. When they are done, they log out to end the session. Administrators also log in to access the system but have more functions. They can manage user and cashier accounts, create cashier accounts, search and delete user accounts, and generate reports. After completing their tasks, the administrator logs out.

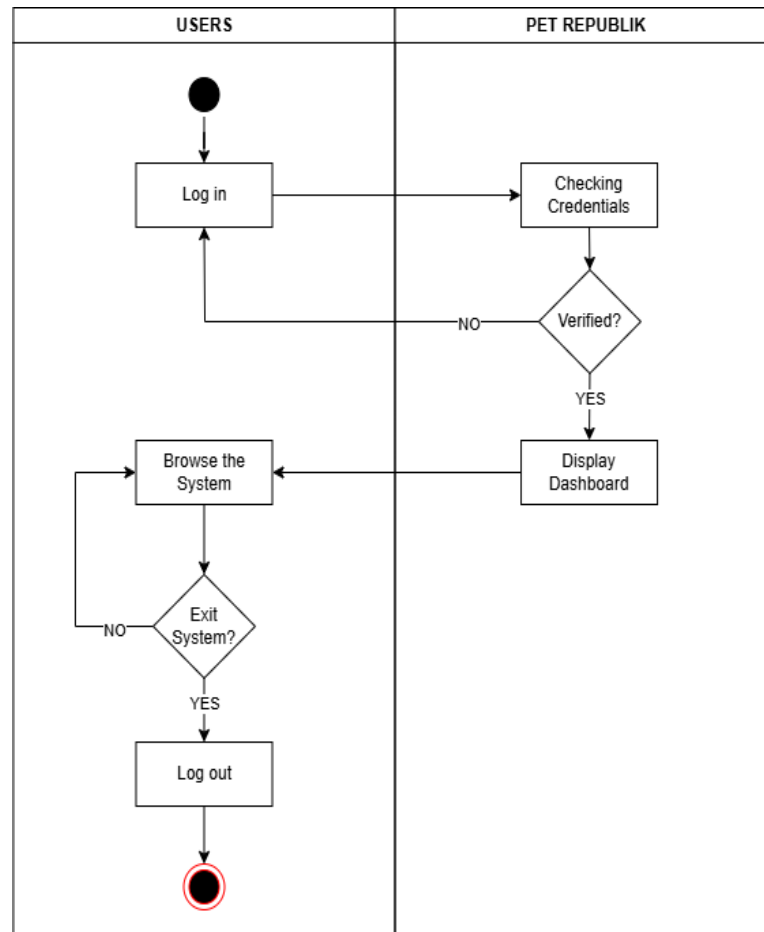


Figure 5. Users log in process

Figure 6 illustrates how the customers place an order through the online system of John & Jai Pet Republik Pet Shop with a Cash-on-Delivery (COD) Model Application. To begin with, the customer accesses the application and browses for pet food or accessories they wish to purchase. If the desired items are available, they can add them to their shopping cart and choose to continue shopping or proceed to check out. Upon reviewing the cart, customers can confirm their

selected items then enter the delivery details, select the Cash-on-Delivery payment method and finalize the order. Once the order is confirmed, the system validates the customer information, processes the request and notifies the store. The pet shop staff or cashier then receives the order details, prepares the items for delivery and dispatches the package. Throughout the process, the system updates the delivery status and keeps the customer informed until the order is successfully delivered and payment is collected. The workflow ensures an efficient and transparent transaction process for both the customer and the business.

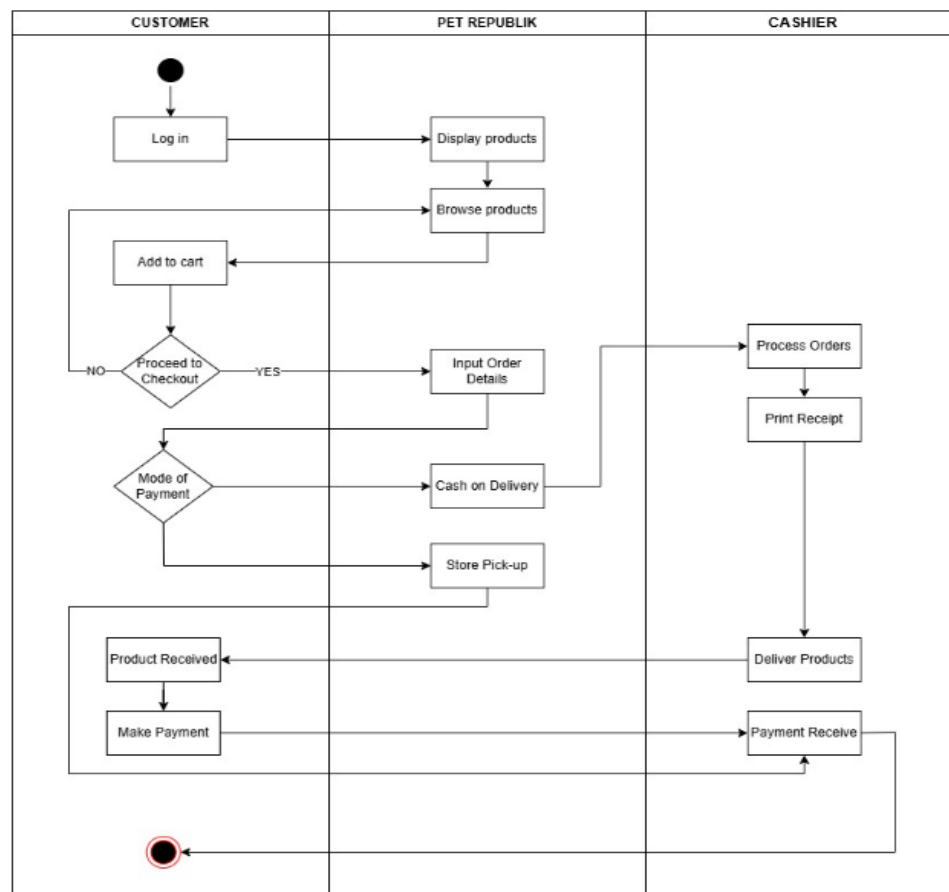


Figure 6. Integrated Online Ordering Process

Figure 7 illustrates the integrated appointment booking process for John & Jai Services, covering pet grooming appointments through a unified digital system. The process begins when the customer logs into the application and accesses the service interface. Within the system, the customer is presented with a selection of

available services along with their corresponding time slots. After browsing, the customer selects their preferred service and desired schedule. The system then validates the selected time for availability. If the chosen slot is not available; the customer may be prompted to choose a different time.

Once a valid slot is confirmed, the customer proceeds to provide necessary personal information such as contact details and service-specific preferences. The system then sends an automated confirmation along with a reminder of the appointment details. On the service provider's side, John & Jai Services receive a notification of the confirmed appointment and begin preparations accordingly. Finally, the customer either attends the scheduled session or drops off their pet for grooming, completing the appointment process. This integrated workflow ensures a streamlined experience, minimizing errors and improving customer convenience through digital scheduling and communication.

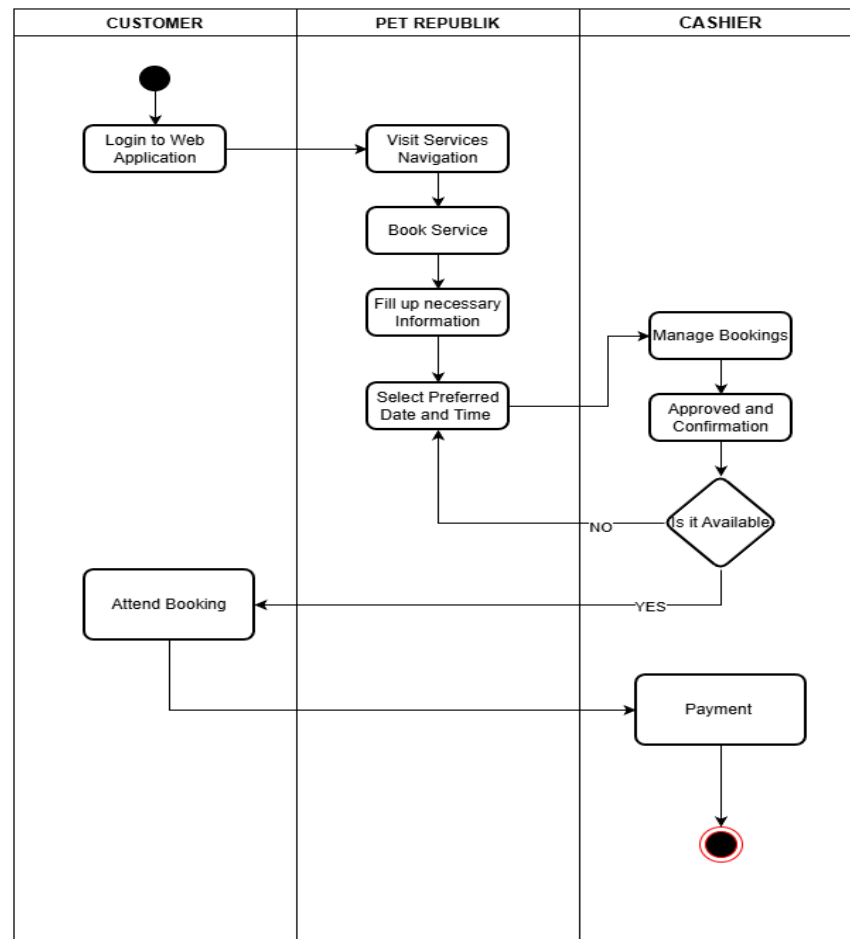


Figure 7. Integrated Appointment Booking Process for JOHN & JAI Services

Development Team

Figure 8 shows the development team led by Project Manager Mark Jhon B. Mamacos (PM) and includes several key roles such as Sponsor, Adviser, Analyst, Designer, and Programmer. With the guidance of the Project Adviser, Mr. Rodrigo Jr. B. Sumuob, the team was able to create effective plans to develop a reliable system that meets user needs. The team demonstrates strategic leadership, technical expertise, and effective communication in achieving the project goals and timelines.

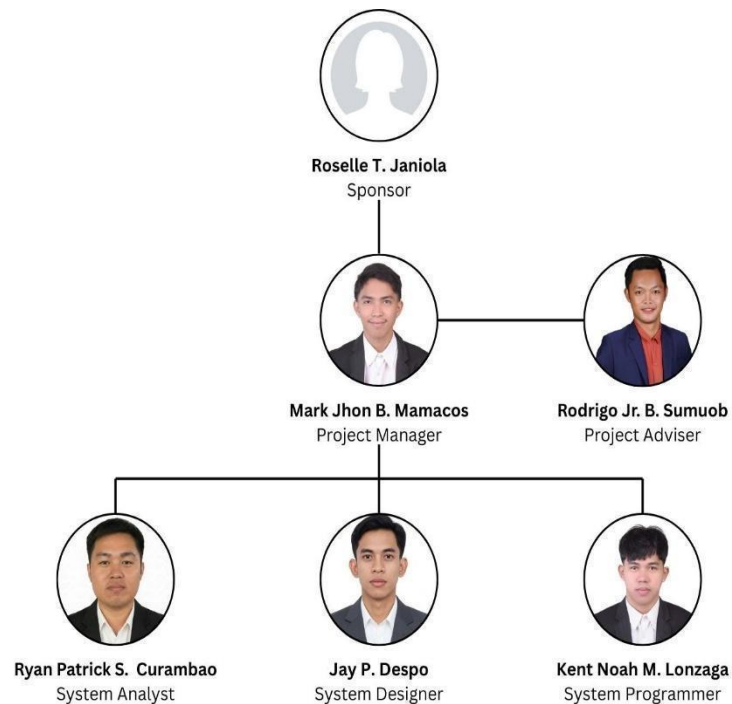


Figure 8. Development Team

Work Breakdown Structure

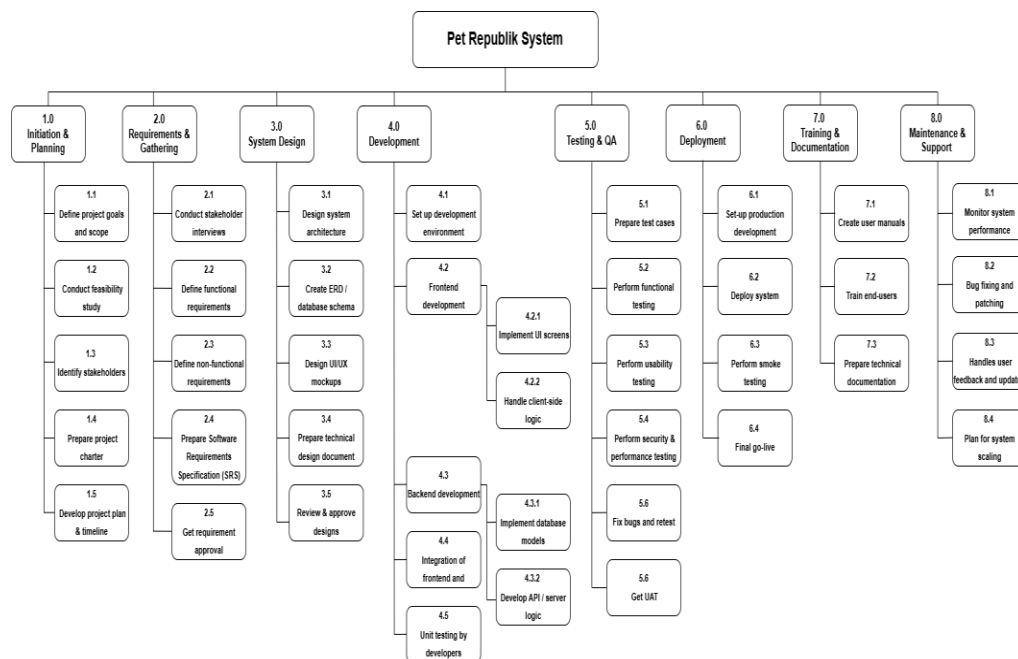
The Work Breakdown Structure (WBS) for the Pet Republik System presents a comprehensive, step-by-step breakdown of the system development process, emphasizes the eight major phases: initiation and planning, requirements gathering, system design, development, testing and quality assurance, deployment, training and documentation, and maintenance and support. It starts with defining the project goals, identifying the stakeholders, and making a timeline for well-organized and time-achievable tasks, then followed by the collection of functional and non-functional requirements. The design phase focuses on system architecture, database schema, and UI/UX mockups, which helps the development of frontend and backend components. Testing ensures that the system meets quality standards through functional, usability, and performance evaluations. Deployment makes the system for operational use, while training and

documentation prepare users for the adoption of the system. The process ends with ongoing maintenance, updates, and scalability planning, ensuring the system remains effective and reliable after launch.

Figure 9. Work Breakdown Structure

WBS Dictionary

The Work Breakdown Structure (WBS) Dictionary of the Pet Republik System is aligned with the Modified Waterfall Model methodology used in the development of the project. The phases presented in the WBS, such as initiation and planning,



requirements gathering, system design, development, testing and quality assurance, deployment, training and documentation, and maintenance and support, reflect the structured and sequential process of the Modified Waterfall Model.

The methodology was chosen because it provides a systematic approach in developing the system while allowing limited revisions and improvements when necessary. Each phase in the WBS Dictionary is organized according to the completion of the previous phase to ensure that the requirements, designs, and

functionalities are properly analyzed, developed, tested, and validated before proceeding to the next stage. Through this approach, the project team was able to maintain proper documentation, minimize development errors, and ensure the quality and reliability of the proposed system.

Furthermore, the methodology is suitable for the Pet Republik System because the project has clearly defined business processes and system requirements. The WBS Dictionary serves as a guide in managing the project tasks, schedules, responsibilities, and deliverables throughout the system development lifecycle.

The following are the meanings of the abbreviated task owner:

PM- Project Manager

P- Programmer

SD- System Designer

PS - Project Sponsor

SA- System Analyst

Table 4. WBS Dictionary

WBS Code	Task Name	Start Date	End Date	Roles	Mandays
1.0	Initiation & Planning				11.50
1.1	Define project goals and scope				2.50
1.1.1	Identify the main purpose of the project	03-Apr-25	04-Apr-25	PM	0.60
1.1.2	Determine project objectives and expected outcomes	04-Apr-25	05-Apr-25	PM	0.30
1.1.3	Define system features and functionalities	06-Apr-25	08-Apr-25	PM	0.60
1.1.4	Identify project limitations and boundaries	09-Apr-25	09-Apr-25	PM	0.40
1.1.5	Gather initial project requirements	10-Apr-25	12-Apr-25	PM	0.30
1.1.6	Document project scope statement	13-Apr-25	14-Apr-25	PM	0.30

1.2	Conduct feasibility study				3.30
1.2.1	Conduct technical feasibility analysis	15-Apr-25	17-Apr-25	SA, PS	1.00
1.2.2	Analyze operational feasibility	17-Apr-25	18-Apr-25	SA, PS	0.70
1.2.3	Evaluate economic/cost feasibility	18-Apr-25	19-Apr-25	SA, PS	0.30
1.2.4	Assess schedule feasibility	19-Apr-25	20-Apr-25	SA, PS	0.50
1.2.5	Identify possible project risks	21-Apr-25	21-Apr-25	SA, PS	0.30
1.2.6	Prepare feasibility study report	22-Apr-25	22-Apr-25	SA, PS	0.50
1.3	Identify stakeholders and form the team				1.60
1.3.1	Identify internal stakeholders	23-Apr-25	23-Apr-25	PM, SA	0.40
1.3.2	Identify external stakeholders	23-Apr-25	23-Apr-25	PM, SA	0.40
1.3.3	Define roles and responsibilities	24-Apr-25	24-Apr-25	PM, SA	0.20
1.3.4	Conduct initial team meeting	24-Apr-25	24-Apr-25	PM, SA	0.40
1.3.5	Establish communication channels	24-Apr-25	24-Apr-25	PM, SA	0.20
1.4	Prepare project charter				1.60
1.4.1	Define project vision and mission	25-Apr-25	25-Apr-25	PM, SA	0.30
1.4.2	Document project objectives and scope	25-Apr-25	25-Apr-25	PM, SA	0.20
1.4.3	Identify key stakeholders and sponsors	25-Apr-25	25-Apr-25	PM, SA	0.30

1.4.4	Define project deliverables	26-Apr-25	26-Apr-25	PM, SA	0.20
1.4.5	Establish project budget and resources	26-Apr-25	26-Apr-25	PM, SA	0.40
1.5.6	Obtain approval/sign-off for project charter	26-Apr-25	26-Apr-25	PM, SA	0.20
1.5	Develop project plan & timeline				2.50
1.5.1	Create Work Breakdown Structure (WBS)	27-Apr-25	27-Apr-25	PM	0.40
1.5.2	Identify project tasks and activities	27-Apr-25	27-Apr-25	PM	0.20
1.5.3	Estimate task durations and resources	27-Apr-25	27-Apr-25	PM	0.20
1.5.4	Develop project schedule and timeline	28-Apr-25	28-Apr-25	PM	0.40
1.5.5	Assign responsibilities to team members	28-Apr-25	28-Apr-25	PM	0.20
1.5.6	Prepare risk management and contingency plans	02-May-25	03-May-25	PM	0.50
1.5.7	Review and finalize project plan	03-May-25	04-May-25	PM	0.60
2.0	Requirements Gathering				14.70
2.1	Conduct stakeholder interviews				2.40
2.1.1	Identify stakeholders to be interviewed	05-May-25	05-May-25	SA	0.30
2.1.2	Prepare interview questionnaires	05-May-25	05-May-25	SA	0.50
2.1.3	Schedule interview sessions	06-May-25	06-May-25	SA	0.40
2.1.4	Conduct stakeholder interviews	07-May-25	07-May-25	SA	0.20

2.1.5	Record and analyze stakeholder feedback	08-May-25	09-May-25	SA	0.60
2.1.6	Summarize gathered requirements	09-May-25	10-May-25	SA	0.40
2.2	Define functional requirements				2.20
2.2.1	Identify core system functionalities	11-May-25	11-May-25	SA, PM	0.50
2.2.2	Define user roles and permissions	12-May-25	12-May-25	SA, PM	0.40
2.2.3	Specify input and output requirements	12-May-15	12-May-25	SA, PM	0.40
2.2.4	Document process workflows	13-May-25	14-May-25	SA, PM	0.20
2.2.5	Create use cases/user stories	15-May-25	16-May-25	SA, PM	0.40
2.2.6	Review functional requirements with stakeholders	18-May-25	20-May-25	SA, PM	0.30
2.3	Define non-functional requirements				2.60
2.3.1	Define system performance requirements	22-May-25	22-May-25	SA	0.30
2.3.2	Identify security requirements	22-May-25	22-May-25	SA	0.30
2.3.3	Define usability and accessibility standards	22-May-25	22-May-25	SA	0.20
2.3.4	Specify scalability and reliability requirements	23-May-25	23-May-25	SA	0.30
2.3.5	Determine compatibility requirements	23-May-25	23-May-25	SA	0.50
2.3.6	Document maintenance and support requirements	24-May-25	24-May-25	SA	1.00
2.4	Prepare Software Requirements Specification (SRS)				3.60

2.4.1	Compile gathered requirements	25-May-25	25-May-25	SA	0.40
2.4.2	Organize functional and non-functional requirements	25-May-25	25-May-25	SA	0.40
2.4.3	Prepare system descriptions and diagrams	26-May-25	26-May-25	SA	1.00
2.4.4	Document assumptions and constraints	27-May-25	27-May-25	SA	0.30
2.4.5	Review and revise the SRS document	27-May-25	27-May-25	SA	0.50
2.4.6	Finalize the SRS documentation	28-May-25	30-May-25	SA	1.00
2.5	Get requirement approval				3.90
2.5.1	Present requirements to stakeholders	01-Jun-25	01-Jun-15	PS, PM	0.30
2.5.2	Conduct review meetings	01-Jun-25	01-Jun-25	PS, PM	0.50
2.5.3	Gather feedback and requested revisions	02-Jun-25	02-Jun-25	PS, PM	1.00
2.5.4	Update requirement documents	03-Jun-25	03-Jun-25	PS, PM	0.20
2.5.5	Obtain stakeholder approval/signatures	03-Jun-25	03-Jun-25	PS, PM	0.60
2.5.6	Archive approved requirements documentation	04-Jun-25	06-Jun-25	PS, PM	1.30
3.0	System Design				11.30
3.1	Design system architecture				2.90
3.1.1	Identify system components and modules	07-Jun-25	07-Jun-25	SD	1.00
3.1.2	Define system architecture type	07-Jun-25	08-Jun-25	SD	0.50

3.1.3	Design data flow and process flow	08-Jun-25	08-Jun-25	SD	0.40
3.1.4	Define integration and communication methods	09-Jun-25	09-Jun-25	SD	0.40
3.1.5	Create system architecture diagrams	09-Jun-25	09-Jun-25	SD	0.40
3.1.6	Review architecture design with the team	10-Jun-25	12-Jun-25	SD	0.20
3.2	Create ERD / database schema				1.90
3.2.1	Identify system entities and attributes	13-Jun-25	13-Jun-25	SD	0.30
3.2.2	Define relationships between entities	14-Jun-25	14-Jun-25	SD	0.40
3.2.3	Create Entity Relationship Diagram (ERD)	15-Jun-25	15-Jun-25	SD	0.20
3.2.4	Design database tables and structures	16-Jun-25	16-Jun-25	SD	0.60
3.2.5	Define primary and foreign keys	16-Jun-25	17-Jun-25	SD	0.20
3.2.6	Review and optimize database schema	17-Jun-25	18-Jun-25	SD	0.20
3.3	Design UI/UX mockups				2.00
3.3.1	Gather UI/UX design requirements	18-Jun-25	18-Jun-25	SD	0.30
3.3.2	Create wireframes for system screens	18-Jun-25	18-Jun-25	SD	0.50
3.3.3	Design user interface layouts	19-Jun-25	20-Jun-25	SD	0.30
3.3.4	Develop interactive mockups/prototypes	21-Jun-25	21-Jun-25	SD	0.20
3.3.5	Apply branding, colors, and typography	22-Jun-25	22-Jun-25	SD	0.40

3.3.6	Conduct usability review and revisions	23-Jun-25	25-Jun-25	SD	0.30
3.4	Prepare technical design document				2.20
3.4.1	Compile architecture and database designs	26-Jun-25	26-Jun-25	SA	0.40
3.4.2	Document system modules and functionalities	26-Jun-25	26-Jun-25	SA	0.40
3.4.3	Define API and integration specifications	27-Jun-25	27-Jun-25	SA	0.30
3.4.4	Include UI/UX design references	27-Jun-25	27-Jun-25	SA	0.20
3.4.5	Document technical standards and tools	28-Jun-25	28-Jun-25	SA	0.30
3.4.6	Review and finalize technical design document	29-Jun-25	30-Jun-25	SA	0.60
3.5	Review & approve designs				2.30
3.5.1	Present system designs to stakeholders	01-Jul-25	01-Jul-25	PM, SD	0.40
3.5.2	Conduct design review meetings	01-Jul-25	01-Jul-25	PM, SD	0.40
3.5.3	Gather feedback and suggestions	02-Jul-25	02-Jul-25	PM, SD	0.30
3.5.4	Revise designs based on feedback	03-Jul-25	03-Jul-25	PM, SD	0.20
3.5.5	Validate final design documents	04-Jul-25	04-Jul-25	PM, SD	0.60
3.5.6	Obtain approval/sign-off for designs	05-Jul-25	06-Jul-25	PM, SD	0.40
4.0	Development				23.50
4.1	Set up development environment				2.30

4.1.1	Install required development tools and software	07-Jul-25	07-Jul-25	SP	0.40
4.1.2	Configure development frameworks and libraries	08-Jul-25	08-Jul-25	SP	0.60
4.1.3	Set up version control repository	09-Jul-25	10-Jul-25	SP	0.50
4.1.4	Configure local development server	11-Jul-25	12-Jul-25	SP	0.30
4.1.5	Set up database environment	13-Jul-25	13-Jul-25	SP	0.30
4.1.6	Test and verify development environment setup	14-Jul-25	15-Jul-25	SP	0.20
4.2	Frontend development				4.20
4.2.1	Analyze UI/UX design specifications	16-Jul-25	16-Jul-25	SP	0.40
4.2.2	Develop frontend project structure	17-Jul-25	24-Jul-25	SP	2.50
4.2.3	Implement responsive user interface components	24-Jul-25	25-Jul-25	SP	0.30
4.2.4	Integrate frontend libraries and dependencies	26-Jul-25	27-Jul-25	SP	0.20
4.2.5	Optimize frontend performance	28-Jul-25	30-Jul-25	SP	0.30
4.2.6	Conduct frontend functionality testing	01-Aug-25	03-Aug-25	SP	0.50
4.2.1	Implement UI screens				2.00
4.2.1.1	Develop login and registration screens	04-Aug-25	05-Aug-25	SP	0.40
4.2.1.2	Create dashboard and navigation screens	05-Aug-25	06-Aug-25	SP	0.20
4.2.1.3	Develop forms and input interfaces	06-Aug-25	07-Aug-25	SP	0.30

4.2.1.4	Implement product/service display pages	08-Aug-25	09-Aug-25	SP	0.60
4.2.1.5	Create user profile and settings pages	10-Aug-25	11-Aug-25	SP	0.30
4.2.1.6	Test UI responsiveness and consistency	12-Aug-25	13-Aug-25	SP	0.20
4.2.2	Handle client-side logic				1.90
4.2.2.1	Implement form validation	13-Aug-25	15-Aug-25	SP	0.30
4.2.2.2	Handle user interactions and events	15-Aug-25	16-Aug-25	SP	0.20
4.2.2.3	Manage frontend state and data flow	17-Aug-25	17-Aug-25	SP	0.40
4.2.2.4	Implement authentication handling	18-Aug-25	19-Aug-25	SP	0.20
4.2.2.5	Connect frontend to APIs	19-Aug-25	20-Aug-25	SP	0.60
4.2.2.6	Debug and optimize client-side scripts	20-Aug-25	22-Aug-25	SP	0.20
4.3	Backend development				5.00
4.3.1	Set up backend project structure	23-Aug-25	23-Aug-25	SP	1.00
4.3.2	Configure server and framework settings	24-Aug-25	26-Aug-25	SP	1.60
4.3.3	Implement business logic and functionalities	26-Aug-25	28-Aug-25	SP	1.50
4.3.4	Configure database connections	28-Aug-25	29-Aug-25	SP	0.20
4.3.5	Implement security and authentication features	29-Aug-25	30-Aug-25	SP	0.50
4.3.6	Perform backend debugging and testing	31-Aug-25	01-Sep-25	SP	0.20

4.3.1	Implement database models				2.40
4.3.1.1	Create database tables and collections	02-Sep-25	03-Sep-25	SP	0.40
4.3.1.2	Define entity relationships	04-Sep-25	05-Sep-25	SP	0.20
4.3.1.3	Implement database constraints and validations	06-Sep-25	08-Sep-25	SP	1.00
4.3.1.4	Create migration scripts	09-Sep-25	12-Sep-25	SP	0.40
4.3.1.5	Seed initial database data	13-Sep-25	17-Sep-25	SP	0.20
4.3.1.6	Test database operations	18-Sep-25	20-Sep-25	SP	0.20
4.3.2	Develop API / server logic				1.80
4.3.2.1	Design API endpoints	21-Sep-25	22-Sep-25	SP	0.30
4.3.2.2	Implement CRUD operations	23-Sep-25	25-Sep-25	SP	0.50
4.3.2.3	Handle server-side validation	26-Sep-25	27-Sep-25	SP	0.20
4.3.2.4	Implement authentication and authorization	28-Sep-25	29-Sep-25	SP	0.40
4.3.2.5	Integrate third-party services/APIs	29-Sep-25	30-Sep-25	SP	0.20
4.3.2.6	Test API functionality and responses	30-Sep-25	01-Oct-25	SP	0.20
4.4	Integration of frontend and backend				1.60
4.4.1	Connect frontend interfaces to backend APIs	05-Oct-25	07-Oct-25	SP, SA	0.20
4.4.2	Test data flow between client and server	08-Oct-25	10-Oct-25	SP, SA	0.40

4.4.3	Handle API response and error management	11-Oct-25	12-Oct-25	SP, SA	0.30
4.4.4	Integrate authentication and session handling	13-Oct-25	18-Oct-25	SP, SA	0.30
4.4.5	Optimize system performance and communication	23-Oct-25	24-Oct-25	SP, SA	0.20
4.4.6	Conduct integration testing	24-Oct-15	26-Oct-25	SP, SA	0.20
4.5	Unit testing by developers				2.30
4.5.1	Prepare unit test cases	27-Oct-25	27-Oct-25	SP	0.30
4.5.2	Test frontend components and functions	27-Oct-25	27-Oct-25	SP	0.40
4.5.3	Test backend modules and APIs	28-Oct-25	28/20/25	SP	0.50
4.5.4	Identify and fix bugs/errors	28-Oct-25	29-Oct-25	SP	0.30
4.5.5	Re-test corrected functionalities	29-Oct-25	30-Oct-25	SP	0.20
4.5.6	Document unit testing results	03-Nov-25	03-Nov-25	SP	0.60
5.0	Testing & QA				10.80
5.1	Prepare test cases and scenarios				1.70
5.1.1	Identify system features to be tested	04-Nov-25	04-Nov-25	SA	0.20
5.1.2	Define testing objectives and scope	04-Nov-25	04-Nov-25	SA	0.20
5.1.3	Create functional test cases	04-Nov-25	04-Nov-25	SA	0.40
5.1.4	Prepare test scenarios and test data	05-Nov-25	05-Nov-25	SA	0.40

5.1.5	Review and validate test cases	05-Nov-25	05-Nov-25	SA	0.20
5.1.6	Organize testing documentation	05-Nov-25	06-Nov-25	SA	0.30
5.2	Perform functional testing				1.60
5.2.1	Execute functional test cases	06-Nov-25	06-Nov-25	SA	0.20
5.2.2	Verify system features and workflows	06-Nov-25	06-Nov-25	SA	0.20
5.2.3	Test input validation and outputs	07-Nov-25	07-Nov-25	SA	0.30
5.2.4	Identify functional defects and issues	08-Nov-25	09-Nov-25	SA	0.30
5.2.5	Record testing results and observations	09-Nov-25	09-Nov-25	SA	0.20
5.2.6	Report functional testing findings	10-Nov-25	10-Nov-25	SA	0.40
5.3	Perform usability testing				1.70
5.3.1	Prepare usability testing criteria	10-Nov-25	10-Nov-25	SA	0.20
5.3.2	Select users/test participants	11-Nov-25	11-Nov-25	SA	0.40
5.3.3	Conduct usability testing sessions	11-Nov-25	12-Nov-25	SA	0.20
5.3.4	Observe user interactions and feedback	12-Nov-25	12-Nov-25	SA	0.20
5.3.5	Identify usability issues and improvements	13-Nov-25	14-Nov-25	SA	0.20
5.3.6	Document usability testing results	14-Nov-25	15-Nov-25	SA	0.50
5.4	Perform security and performance testing				1.50

5.4.1	Conduct authentication and access control testing	15-Nov-25	16-Nov-15	SA	0.20
5.4.2	Test data security and protection measures	17-Nov-25	19-Nov-25	SA	0.30
5.4.3	Perform vulnerability assessment	20-Nov-25	22-Nov-25	SA	0.30
5.4.4	Conduct load and stress testing	22-Nov-25	23-Nov-25	SA	0.20
5.4.5	Measure system response time and performance	24-Nov-25	25-Nov-25	SA	0.20
5.4.6	Document security and performance test results	26-Nov-25	28-Nov-25	SA	0.30
5.5	Fix bugs and retest				1.70
5.5.1	Review reported bugs and issues	03-Dec-25	05-Dec-25	PS, SA	0.20
5.5.2	Prioritize defects based on severity	06-Dec-25	08-Dec-25	PS, SA	0.30
5.5.3	Fix identified system bugs	10-Dec-25	14-Dec-25	PS, SA	0.20
5.5.4	Perform regression testing	15-Dec-25	16-Dec-25	PS, SA	0.40
5.5.5	Retest corrected functionalities	20-Dec-25	24-Dec-25	PS, SA	0.40
5.5.6	Update bug tracking documentation	26-Dec-25	28-Dec-25	PS, SA	0.20
5.6	Get UAT (User Acceptance Testing) approval				2.60
5.6.1	Prepare UAT environment and materials	12-Jan-26	16-Jan-26	PS, SA	0.20
5.6.2	Conduct User Acceptance Testing sessions	18-Jan-26	20-Jan-26	PS, SA	0.20
5.6.3	Gather user feedback and observations	22-Jan-26	26-Jan-26	PS, SA	0.40

5.6.4	Resolve final issues and revisions	02-Feb-26	10-Feb-26	PS, SA	0.70
5.6.5	Present final system for approval	12-Feb-26	19-Feb-26	PS, SA	0.50
5.6.6	Obtain UAT sign-off and approval documentation	20-Feb-26	28-Feb-26	PS, SA	0.60
6.0	Deployment				40.40
6.1	Set up production environment				5.00
6.1.1	Prepare production server and hosting environment	09-Mar-26	09-Mar-26	SP	0.50
6.1.2	Configure domain and network settings	09-Mar-26	09-Mar-26	SP	0.50
6.1.3	Set up production database	10-Mar-26	10-Mar-26	SP	0.50
6.1.4	Configure security settings and backups	10-Mar-26	10-Mar-26	SP	0.50
6.1.5	Install required software and dependencies	11-Mar-26	11-Mar-26	SP	1.00
6.1.6	Verify production environment readiness	11-Mar-26	12-Mar-26	SP	2.00
6.2	Deploy system				5.00
6.2.1	Prepare deployment package/build files	13-Mar-26	13-Mar-26	SP	1.00
6.2.2	Upload system files to production server	14-Mar-26	14-Mar-26	SP	0.50
6.2.3	Configure application settings in production	14-Mar-26	14-Mar-26	SP	0.50
6.2.4	Connect production database and services	15-Mar-26	15-Mar-26	SP	1.00
6.2.5	Perform deployment verification checks	16-Mar-26	16-Mar-26	SP	1.00

6.2.6	Document deployment procedures and status	17-Mar-26	17-Mar-26	SP	1.00
6.3	Perform smoke testing				4.40
6.3.1	Prepare smoke test checklist	18-Mar-26	18-Mar-26	SA	0.50
6.3.2	Verify system startup and accessibility	18-Mar-26	18-Mar-26	SA	0.50
6.3.3	Test critical system functionalities	19-Mar-26	19-Mar-26	SA	0.40
6.3.4	Check database and API connectivity	19-Mar-26	19-Mar-26	SA	1.00
6.3.5	Identify and resolve deployment issues	19-Mar-26	19-Mar-26	SA	1.00
6.3.6	Document smoke testing results	19-Mar-26	20-Mar-26	SA	1.00
6.4	Final go-live				26.00
6.4.1	Announce system launch schedule	21-Mar-26	23-Mar-26	PM, PS	3.00
6.4.2	Enable public/user access to the system	24-Mar-26	27-Mar-26	PM, PS	4.00
6.4.3	Monitor system performance after launch	28-Mar-26	30-Mar-26	PM, PS	3.00
6.4.4	Provide initial technical support	02-Apr-26	14-Apr-26	PM, PS	13.00
6.4.5	Address immediate post-launch issues	15-Apr-26	15-Apr-26	PM, PS	1.00
6.4.6	Confirm successful system go-live and operations	16-Apr-26	17-Apr-26	PM, PS	2.00
7.0	Training & Documentation				7.30
7.1	Create user manuals				3.30

7.1.1	Identify user guide requirements	18-Apr-26	18-Apr-26	SA	0.30
7.1.2	Prepare step-by-step system instructions	18-Apr-26	19-Apr-26	SA	0.40
7.1.3	Create screenshots and visual references	19-Apr-26	20-Apr-26	SA	0.40
7.1.4	Document common issues and troubleshooting steps	20-Apr-26	21-Apr-26	SA	1.00
7.1.5	Review and revise user manuals	22-Apr-26	22-Apr-26	SA	1.00
7.1.6	Finalize and distribute user manuals	22-Apr-26	23-Apr-26	SA	0.20
7.2	Train end-users				2.20
7.2.1	Prepare training materials and presentations	24-Apr-26	24-Apr-26	SA	0.20
7.2.2	Schedule user training sessions	24-Apr-26	24-Apr-26	SA	0.20
7.2.3	Demonstrate system functionalities	24-Apr-26	24-Apr-26	SA	0.20
7.2.4	Conduct hands-on practice activities	25-Apr-26	25-Apr-26	SA	0.50
7.2.5	Answer user questions and concerns	25-Apr-26	25-Apr-26	SA	0.50
7.2.6	Evaluate training effectiveness and feedback	25-Apr-26	26-Apr-26	SA	0.60
7.3	Prepare technical documentation				1.80
7.3.1	Document system architecture and design	27-Apr-26	27-Apr-26	SA	0.20
7.3.2	Prepare database and API documentation	27-Apr-26	27-Apr-26	SA	0.20
7.3.3	Document installation and deployment procedures	28-Apr-26	28-Apr-26	SA	0.30

7.3.4	Create maintenance and troubleshooting guides	28-Apr-26	28-Apr-26	SA	0.30
7.3.5	Review and organize technical documents	29-Apr-26	29-Apr-26	SA	0.40
7.3.6	Finalize and archive technical documentation	30-Apr-26	30-Apr-26	SA	0.40
8.0	Maintenance & Support				5.70
8.1	Monitor system performance				1.60
8.1.1	Track system uptime and availability	05-May-26	05-May-26	SP	0.20
8.1.2	Monitor server and database performance	05-May-26	05-May-26	SP	0.20
8.1.3	Analyze system logs and error reports	06-May-26	06-May-26	SP	0.30
8.1.4	Check application response times	06-May-26	06-May-26	SP	0.30
8.1.5	Identify performance bottlenecks	07-May-26	08-May-26	SP	0.40
8.1.6	Prepare performance monitoring reports	09-May-26	10-May-26	SP	0.20
8.2	Bug fixing and patching				1.90
8.2.1	Receive and document reported bugs	11-May-26	11-May-26	SP	0.30
8.2.2	Analyze root causes of issues	12-May-26	12-May-26	SP	0.20
8.2.3	Develop fixes and security patches	13-May-26	14-May-26	SP	0.40
8.2.4	Test implemented fixes	15-May-26	16-May-26	SP	0.40
8.2.5	Deploy patches to production environment	17-May-26	18-May-26	SP	0.20

8.2.6	Update maintenance and patch records	19-May-26	20-May-26	SP	0.40
8.3	Handle user feedback and updates				1.30
8.3.1	Collect user feedback and suggestions	20-May-26	21-May-26	SA	0.20
8.3.2	Review and prioritize requested improvements	22-May-26	22-May-26	SA	0.20
8.3.3	Plan and implement system updates	22-May-26	22-May-26	SA	0.20
8.3.4	Notify users about updates and changes	23-May-26	23-May-26	SA	0.30
8.3.5	Evaluate user satisfaction after updates	24-May-26	24-May-26	SA	0.20
8.3.6	Document feedback and update history	25-May-26	26-May-26	SA	0.20
8.4	Plan for system scaling or future enhancements				0.90
8.4.1	Identify future feature requirements	25-May-26	25-May-26	PM, PS	0.20
8.4.2	Plan infrastructure and scalability improvements	26-May-26	27-May-26	PM, PS	0.30
8.4.3	Evaluate new technologies and integrations	28-May-26	28-May-26	PM, PS	0.20
8.4.4	Review and approve future development plans	29-May-26	30-May-26	PM, PS	0.20

Gantt Chart

Figure 10 shows the project's Gantt chart, which presents the project phases, schedules, start dates, finish dates, and duration of each activity involved in the system development process. The Gantt chart serves as a visual tool for project management, allowing the team to monitor the progress and sequence of tasks throughout the project timeline.

The chart outlines the project schedule from April 2025 to May 2026 and

consists of eight major phases: Initiation and Planning, Requirements Gathering, System Design, Development, Testing and Quality Assurance, Deployment, Training and Documentation, and Maintenance and Support.

The figure shows these activities throughout the project duration, with the Development phase being the most extended stage, running from July 2025 to November 2025, followed by the Testing and QA phase. The project concludes with Deployment, Training and Documentation, and Maintenance and Support activities, which ensure successful implementation, operation, and continuous improvement of the system.

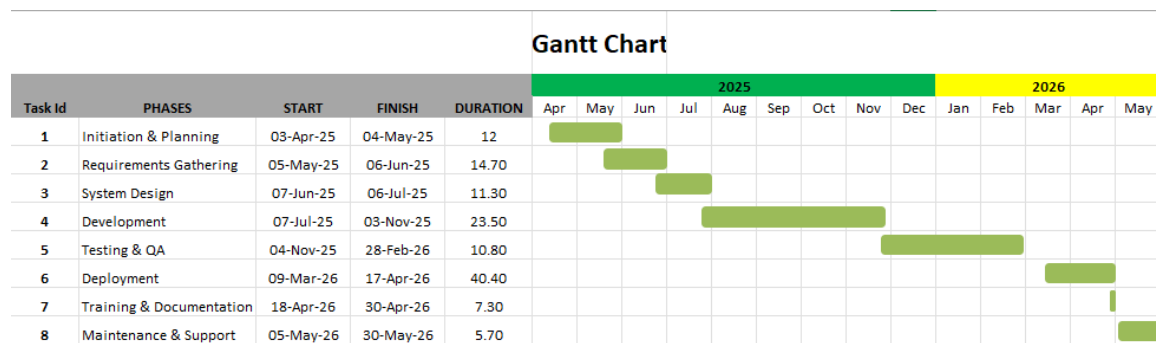


Figure 10. Gantt Chart

C. Financial Plan and Economic Feasibility

The Financial Plan and Economic Feasibility section presents the estimated budget allocation and financial requirements needed for the development and implementation of the proposed Pet Republik Information System. It outlines the planned distribution of resources, including manpower, hardware, software, and other operational expenses, to ensure the successful completion of the project within the allocated schedule and budget. This section also evaluates the economic feasibility of the system by determining whether the proposed project is financially practical, sustainable, and beneficial for John & Jai Pet Republik Pet Shop in improving its business operations and services.

Budget Allocation and Schedule of the Proposed Information System

Table 5 presents a detailed monthly breakdown of the project costs across categories such as WBS, technology, auditing cost and effort costs, totaling a ₱388,241.65 over 14 months. It shows how the resources are being distributed across time, helping monitor the cumulative expenses to ensure that the project stays within budget. The table provides a clear overview for its stakeholders to track financial progress, anticipate upcoming costs and align its budgeting with project milestones.

Table 5. Budget Allocation and Schedule

Year	Month	WBS Cost	Technology Cost	Audit Cost	Effort Cost	Total Cost	Cumulative Cost
2025	April	₱26,429.30		₱5,000.00	₱2,642.93	₱34,072.23	₱34,072.23
	May	₱24,302.40	₱1,830.00	₱5,000.00	₱2,430.24	₱33,562.64	₱67,634.87
	June	₱25,524.20		₱5,000.00	₱2,552.42	₱33,076.62	₱100,711.49
	July	₱17,907.40		₱5,000.00	₱1,790.74	₱24,698.14	₱125,409.63
	August	₱12,512.00		₱5,000.00	₱1,251.20	₱18,763.20	₱144,172.83
	September	₱5,712.00		₱5,000.00	₱571.20	₱11,283.20	₱155,456.03
	October	₱7,232.00		₱5,000.00	₱723.20	₱12,955.20	₱168,411.23
	November	₱10,858.50		₱5,000.00	₱1,085.85	₱16,944.35	₱185,355.58
	December	₱2,626.50		₱5,000.00	₱262.65	₱7,889.15	₱193,244.73
2026	January	₱1,236.00			₱123.60	₱1,359.60	₱194,604.33
	February	₱2,781.00			₱278.10	₱3,059.10	₱197,663.43

	March	₱43,478.00		₱5,000.00	₱4,347.80	₱52,825.80	₱ 250,489.23
	April	₱48,206.50	₱ 64,995.00	₱5,000.00	₱4,820.65	₱123,022.15	₱ 373,511.38
	May	₱ 8,845.70		₱5,000.00	₱ 884.57	₱14,730.27	₱ 388,241.65
	TOTAL	₱237,651.50	₱ 66,825.00	₱60,000.00	₱23,765.15	₱388,241.65	

Effort Cost

Table 6 details the work required to complete the task of the project, with particular emphasis on activities such as reporting, meetings and communication that require human labor. Table 6 also shows the work of the system designer, system analyst, system programmer and project manager, all of whom are contributing in some way to the achievement of the project goal. The total effort cost, ₱23,765.15, which includes both the anticipated investment as well as the necessary associated teamwork and project management, was excluded from the overall cost estimate even though the project sponsor's participation is entirely voluntary and free of financial commitment. This cost table shows the breakdown and stresses the important role of financial commitment and human resources in making the project goals come true by including all relevant aspects.

Table 6. Effort Cost

Resources	Communication	Meeting	Reporting	Total
Project Manager	₱ 6,274.88	₱ 3,137.44	₱ 627.49	₱ 10,039.80
Project Sponsor	₱ -	₱ -	₱ -	₱ -
System Programmer	₱ 3,145.00	₱ 1,572.50	₱ 314.50	₱ 5,032.00
System Designer	₱ 1,097.69	₱ 548.84	₱ 109.77	₱ 1,756.30

System Analyst	₱ 4,335.66	₱ 2,167.83	₱ 433.57	₱ 6,937.05
			Total	₱ 23,765.15

Hardware Estimated Cost Vs Actual Cost

Table 7 contains a comparison of the estimated cost and actual cost of hardware components and describes quantities, estimated costs and actual costs of each item, although the estimated cost is ₱64,995.00 both items are specified as provided by the company and there are no additional costs.

Table 7. Hardware Estimated Cost Vs Actual Cost

Hardware	Quantity	Unit Cost	Actual Cost
DELL Inspiron 3030-I714700 Intel® Core™ i7 14th Gen. Desktop	1	₱ 64,995.00	Provided
		Total Cost	

Software Estimated Cost Vs Actual Cost

Table 8 presents a comparison between the estimated cost and the actual cost of software components. An estimated cost of ₱ 1,830.00, the same as the actual cost. Therefore, the total expenditure remains unchanged at ₱ 1,830.00, ensuring reliable hosting for project operations.

Table 8. Software Cost Estimates Vs Actual Cost

Software	Quantity	Unit Price	Actual Cost
XAMPP	1	Free	Free
Laravel Framework	1	Free	Free

Node.js	1	Free	Free
Git	1	Free	Free
Composer	1	Free	Free
PHP	1	Free	Free
Javascript	1	Free	Free
Domain Name	1	60	60
Web Hosting	2	885/Month	₱ 1,770.00
		TOTAL COST	₱ 1,830.00

D. Project Management Plan and Tracking

Stakeholder Analysis

Table 9 shows the list of key stakeholders that benefit the project within the organization. The stakeholders serve as a bridge of the project to improve and maintain the services.

Table 9. Stakeholders Analysis

Stakeholder Name	Impact	Influence	What's important to the stakeholder?	How could the stakeholder contribute to the project?	How could the stakeholder block the Project?	Strategy for engaging the stakeholder?
Roselle T. Janiola Owner	High	High	Provides financial support, establishes strategic goals and makes critical decisions to manage the success and growth of the pet store business.	Approves the project budgeting, defines project goals and ensures alignment with the overall objectives of the pet store business.	She may delay funding or approvals and might introduce changes to project goals during implementation.	Engage the owner through clear and consistent communication by aligning with their business goals, involving them early in the project's plan, setting realistic expectations, providing regular progress updates, seeking feedback and also promptly addressing there any concerns or issues.

Dymple Tacder Danela Carmonas Staff	High	High	Provides the day-to-day process and services, engages with customers, and ensures smooth and efficient operations of the pet store.	Offers realistic feedback on system usability, participates in system testing and also adapts to new workflows and processes to support the successful implementation.	offer inaccurate or incomplete feedback, or neglect to participate in necessary training and potentially hinder the project's progress.	involvement through hands-on training, ongoing communication by recognizing and also incorporating their input to enhance the system progress.
Customers	High	High	Serve as the primary source of revenue; their satisfaction is essential for driving the business growth and long-term success.	Provides feedback on the online ordering system to enhance the users' experience and ensure smoother service delivery.	Might reject or overlook the new proposed system if it is not user-friendly, impacting on its overall success and adoption.	Involve by rolling out the system gradually, gathering its user feedback and also continuously refining the interface to improve the customer's experience.

Project Charter

A project charter for pet store system development is a formal document detailing all important project details and boundaries. The purpose of the project is the improvement of pet store operations through such aspects as online purchasing, customer messages, appointment scheduling, and sales management. The project charter also defines the objectives, scope, stakeholders, deliverables, time frame & required resources. The project charter ensures that

everyone involved with the pet store system development project understands the objectives, limitations, risks, and expected results.

Table 10. Project Charter

Project Name	Pet Republik: A Web-Based Integrated Platform for Online Ordering and Service Appointment Management for John & Jai Company
Project Description	The Pet Republik is a web-based system designed to digitize various aspects of the John & Jai Pet Republik Pet Shop Company. It includes features of an online ordering and appointment booking and pet grooming. This system improves customer communication and overall pet care services. Additionally, it creates an opportunity for the company to boost overall sales.
Project Objectives	The project aims to develop a web-based platform that integrates an online pet store and other services. It aims to enhance customer experience through automated notification, while streamlining their operation and services. By combining e-commerce functionality with the company's services, it greatly improves operational efficiency and deepens the customer relationship that supports future growth of the company.
Project Considered Successfully when:	The project will be considered successful when the Pet Republik system for John & Jai Pet Republik Pet Shop ensures the positive feedback from the project sponsor when implemented and also has a full functionality (online ordering, and appointment booking for pet grooming), and lastly it will be considered successful when it meets the project sponsors needs for their company.

Project Participants	Title: Project Sponsor Project Manager System Analyst System Designer System Programmer	Name: Roselle T. Janiola Mark Jhon B. Mamacos Ryan Patrick S. Curambao Jay P. Despo Kent Noah M. Lonzaga
Available Resources	The project has an allocated budget of ₱237,651.50 to support manpower, software, hardware, development, testing, and deployment activities.	
Potential Risk	System Downtime, Internet Dependency, Security Breaches, User Resistance, data loss and Data Privacy Regulations	
Approval:	Title and Name: Project Sponsor: Roselle T. Janiola _____ Signature Project Manager: Mark Jhon B. Mamacos _____ Signature	

RACI Chart

The RACI is an acronym for Responsible, Accountable, Consulted and Informed. Each individual tasked with a specific task is deemed responsible. The

one who assigns these tasks to the team is called accountable. A person offering advice, feedback, or suggestions is referred to as a consultant. And lastly, the individual who will be informed after completing the task, typically the business owner or advisor, is called the informed.

Table 11. RACI Legends

LEGEND	
Responsible	Tasked with the responsibility of fulfilling the assigned assignment
Accountable	The individual possesses the authority for decision-making and accounting. Only one person is assigned to teach a task.
Consulted	Individuals who advise and have a stake in decision-making process
Informed	It is necessary to have knowledge regarding the decision and progress of the related task, also, the action that has taken

Table 12. RACI Chart

PET REPUBLIK: A WEB-BASED INTEGRATED PLATFORM FOR ONLINE ORDERING AND SERVICE APPOINTMENT MANAGEMENT FOR JOHN & JAI COMPANY IN PANABO CITY					
Main Activity/Task and Milestone	Project Manager	System Analyst	System Designer	System Programmer	Project Sponsor
Initiation & Planning					

Define project goals and scope	A	R	I	I	I
Conduct feasibility study	A	R	I	I	I
Identify stakeholders and form the team	A	R	I	I	I
Prepare project charter	A	R	I	I	I
Develop project plan & timeline	A	R	I	I	c
Requirements Gathering					
Conduct stakeholder's interviews	C	I	R	A	I
Define functional requirements	C	I	R	A	I
Define non-functional requirements	C	I	R	A	I
Prepare software requirements specification	C	I	R	A	I
Get requirement approval	C	I	R	A	I
System Design					
Design system architecture	C	I	A	R	I
Create ERD / database schema	C	I	A	R	I
Design UI/UX mockups	C	I	A	R	I
Prepare technical design document	C	I	A	R	I
Review & approve designs	C	I	A	R	I
Development					

Set up development environment	A	C	R	I	I
Frontend development	A	C	R	I	I
Implement UI screens	A	C	R	I	I
Handle client-side logic	A	C	R	I	I
Backend development	A	C	R	I	I
Implement database models	A	C	R	I	I
Develop API / server logic	I	C	I	A	I
Integration of frontend and backend	I	I	A	A	I
Unit testing by developers	I	R	C	R	I
Testing & QA					
Prepare test cases and scenarios	R	I	C	A	I
Perform functional testing	A	C	I	R	I
Perform usability testing	A	R	I	I	I
Perform security and performance testing	A	C	I	A	C
Fix bugs and retest	I	C	I	R	I
Get UAT	I	I	I	I	I
Deployment					
Set up production environment	I	I	I	A	I
Deploy system	R	C	I	R	C
Perform smoke testing	I	R	I	R	I
Final go-live	A	I	I	C	I

Training & Documentation					
Create user manuals	A	I	I	I	I
Train end-users	R	C	I	I	I
Prepare technical documentation	A	R	I	C	I
Maintenance & Support					
Monitor system performance	I	A	I	C	C
Bug fixing and patching	C	I	I	R	I
Handle user feedback and updates	R	R	I	I	I
Plan for system scaling or future developments	A	R	I	C	I

E. Risk Management Plan

This section outlines the process for identifying and documenting potential risks and their impacts on the system development. In developing the Pet Republik Web-based System for John & Jai Pet Republik Pet Shop, effective risk management is essential to the project's success. When a risk is identified, the project team should assess the possibility of impacts, create a plan to reduce it and consistently monitor and update this plan as necessary. Furthermore, the team should look for opportunities for improvement to prevent similar risks in the future.

Table 13. Risk Management Plan

No.	The Objective of the Project	Issue/s	Risk/s	Type of Risk	Likelihood	Severity	Risk Factor	Proposed Action
1	Create a registration module that allows Admin and Cashier to create and manage customer accounts and information.	Internet outage or power failure	Interrupted registration process	Technical	2	4	12	Implement auto- save functionality and display order confirmation only after full submission.
2	Create an application module that allows user to purchase products on the system.	Product out of stock during checkout	Failed transactions	Operational	4	6	16	Apply Real-time tracking (product, item) updates and alerts seller for out of stocks
3	Develop an application module for service booking in the system.	Double booking of time slots	Customer dissatisfaction	Functional	4	6	10	Implement real- time time slot availability checks with Lock time slots during the booking process

4	Create an application module for product listing.	Duplicate product entries	Confusion for customers	Operational	2	4	8	Implement unique product identifiers and Use automated duplicate detection algorithms
5	Develop a reporting module for real-time purchase notification.	Notification delay or failure	Missed customer engagement	Functional	2	6	12	Implement reliable notification infrastructure alerts
6	Deploy the system in the specific organization.	Insufficient training for staff	User errors	Technical	2	5	8	Conduct hands- on training sessions to the staffs
7	Provide customer access via web system that is accessible in cellphone.	Mobile responsiveness issue	Poor user experience	Technical	4	8	9	Use a responsive front-end framework such as Bootstrap or Tailwind CSS and conduct testing across major browsers and mobile devices to

								ensure system usability on smartphones.
8	Protect customer data, order history, and login credentials.	Data breaches or leaks due to poor database security	Data theft or breach	Security	3	6	10	Apply data encryption, HTTPS, and validation for all user inputs and login sessions.

F. Project Closure Plan

The Project Closure Plan summarizes the development and completion status of the Pet Republik Web System for John & Jai Pet Republik Pet Shop. The report provides essential information about the project status, including a brief description, project sponsor details, the team responsible for the project and the performance of the various objectives.

Table 14. Project Closure Plan

Project Closure Report				
Project Name	Pet Republik: A Web-Based Integrated Platform For Online Ordering and Service Appointment Management for John & Jai Company in Panabo City			
Brief Description	This project Pet Republik: A Web-Based Integrated Platform for Online Ordering and Service Appointment Management for John & Jai Company in Panabo City, aims to streamline various services including pet appointment booking and the online ordering of products.			
Project Sponsor	Roselle T. Janiola			
Prepared by:	BSIS 4 TH YEAR STUDENTS PM - Mark Jhon B. Mamacos SA - Ryan Patrick S.Curambao SDES – Jap P. Despo SDEV – Kent Noah M. Lonzaga			
Objectives Achievement				
Objective	Description	Performance	Not Achieved	Achieved
Online Ordering System	Allows users to sell products, browse products and add it in to cart, check out, and track orders	Very Efficient		✓

Online Service Booking	Allows user to book services (pet appointment)	Very Efficient		✓
Cash On-Delivery Process	Every online transaction will be undergoing Cash On-Delivery	Very Efficient		✓

G. System Analysis

System Requirements (Functional and Nonfunctional)

This section primarily discusses the system requirements, focusing on functional and nonfunctional requirements. Functional requirements define what the system should do, while nonfunctional requirements describe how the system should perform and maintain its quality.

Functional Requirements

- The system shall allow users and admins to register and create accounts.
- The system should authenticate users using a username and password.
- The system shall allow users to recover or reset passwords.
- The system shall allow users to update their name, gmail & passwords.
- Cashiers shall be able to upload, update, and delete products.
- Customers shall be able to browse pet products and services.
- Customers shall be able to add items to a cart and complete purchases.
- The system shall provide a secure checkout and payment process.
- The system shall allow customers to request pet grooming and

delivery services.

- The system shall allow scheduling of service appointments.
- Cashiers shall be able to view, approve, and manage service booking requests.
- The system shall provide real-time notifications for events and updates.
- Customers shall be able to chat with cashiers regarding purchases and inquiries.
- The admin shall be able to manage and delete user accounts.
- The admin shall be able to create cashier accounts.
- The system shall generate reports on transactions, user activity, and service performance.

Non-Functional Requirements

- **Functional Suitability:** The system should provide complete and appropriate features for managing products, services, bookings, transactions, and customer information effectively.
- **Performance Efficiency:** The system must process transactions, administrative tasks, bookings, and reports quickly without delays, even during continuous or multiple-user access.
- **Usability:** The system interface should be user-friendly, easy to understand, and provide clear navigation, labels, and instructions for customers, cashiers, and administrators.
- **Reliability:** The system must consistently record and store transactions, inventory records, bookings, and activity logs accurately while maintaining stable operation.
- **Security:** Security: The system must protect data and allow access only to authorized users.
- **Maintainability:** The system should support easy updates and management without disrupting operations.

Process Model: Use Case Diagram and Description

The use case diagram below represents the core functionalities of the Pet Republik system and the interactions between various types of users and the system itself. The diagram outlines the relationships between actors, namely Admin, Customer, and cashier and their respective use cases within the application. This visualization helps in understanding the scope of the system and how each user's role contributes to the workflow of the platform.

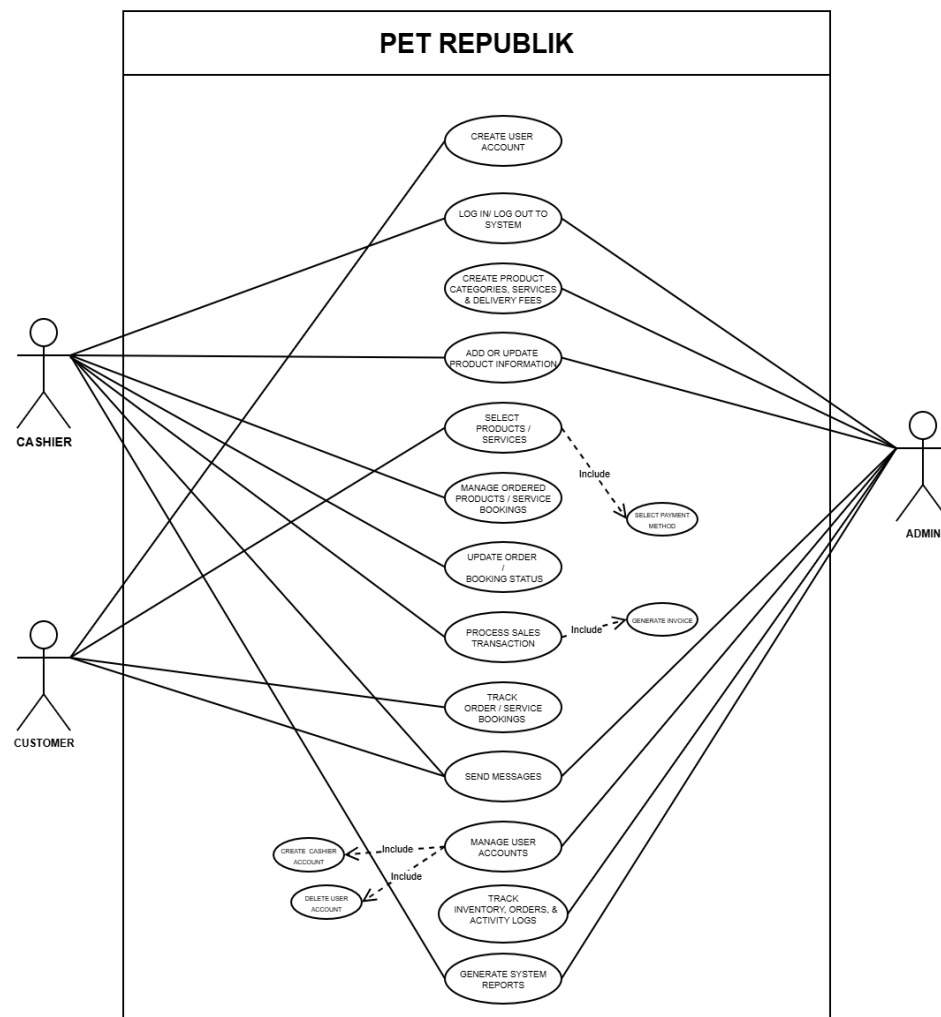


Figure 11. Use Case Diagram for Pet Republik

The table below shows how a user can create an account in the system. It explains the steps, starting from opening the registration page to successfully making a new account. The table also lists who uses this feature (Customer, Admin), what happens during the process, when it starts (entry condition), when it ends (exit condition), and what the system must do to make sure it works properly (quality requirements).

Table 15. Use Case Description for Create Account

Use case name	Create User Account
Participating actors	• Customer • Admin
Flow of events	For Customer: 1. Customer accesses the registration page. 2. The system displays a registration form. 3. Customer fills in required details and submits. 4. The system validates input and creates the account. 5. A confirmation message is shown, and the user is redirected to login. For Admin (Creating Cashier Accounts): 1. Admin logs into the system. 2. Admin accesses the cashier management or user management module. 3. Admin selects "Create Cashier Account." 4. The system displays a form for cashier details (e.g., name, role, email, password). 5. Admin fills out and submits the form. 6. The system validates the input and creates the cashier account. 7. A confirmation message is displayed.

Entry Condition	• Customer accesses the registration page. • Admin accesses the cashier account creation module.
Exit Condition	• A new customer or cashier account is created and saved in the system.
Quality Requirements	• All input fields must be validated before submission. • Admin access to cashier creation must be restricted and secure. • Confirmation messages must be shown after successful account creation. • The system must ensure secure data handling and password encryption.

The table below explains how users can log in and log out of the system. It shows the steps from entering login details to either getting access or ending the session. The users who can use this feature include the Customer, cashier, and Admin. It also shows when the process starts (opening the login page), when it ends (after login or logout), and the system's need to keep the login process secure.

Table 16. Use Case Description for Login/Logout

Use case name	Login/Logout to the system
Participating actors	• Customer • Cashier • Admin
Flow of events	1. User enters login credentials. 2. The system authenticates users. 3. Upon success, access is granted based on user role. 4. Logout ends the session.
Entry Condition	• The user navigates to the login page.
Exit Condition	• Users are logged into their dashboard or logged out successfully.

Quality Requirements	Secure authentication and session management.
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The table below explains how the admin manages the creation of product categories, services, and delivery fees within the system. Product categories are organized to help the cashier easily add, locate, and identify items during transactions, improving efficiency and reducing errors. The admin also creates and manages service offerings for customer bookings, ensuring a structured and organized scheduling process. In addition, delivery fees are defined based on specific criteria such as location or distance to ensure accurate and consistent charges, making the ordering and delivery process more efficient and transparent.

Table 17. Use Case Description for creating product categories, services, delivery fees.

Use case name	Create product categories, services & delivery fees
Participating actors	- Admin - Cashier
Flow of events	- The admin logs into the system and accesses the management settings. - The admin creates or updates product categories to properly organize items for easier identification and processing by the cashier. - The admin adds and defines services available for customer booking, including relevant details such as type of service, or description. - The admin sets delivery fees based on specific criteria (e.g., location or distance). - All created categories, services, and delivery fees are saved and applied in the system. - The cashier and customers can then use these settings for product transactions, service bookings, and delivery charge calculations.

Entry Condition	The admin has valid login credentials and successfully accesses the system's management module. The system is operational, and the admin has the required permissions to create or modify product categories, services, and delivery fee settings.
Exit Condition	Product categories, services, and delivery fees are successfully created or updated and properly saved in the system. The configured settings are available for use by the cashier for product transactions, by customers for service bookings, and the calculation of delivery charges.
Quality Requirements	- Accuracy – The system must correctly store and apply product categories, services, and delivery fees without calculation or classification errors. - Consistency – Data should remain uniform across all modules, ensuring the same categories, services, and delivery charges appear correctly for all users. - Usability – The interface must be clear and user-friendly, allowing administrators to easily create, edit, and manage entries. - Reliability – The system should function without crashes or data loss and must save updates successfully every time. - Security – Only authorized administrators can create or modify categories, services, and delivery fee settings. - Validation Control – The system must prevent duplicate, incomplete, or invalid entries before saving data. - Performance Efficiency – Updates and retrieval of data should happen quickly to support smooth transactions and bookings. - Data Integrity – All records must be properly stored and protected to ensure long-term accuracy and traceability.

The table below shows that both the cashier and the administrator have the ability to add products into the system. This shared access allows the administrator to manage and maintain the overall product list, such as adding new items,

updating details, or organizing inventory, while the cashier can quickly add products as needed during daily operations. Allowing both roles to perform this function helps ensure that product information remains updated, transactions run smoothly, and inventory records stay accurate and complete.

Table 18. Use Case Description for Add and Update Products.

Use case name	Add or Update Product Information
Participating actors	- Admin - Cashier
Flow of events	- The user (cashier or admin) logs into the system with authorized access. - The user navigates to the product management or inventory section. - The user selects the option to add a new product. - The user enters the required product details, such as product name, category, price, quantity, and other relevant information. - The system validates the entered information to ensure completeness and accuracy. - The user saves the product information. - The system stores the new product and updates the inventory records, making it available for transactions and monitoring.
Entry Condition	The cashier or administrator has valid login credentials and successfully accesses the system. The system is operational, and the user has the required permissions to add products. Necessary product information (such as name, category, price, and quantity) is available before starting the process.
Exit Condition	The product is successfully added and saved in the system, and the inventory records are automatically updated. The newly added product is available for viewing, transactions, and monitoring, with no validation errors or data issues.

Quality Requirements	• Accuracy – Product information must be correctly recorded, including name, category, price, and quantity. • Data Validation – The system must prevent incomplete, duplicate, or invalid product entries before saving. • Usability – The product entry interface should be simple, clear, and easy for both cashier and administrator to use. • Reliability – The system must save products successfully without crashes, errors, or data loss. • Security – Only authorized users (cashier and administrator) can add products. • Performance Efficiency – Product data should be processed and saved quickly to support smooth operations. • Data Integrity – All product records must remain consistent, properly stored, and retrievable for future transactions and monitoring.
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The table below explains how a customer can book services, such as pet grooming, and place orders through the system. The process begins when the customer accesses the services or products section, selects a specific service or item, and chooses a preferred date and time. For orders, the customer selects products, reviews the product details, and proceeds to checkout, including delivery or payment options if applicable. The system then checks service availability, prevents scheduling conflicts, calculates order details, and verifies all required information. The process ends once the appointment or order is confirmed and securely stored in the system, with confirmation provided to ensure a smooth and reliable booking and ordering experience.

Table 19. Use Case Description for Select Products or Services.

	Select Products or Services
Participating actors	• Customer

Flow of events	1. The customer logs into the system and accesses the services or products section. 2. The customer selects the desired service (e.g., grooming) or chooses products to order. 3. For service booking, the customer selects the preferred date and time. 4. For product orders, the customer reviews selected items and proceeds to check out. 5. The system checks service availability, prevents schedule conflicts, and verifies product availability. 6. The system calculates the total cost, including applicable delivery fees or service charges. 7. The customer confirms the booking or order details and submits the request. 8. The system records the appointment or order and updates the schedule or inventory. 9. The system provides confirmation to the customer, completing the booking or ordering process.
Entry Condition	• The customer has a registered account and successfully logs into the system. The system is operational, and available services and products are displayed. Required information such as service details, preferred schedule, product selections, delivery information, and payment method (if applicable) is ready before proceeding.
Exit Condition	• The service booking or product order is successfully confirmed and recorded in the system. The appointment schedule or inventory is updated accordingly, and any applicable delivery or payment details are processed. A confirmation is provided to the customer, and all transaction records are securely stored without errors.

Quality Requirements	• Accuracy – The system must correctly record booking details, order information, schedules, and payment or delivery data. • Availability Checking – The system must verify service schedules and product stock to prevent conflicts or unavailable selections. • Usability – The booking and ordering process must be simple, clear, and easy for customers to follow. • Reliability – The system must process bookings and orders successfully without errors, crashes, or data loss. • Security – Customer information, transactions, and payment details must be securely stored and protected from unauthorized access. • Real-Time Updates – Service schedules and product inventory must update immediately after confirmation. • Data Integrity – All booking and order records must remain accurate, consistent, and retrievable for future reference.
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The table below explains how the cashier manages both ordered products and service bookings made by customers. It shows the process of reviewing and handling customer transactions, including verifying product orders, checking payment status, and preparing items for release or delivery. For service bookings, the cashier confirms appointment details, updates the schedule if needed, and ensures that the requested service is properly recorded and coordinated. This process helps maintain accurate transaction records, ensures timely order fulfillment and service delivery, and supports smooth daily operations by keeping both product sales and service appointments well organized.

Table 20. Use Case Description for Manage Ordered Products and Service Booking

Use case name	Manage Ordered Products and Service Bookings
Participating actors	Cashier
Flow of events	- The cashier logs into the system with authorized access. - The cashier views the list of customer product orders and service bookings. - The cashier reviews order details, including items, quantities, payment status, and delivery information (if applicable). - The cashier verifies and processes product orders for release, pickup, or delivery. - The cashier checks service booking details, confirms appointment schedules, and ensures availability. - The cashier updates the status of orders and bookings. - The system records all updates and automatically adjusts inventory and service schedules. - The transaction and booking status are finalized and stored in the system.
Entry Condition	The cashier has valid login credentials and successfully accesses the system. Customer product orders and service bookings have already been submitted and recorded. The system is operational, and the cashier has the required permission to view, verify, and manage orders and appointments.
Exit Condition	Customer product orders and service bookings are successfully reviewed, verified, and updated by the cashier. Order statuses and appointment details are properly recorded, inventory and schedules are updated if necessary, and all changes are saved in the system without errors.

Quality Requirements	• Accuracy – The system must correctly display and update order details, booking information, payment status, and schedules. • Reliability – Order and booking processing must be completed without system errors, crashes, or data loss. • Real-Time Updates – Inventory levels and service schedules must update immediately after the cashier processes transactions. • Usability – The interface should be clear and easy for the cashier to navigate when reviewing and managing orders and bookings. • Security – Only authorized cashiers can access and manage customer orders and service appointments. • Data Validation – The system must prevent incorrect or incomplete status updates and ensure all required information is verified. • Performance Efficiency – The system should load, process, and save updates quickly to support smooth daily operations • Data Integrity – All transaction and booking records must remain accurate, consistent, and securely stored for tracking and reporting.
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The table below explains how cashiers update the status of customer product orders and service bookings throughout the transaction process. It outlines how the cashier monitors each order or appointment and modifies its status based on its current progress. For product orders, this may include marking them as confirmed, being prepared, ready for pickup, out for delivery, completed, or canceled. For service bookings, the cashier may update the appointment to show confirmation, rescheduling, completion, or cancellation depending on the situation.

Table 21. Use Case Description for Update Order or Booking Status.

Use case name	Update Order or Booking Status
Participating actors	Cashier
Flow of events	- The cashier logs into the system with authorized access. - The cashier navigates the orders and bookings management section. - The cashier selects a specific customer order or service booking to review. - The cashier checks the current details and progress of the selected transaction or appointment. - The cashier updates the status based on its current stage (e.g., confirmed, preparing, ready, completed, or canceled). - The system validates the update to ensure it is correct and allowed. - The cashier saves the updated status. The system records the changes, updates related inventory or schedules if needed and stores the updated information. - The updated status becomes visible for tracking and reference by the system and relevant users.
Entry Condition	The cashier has valid login credentials and successfully accesses the system. Customer orders or service bookings already exist and are recorded in the system. The system is operational, and the cashier has the required permissions to view and update order and booking statuses.
Exit Condition	The order or service booking status is successfully updated and saved in the system. All related records, such as inventory or service schedules, are adjusted if necessary. The updated status is accurately reflected for tracking and monitoring, with no errors in data processing or storage.
Quality Requirements	- Accuracy – The system must correctly update and display the current status of orders and service bookings. - Real-Time Synchronization – Status changes must immediately reflect across inventory, schedules, and tracking records.

	• Reliability – The system must process and save status updates without errors, crashes, or data loss. • Usability – The interface must be simple and clear, allowing cashiers to update statuses quickly and correctly. • Security – Only authorized cashiers can modify order and booking statuses. • Validation Control – The system must ensure that only valid and appropriate status changes are allowed. • Performance Efficiency – Status updates must be processed and saved quickly to support smooth operations. • Data Integrity – All status changes must be accurately recorded and securely stored for tracking and reporting.
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The table below explains how the cashier uses the POS (Point of Sale) system to process and manage customer transactions, including service bookings, product orders, and walk-in purchases. It shows how the cashier accesses booking or order details, verifies the information, and completes the payment process using the POS. For walk-in customers, the cashier directly selects products or services, calculates the total amount, and records the transaction in real time. The system then updates inventory, confirms service fulfillment, and generates transaction records or receipts. This process helps ensure fast, accurate, and organized handling of all customer transactions, whether they are pre-booked, pre-ordered, or made on the spot.

Table 22. Use Case Description for Point of Sale.

Use case name	Process Sale Transaction
Participating actors	• Cashier

Flow of events	1. The cashier logs into the POS (Point of Sale) system with authorized access. 2. The cashier identifies the type of transaction service booking, product order, or walk-in purchase. 3. For bookings or pre-orders, the cashier retrieves the customer's record and verifies the details. 4. For walk-in customers, the cashier selects the products or services directly in the POS system. 5. The system calculates the total amount, including applicable charges or discounts. 6. The cashier confirms the transaction details with the customer. 7. The cashier processes the payment using the selected payment method. 8. The system records the transaction, updates inventory and service status if applicable, and generates a receipt. 9. The transaction is completed and stored in the system for tracking and reporting.
Entry Condition	• The cashier has valid login credentials and successfully accesses the POS (Point of Sale) system. Customer bookings or product orders are already recorded in the system, or a walk-in customer is present for a transaction. The POS system is operational, and the cashier has the required permissions to process payments and complete transactions. Relevant transaction details, such as selected products, services, or booking information, are available before processing.
Exit Condition	• The transaction is successfully completed and recorded in the POS system. Payment is processed, a receipt is generated, and the status of the order or service booking is updated if applicable. Inventory and service records are automatically adjusted, and all transaction details are securely stored without errors.

Quality Requirements	• Accuracy – The POS system must correctly calculate totals, process payments, and record transaction details for bookings, orders, and walk-in customers. • Reliability – Transactions must be completed and saved without system errors, crashes, or data loss • . Real-Time Updates – Inventory levels, booking status, and transaction records must update immediately after payment is processed. • Usability – The POS interface must be simple, clear, and easy for cashiers to operate during transactions. • Security – Payment information and transaction records must be securely processed and accessible only to authorized users. • Performance Efficiency – The system must process transactions quickly to avoid delays and support smooth customer service. • Receipt Generation – The system must automatically generate accurate receipts after each completed transaction. • Data Integrity – All transaction data must be properly stored, consistent, and retrievable for monitoring and reporting.
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The table below explains how customers can track the status of their product orders and service bookings through the system. It shows how customers can view real-time updates on their transactions, including order processing progress, delivery status, or upcoming service appointment details. By accessing the tracking feature, customers can monitor important information such as confirmation, preparation, scheduling, completion, or any changes to their order or booking. This process helps keep customers informed, improves transparency, and allows them to stay updated on the progress of their purchases and scheduled services without needing direct assistance from staff.

Table 23. Use Case Description for Track Order and Service Bookings.

Use case name	Track Order and Service Bookings
Participating actors	Customer
Flow of events	- The customer logs into the system using their account. - The customer navigates to the order and booking tracking section. - The system displays a list of the customer's active and past orders and service bookings. - The customer selects a specific order or booking to view detailed information. - The system shows the current status, progress updates, schedule, or delivery details. - The system updates the information in real time if any changes occur. - The customer reviews the details and exits the tracking section.
Entry Condition	The customer has a registered account and successfully logs into the system. At least one product order or service booking has already been placed and recorded. The system is operational, and the tracking feature is available for the customer to access.
Exit Condition	The customer successfully views the current status and details of their order or service booking. All displayed information is accurate and up to date, and no changes are made to the records. The customer exits the tracking section after reviewing the information.
Quality Requirements	- Accuracy – The system must display correct and up-to-date information about order status, delivery progress, and service booking schedules. - Real-Time Updates – Tracking information must refresh automatically to reflect any changes in order or booking status. - Usability – The tracking interface must be simple, clear, and easy for customers to navigate and understand. - Reliability – The system must consistently load and display tracking details without errors, delays, or data

	loss. • Security – Only authorized customers can view their own order and booking information. • Performance Efficiency – Tracking data should load quickly to provide a smooth user experience. • Transparency – The system must clearly show progress updates, status changes, and relevant details for better customer awareness. • Data Integrity – All tracking records must remain accurate, consistent, and securely stored for reference.
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The table below explains how communication works within the system through a structured messaging feature with defined roles. Customers can only send messages to the cashier for inquiries about orders, bookings, or other concerns, making the cashier the main point of contact. The cashier can respond directly to customers and can also communicate with the administrator when issues require higher-level action, approval, or system management. The cashier and administrator can exchange messages as needed, and the cashier relays any final updates or decisions back to the customer, ensuring clear, organized, and efficient communication throughout the system.

Table 24. Use Case Description for Send Messages.

Use case name	Send Messages
Participating actors	• Admin • Cashier • Customer

Flow of events	1. The customer logs into the system and accesses the messaging feature. 2. The customer sends a message to the cashier regarding an inquiry, concern, or request. 3. The cashier receives and reviews the customer's message. 4. The cashier responds directly to the customer if the concern can be resolved at their level. 5. If the issue requires higher authority, approval, or system management, the cashier sends a message to the administrator. 6. The administrator reviews the message and replies with instructions, decisions, or clarifications. 7. The cashier receives the administrator's response and communicates the final update or resolution to the customer. 8. The conversation is recorded and stored in the system for reference.
Entry Condition	• The customer, cashier, or administrator has a registered account and successfully logs into the system. The messaging feature is available and operational. The customer has an inquiry or concern to send to the cashier, or the cashier and administrator need to communicate regarding transactions, issues, or system-related matters.
Exit Condition	• The message is successfully sent, received, and recorded in the system. The inquiry or concern is addressed through communication between the customer and cashier, and if needed, between the cashier and administrator. Any responses or resolutions are delivered, and the conversation is stored for reference without errors.
Quality Requirements	• Message Accuracy – The system must correctly send, receive, and display messages without loss or alteration of content. • Real-Time Delivery – Messages should be delivered and received promptly to support timely communication. • Reliability – The messaging feature must function consistently without failures, delays, or missing messages. • Usability – The messaging interface must be simple, clear, and easy for customers, cashiers, and administrators to use.

	• Role-Based Access Control – Communication must follow defined rules (customer to cashier only; cashier and administrator can message each other). • Security and Privacy – Messages must be securely stored and accessible only to authorized users. • Data Storage – All conversations must be properly recorded and retrievable for tracking and reference. • Performance Efficiency – The system must handle message sending, receiving, and loading quickly to ensure smooth communication.
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The table below explains how the admin manages user accounts within the system, including creating cashier accounts and deleting existing user accounts when necessary. It shows how the admin oversees account access by adding new cashier profiles with appropriate permissions to perform their roles and responsibilities. The administrator can also remove or deactivate accounts that are no longer needed, such as inactive users or staff who no longer have access rights. This process helps maintain system security, ensures proper role assignment, and keeps user records organized and up to date for efficient system management.

Table 25. Use Case Description for Manage User Accounts.

Use case name	Manage User Accounts
Participating actors	• Admin
Flow of events	1. The administrator logs into the system with authorized access. 2. The administrator navigates to the user account management section. 3. To create a cashier account, the administrator selects the option to add a new user. 4. The administrator enters the required account details and assigns the cashier role and permissions. 5. The system validates the information and saves the new cashier account. 6. To delete an account, the administrator selects an existing user from the account list.

	7. The administrator reviews the account details and confirms the deletion or deactivation. 8. The system removes or disables the account and updates the user records. 9. All changes are saved and recorded in the system for monitoring and reference.
Entry Condition	• The admin has valid login credentials and successfully accesses the system. The user management module is operational, and the administrator has the necessary permissions to create, modify, or delete user accounts. Required information for creating or managing accounts, such as user details and roles, is available.
Exit Condition	• The User accounts are successfully created, updated, or deleted as intended. The system accurately reflects all changes, including newly added cashier accounts or removed/deactivated accounts. All modifications are securely saved, and the user records remain organized, up-to-date, and accessible for future reference or auditing.
Quality Requirements	• Accuracy – User account information must be correctly recorded, including roles, permissions, and personal details. • Reliability – Account creation, modification, and deletion must be processed without system errors, crashes, or data loss. • Security – Only authorized administrators can manage user accounts, and sensitive account information must be protected. • Usability – The user management interface must be clear and easy to navigate for creating, editing, or deleting accounts. • Validation Control – The system must prevent duplicate accounts, incomplete information, or invalid role assignments. • Real-Time Updates – Changes to user accounts must reflect immediately across the system. • Data Integrity – All user records must remain consistent, accurate, and securely stored for

	monitoring and auditing. • Performance Efficiency – Account management actions should be processed quickly to support smooth system administration.
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The table below explains how the admin monitors and manages key operational aspects of the system, including inventory levels, stock availability, customer orders, and the activity log. It shows how the admin can track inventory to ensure that products are sufficiently stocked, identify low-stock items, and prevent shortages. Orders from customers are monitored to verify processing status, delivery, and fulfillment. The activity log provides a detailed record of user actions, such as account changes, transaction updates, and system modifications, allowing the administrator to review, audit, and ensure accountability. By overseeing these areas, the admin can maintain efficient operations, prevent errors, and ensure that the system runs smoothly and securely.

Table 26. Use Case Description for Tracks Inventory, Orders, and Activity Log.

Use case name	Tracks Inventory, Orders, and Activity Logs
Participating actors	• Admin
Flow of events	1. The admin logs into the system with authorized access. 2. The admin navigates to the inventory and order management section. 3. The system displays current inventory levels, stock availability, and customer orders. 4. The admin reviews inventory to identify low-stock or out-of-stock items. 5. The admin monitors orders to track their processing, fulfillment, and delivery status. 6. The administrator accesses the activity log to review user actions, system updates, and transaction records.

	7. Any necessary adjustments, such as restocking items or addressing order issues, are made by the administrator. 8. The system updates all records in real time, reflecting changes in inventory, orders, and activity logs. 9. The admin logs out after completing monitoring and management tasks.
Entry Condition	• The admin has valid login credentials and successfully accesses the system. The inventory, order, and activity log modules are operational. Relevant data, such as current stock levels, customer orders, and recorded user activities, are available for review and monitoring.
Exit Condition	• The admin has successfully reviewed and tracked inventory levels, stock availability, customer orders, and activity logs. Any necessary updates or adjustments have been applied, and all records are accurately saved in the system. The system reflects real-time changes, and the admin completes the monitoring task without errors.
Quality Requirements	• Accuracy – Inventory levels, stock data, order details, and activity logs must be correctly recorded and displayed. • Real-Time Updates – Changes in stock, orders, or system activities must update immediately for accurate monitoring. • Reliability – The system must function consistently without errors, crashes, or data loss while tracking records. • Usability – The interface must be clear and easy for the administrator to navigate and review inventory, orders, and activity logs. • Security – Only authorized administrators can access and manage inventory, orders, and activity logs. • Data Integrity – All records must remain consistent, complete, and securely stored for auditing and reference. • Performance Efficiency – The system must quickly process and display updates to support efficient

	monitoring and management. • Traceability – The activity log must provide detailed records of all user actions for accountability and auditing purposes.
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The table below explains how both the admin and cashier can generate and monitor reports within the system. It shows how the admin and cashier access the reporting module to review key operational data, including sales, orders, service bookings, inventory levels, and user activity. These reports help them analyze performance, identify trends, and make informed decisions for better management of daily operations. The system allows them to filter data by date, category, or other relevant criteria, and generate summaries or detailed reports. By using these reports, the admin and cashier can track business performance, monitor transactions, ensure accountability, and maintain organized records for future reference and auditing purposes.

Table 27. Use Case Description for Generate Reports

Use case name	Generate System Reports
Participating actors	• Admin • Cashier
Flow of events	1. The admin or cashier logs into the system with authorized access. 2. They navigate to the reporting or analytics section of the system. 3. The user selects the type of report to generate, such as sales, orders, service bookings, inventory, or activity logs. 4. The user sets the desired filters or criteria, such as date range, product category, or transaction type. 5. The system processes the request and compiles the relevant data. 6. The generated report is displayed on the screen for review, with options for summary or detailed views. 7. The user can export, print, or save the report as needed. The system records the report generation activity in the activity log for tracking and accountability. 8. The user logs out or exits the reporting module after reviewing or saving the report.
Entry Condition	• The admin or cashier has valid login credentials and successfully accesses the system. The reporting module is operational, and relevant data, such as sales, orders, service bookings, inventory, or activity records, is available for generating reports. The user has the necessary permissions to access and generate reports.
Exit Condition	• The report is successfully generated, reviewed, and optionally saved, exported, or printed. All relevant data is accurately compiled, and the system records the report generation activity in the activity log. The administrator or cashier exits the reporting module with all changes and actions properly saved.

Quality Requirements	• Accuracy – Reports must correctly reflect data on sales, orders, service bookings, inventory, and user activities. • Completeness – All relevant information within the selected criteria must be included in the report. • Real-Time Data – Reports should use the most up-to-date information available in the system. • Usability – The reporting interface must be clear, easy to navigate, and allow filtering, sorting, and selecting report types. • Reliability – The system must generate reports consistently without errors, crashes, or data loss. • Security – Only authorized users (admin or cashier) can access and generate reports. • Performance Efficiency – Reports must be generated quickly, even when handling large datasets. • Traceability – All report generation activities should be recorded in the system activity log for auditing and accountability. • Export and Sharing – Reports should be easily exportable or printable in standard formats like PDF or Excel for reference or documentation.
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Process Model: Activity Diagram

The activity diagram for Pet Republik shows the overview of the processes and the steps involved in the software application as a comprehensive guide to support pet services and e-commerce in Panabo city. It consists of three major parts, the Customer, cashier, and Admin activity diagram. Each segment corresponds to a different part of the work that users do in the system and shows a sequence of actions and choices users must select to initiate a certain process. This overarching model facilitates the visualization of the different users' use of the app, enables stakeholders to understand clearly how the system will work, supports communication between stakeholders and facilitates the design and development of effective systems.

The figure 12 activity diagram outlines the process of ordering and delivering products in the Pet Republik System, involving the cashier, System, and Customer. It begins with the cashier logging in and uploading products and services, which are then displayed to customers. Customers view and order products through the dashboard. The system shows the order to the cashier, who can either confirm or reject it. Confirmed orders are prepared and delivered, while rejected orders prompt a rejection notice to the customer. The process concludes with the customer receiving the product and the system confirming a successful delivery.

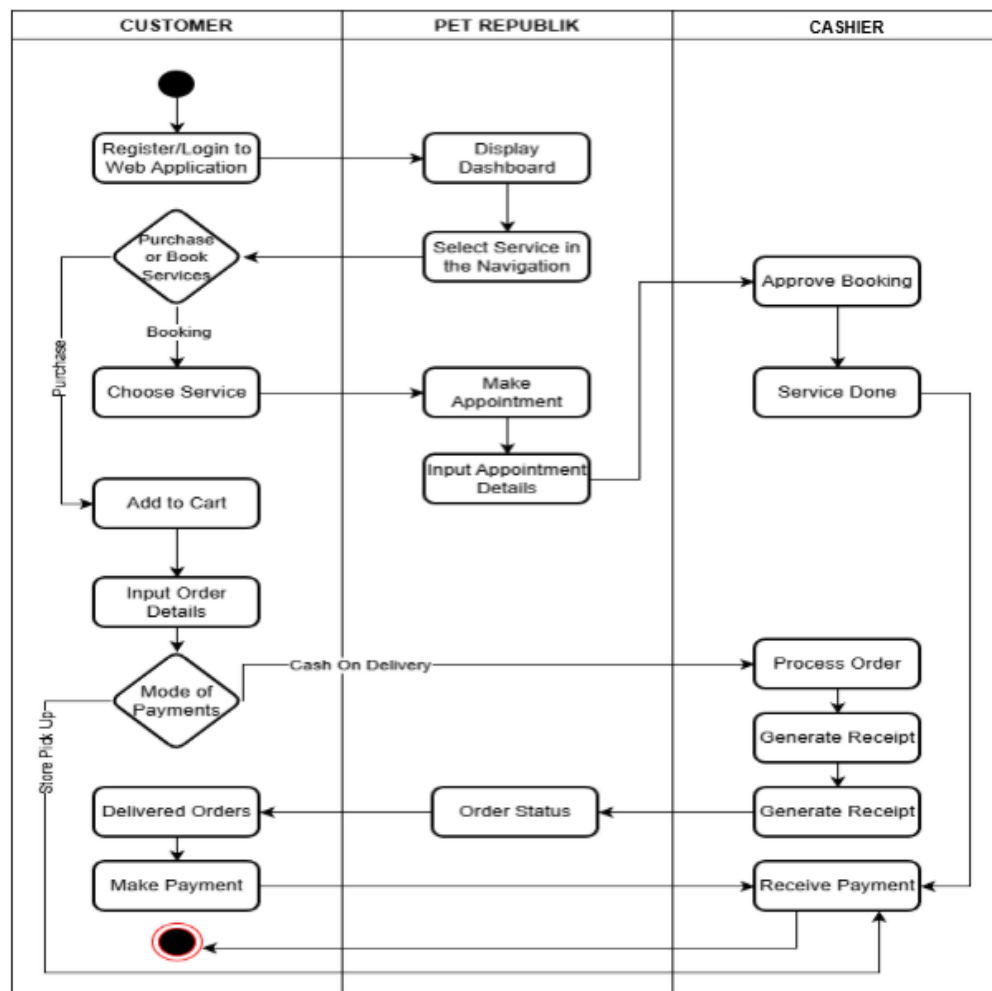


Figure 12. Activity Diagram for Customer

The figure 13 activity diagram illustrates the user interaction flow within the Pet Republik System, involving three main participants: the Customer, the System itself, and the cashier. It begins with the Customer logging in or registering, after which the system verifies credentials and displays available services. The user then decides whether to make an appointment or purchase a product. For purchases, the process includes adding items to the cart, placing an order, making a payment, and waiting for the cashier to prepare and deliver the product. For appointments, the user selects services, the system processes the appointment, and the cashier receives the appointment request. Both processes conclude with confirmation messages and the user being redirected to the dashboard before logging out. The diagram effectively highlights the flow of actions and responsibilities across all roles.

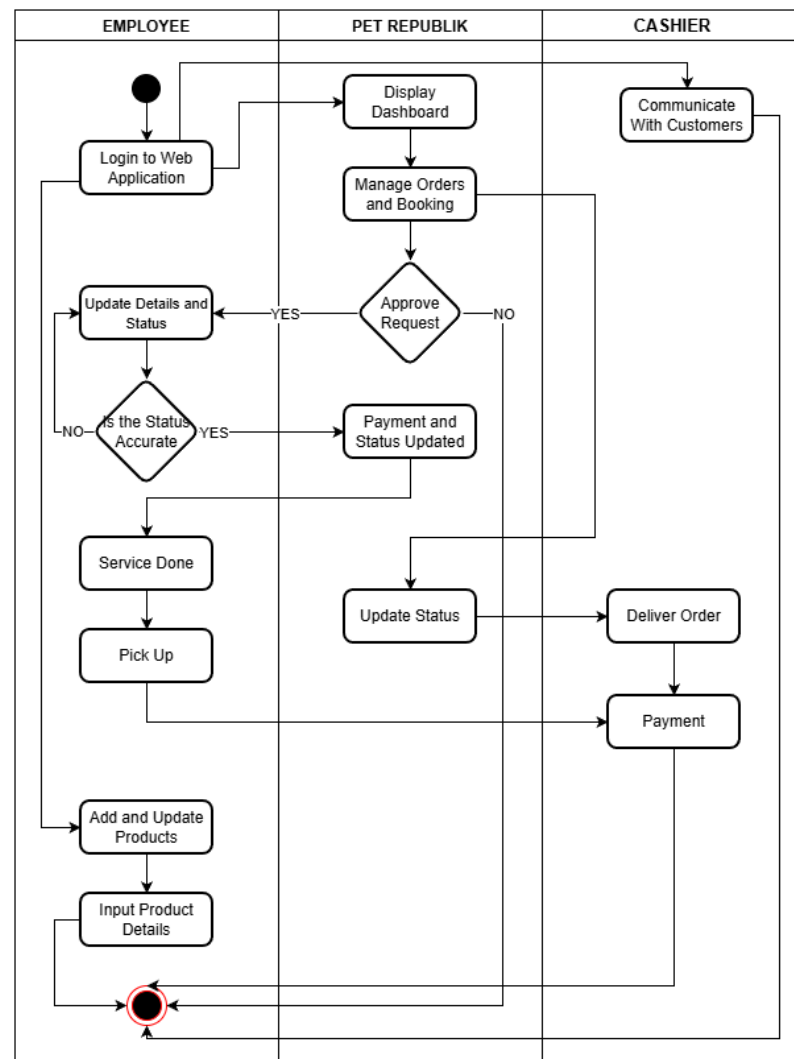


Figure 13. Activity diagram for Cashier

This diagram shows the admin account management process for the Pet Republik system. The admin first logs in and authenticates themselves. Then approves Pet Store accounts and moves them forward in the system, then approves them and moves them up in the system. If needed, the admin can delete accounts and move them out of the platform.

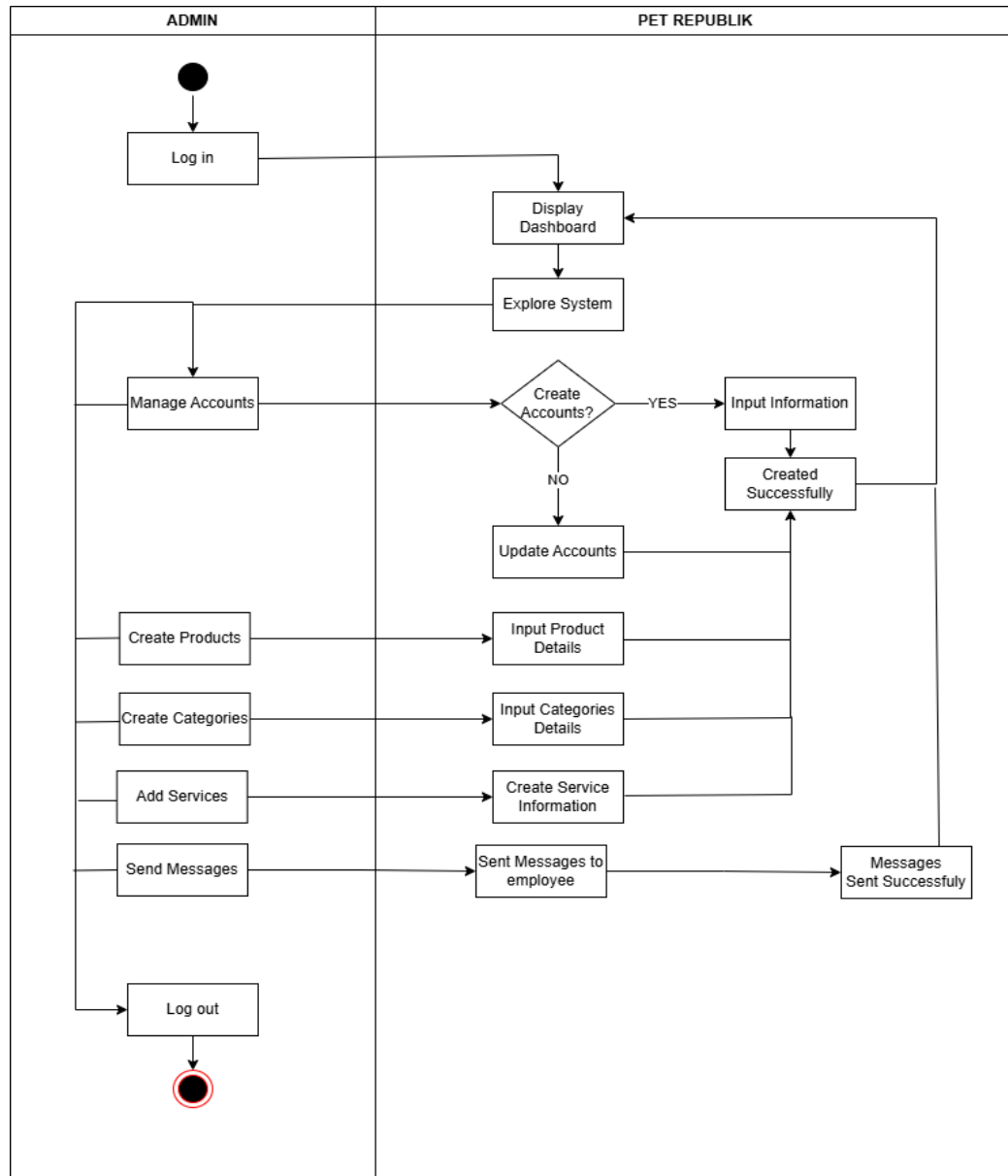


Figure 14. Activity Diagram for Admin

Data Model: Context Data Flow Diagram

The Context Data Flow Diagram for Pet Republik illustrates how the system interacts with its three primary users: Customers, Cashiers, and Admins. The pet shop inputs the product and account information into the system and receives updates on orders and detailed sales reports in return. Customers can sign up,

place orders, book appointments, and make payments. They also receive order confirmations, tracking updates, and search results. Admins manage all user accounts, with the ability to create cashier profiles, delete profiles, and generate important reports to keep everything running smoothly.

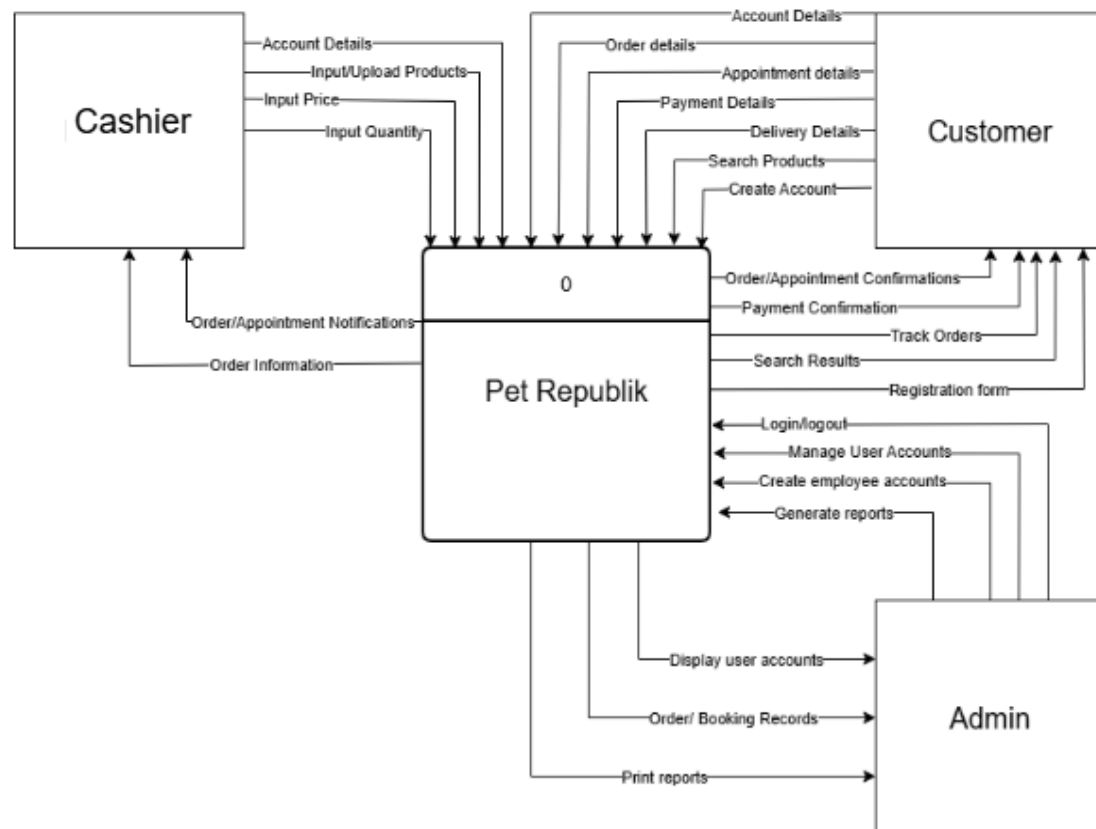


Figure 15. Data Model: Context Data Flow Diagram

Data Model: Level 1

This Data Flow Diagram (DFD) presents a structured view of how data flows within the Pet Republik, a web-based system and how different users interact with its core functionalities. The primary external entities involved are the Customer, Cashier, and Admin, each playing a specific role in the operation of the system. Customers provide account information, view product and service listings, place

orders, make payments, and book appointments. Cashiers handle order updates, manage product and service details, and monitor transaction-related notifications. The system is composed of several major processes, including Account Management, Product and Service Management, Order & Booking Management, Payment & POS, Tracking Management, Message System, and Inventory & Reports Management. All processes are supported by a centralized database that stores essential data such as account records, product and service information, sales data, and booking details. This diagram clearly illustrates how data is exchanged between users, processes, and the database to ensure accurate transactions, efficient service scheduling, and effective administrative control.

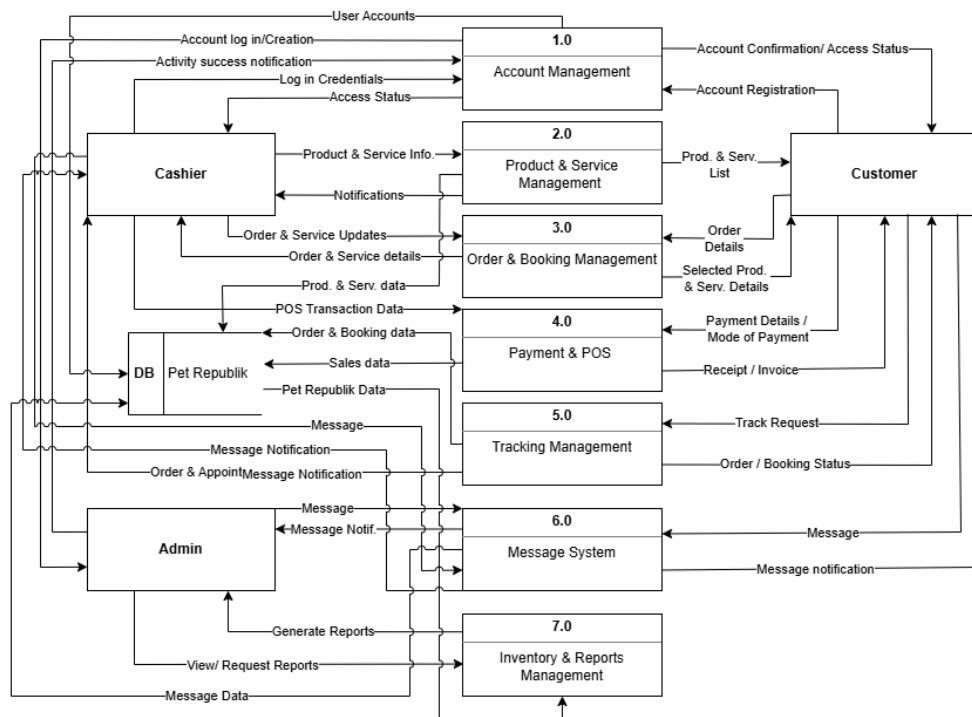


Figure 16. Level 1 Data Flow Diagram

H. System Design

The System Design section presents the overall structure and design of the proposed Pet Republik Information System. It describes how the different components of the system are organized and integrated to support the

functionalities of online ordering, appointment booking, inventory monitoring, report generation, and user management. This section includes the database structure, system architecture, data flow, and interface design that serve as the foundation for the development of the system. Through a well-planned system design, the proposed web-based platform ensures efficient data processing, reliable system performance, and a user-friendly experience for the administrators, cashiers, and customers of John & Jai Pet Republik Pet Shop.

Database Design: Entity Relationship Diagram

This section presents the Entity-Relationship Diagram (ERD) that illustrates the database structure and relationships between entities within the Pet Republik Pet Shop Management System. The diagram demonstrates how different components of the system interact to support efficient management of pet shop operations, including user management, product inventory, sales transactions, bookings, services, and customer communication. The ERD contains several key entities such as Users, Products, Categories, Transactions, Bookings, Services, Messages, Orders, Attendance, Notifications, and Activity Logs. Each entity includes important attributes that allow the system to store, organize, and process data accurately. The relationships among these entities ensure smooth system functionality, such as linking customers to bookings and transactions, employees to attendance records, and products to categories and sales records.

Figure 17 shows the ERD design, which serves as the foundation of the system's database architecture and reflects the core functionalities of Pet Republik Pet Shop. The Entity-Relationship Model provides a clear representation of real-world processes and data relationships, enabling efficient data management, system integration, and reliable operation of the shop management system.

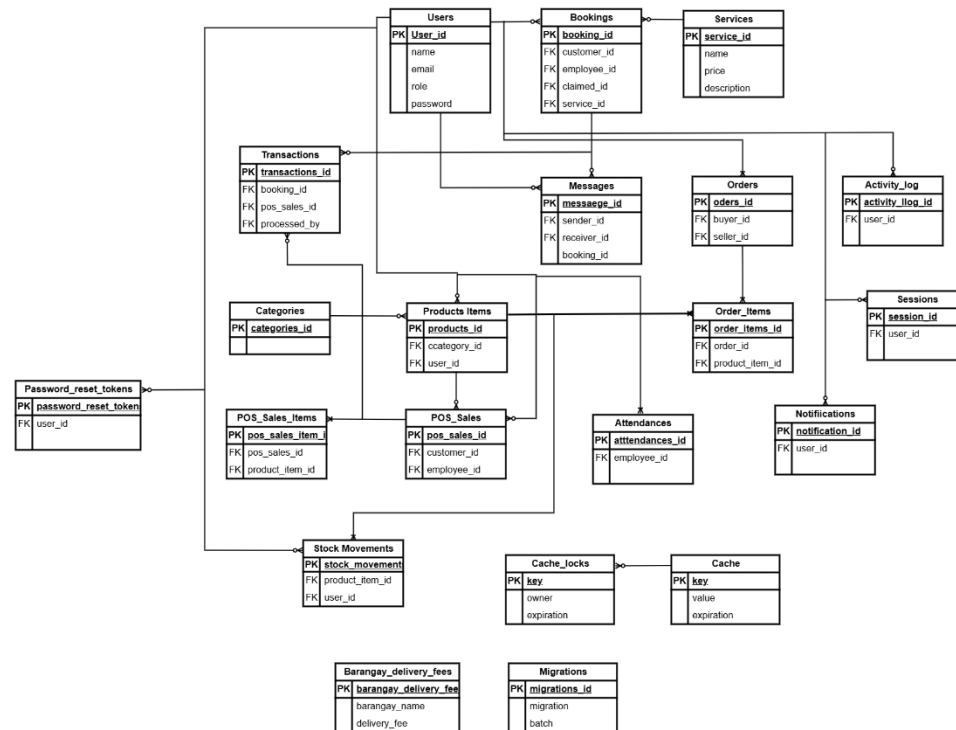


Figure 17. Entity Relationship Diagram

Data Dictionary

This data dictionary provides a comprehensive description of the Pet Republik database used in the development of the system. It presents all database tables along with their corresponding attributes, data types, and brief explanations to clearly define how data is structured, stored, and managed within the application. The purpose of this data dictionary is to serve as a reference for understanding the database design and the relationships among different data entities. The document supports system development by ensuring consistency in data usage and promoting accurate implementation of database operations. It also assists the project team, and system administrators in maintaining data integrity, troubleshooting database-related issues, and implementing future system enhancements. Overall, this data dictionary plays an important role in documenting the database architecture and ensuring the reliability and maintainability of the Pet Republik system.

Table 28. Activity Log

Attribute	Data Type	Description
id	BIGINT	Primary key identifier
log_name	VARCHAR	Log category or channel
description	TEXT	Description of the activity
subject_type	VARCHAR	Model or entity affected
event	VARCHAR	Type of event
subject_id	BIGINT	Affected record ID
causer_type	VARCHAR	Action initiator type
causer_id	BIGINT	Action initiator ID
Properties	JSON	Additional metadata
batch_uuid	UUID	Batch identifier
created_at	TIMESTAMP	Creation time
updated_at	TIMESTAMP	Last update time

Table 29. Attendances

Attribute	Data Type	Description
id	BIGINT	Primary key
employee_id	BIGINT	Employee reference
date	DATE	Attendance date
clock_in_time	TIME	Clock in
clock_out_time	TIME	Clock out
status	VARCHAR	Attendance status
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

Table 30. Barangay Delivery Fees

Attribute	Data Type	Description
id	BIGINT	Primary key
name	VARCHAR	Barangay name
delivery_fee	DECIMAL (10,2)	Delivery fee
created_at	TIMESTAMP	Created time

updated_at	TIMESTAMP	Updated time
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Table 31. Users

Attribute	Data Type	Description
id	BIGINT	Primary key
name	VARCHAR	Full name
email	VARCHAR	Email address
password	VARCHAR	Hashed password
role	VARCHAR	User role
phone	VARCHAR	Phone number
address	TEXT	Address
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

Table 32. Services

Attribute	Data Type	Description
id	BIGINT	Primary key
name	VARCHAR	Service name
description	TEXT	Service description
price	DECIMAL (10,2)	Service price
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

Table 33. Booking

Attribute	Data Type	Description
id	BIGINT	Primary key
booking_number	VARCHAR	Booking reference
customer_id	BIGINT	Customer ID
employee_id	BIGINT	Employee ID
service_id	BIGINT	Service ID
pet_name	VARCHAR	Pet name
pet_type	VARCHAR	Pet type

booking_date	DATE	Booking date
total_amount	DECIMAL (10,2)	Total amount
status	VARCHAR	Booking status
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

Table 34. Product Items

Attribute	Data Type	Description
id	BIGINT	Primary key
sku	VARCHAR	SKU
name	VARCHAR	Product name
price	DECIMAL(10,2)	Price
stock_quantity	INT	Stock
expiry_date	DATE	Expiry date
status	VARCHAR	Product status
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

Table 35. Orders

Attribute	Data Type	Description
id	BIGINT	Primary key
order_number	VARCHAR	Order reference
buyer_id	BIGINT	Buyer ID
seller_id	BIGINT	Seller ID
total_amount	DECIMAL(10,2)	Total amount
status	VARCHAR	Order status
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

Table 36. Transactions

Attribute	Data Type	Description
id	BIGINT	Primary key

booking_id	BIGINT	Booking reference
transaction_type	VARCHAR	Transaction type
payment_method	VARCHAR	Payment method
amount_paid	DECIMAL(10,2)	Amount paid
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

Table 37. Stock Movements

Attribute	Data Type	Description
id	BIGINT	Primary key
product_id	BIGINT	Product reference
quantity	INT	Quantity moved
movement_type	VARCHAR	IN or OUT
created_at	TIMESTAMP	Created time

Table 38. POS Sales

Attribute	Data Type	Description
id	BIGINT	Primary key
sale_number	VARCHAR	POS sale number
total_amount	DECIMAL(10,2)	Sale total
created_at	TIMESTAMP	Created time

Table 39. POS Sales Items

Attribute	Data Type	Description
id	BIGINT	Primary key
pos_sale_id	BIGINT	POS sale reference
product_id	BIGINT	Product reference
quantity	INT	Quantity
price	DECIMAL(10,2)	Item price

Table 40. Notifications

Attribute	Data Type	Description
id	BIGINT	Primary key
user_id	BIGINT	User reference
title	VARCHAR	Title
message	TEXT	Message
read_at	TIMESTAMP	Read time
created_at	TIMESTAMP	Created time

Table 41. Messages

Attribute	Data Type	Description
id	BIGINT	Primary key
sender_id	BIGINT	Sender
receiver_id	BIGINT	Receiver
message	TEXT	Message content
created_at	TIMESTAMP	Created time

Table 42. Password Reset Tokens

Attribute	Data Type	Description
email	VARCHAR	User email
token	VARCHAR	Reset token
created_at	TIMESTAMP	Created time

Table 43. Migrations

Attribute	Data Type	Description
id	INT	Primary key
migration	VARCHAR	Migration name
batch	INT	Batch number

Table 44. Sessions

Attribute	Data Type	Description
id	VARCHAR	Session ID
user_id	BIGINT	User reference
ip_address	VARCHAR	IP address
user_agent	TEXT	User agent
last_activity	INT	Last activity timestamp

Table 45. Cache

Attribute	Data Type	Description
key	VARCHAR	Cache key
value	TEXT	Cached value
expiration	INT	Expiration time

System Architecture

The System Architecture of the Pet Republik System presents a high-level overview of how the system components interact to support business operations. The Pet Republik System serves as the central platform where customers access services, and order pet products through a device, cashiers process the transactions, and admin has the overview of the system to receive generated reports for monitoring and management. The system communicates directly with a centralized database to store and retrieve data efficiently. This architecture ensures organized data flow, secure information management, and smooth coordination among users and system components.

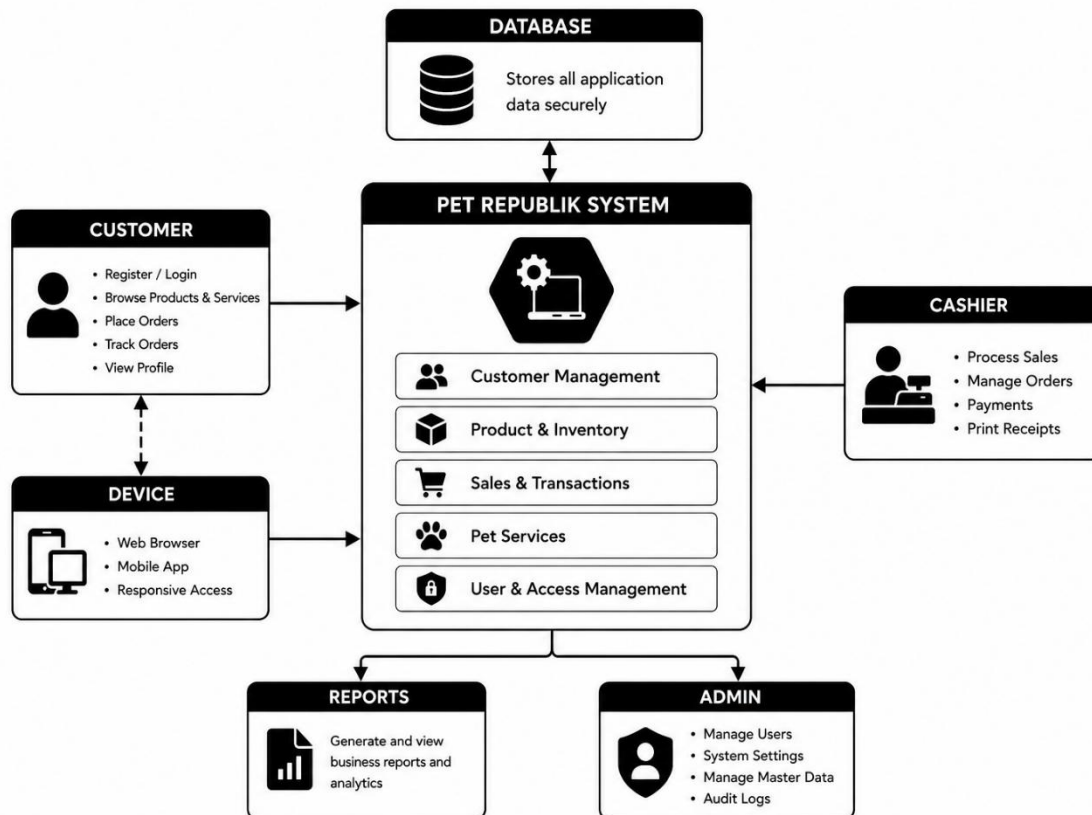


Figure 18. System Architecture

I. System Implementation

The System Implementation section discusses the actual deployment and execution of the proposed Pet Republik Information System. It covers the processes involved in testing, evaluating, and applying the developed system within the operations of John & Jai Pet Republik Pet Shop. This section ensures that the system functions according to the specified requirements and is capable of supporting online ordering, appointment booking, inventory management, and report generation effectively. Furthermore, it highlights the testing procedures conducted, identified errors and improvements, and the overall performance of the system during the implementation phase to ensure reliability, efficiency, and user satisfaction.

Testing Plan

The system was tested through actual deployment in John & Jai Pet Shop Panabo City. The testing approach followed a real-time usage strategy, where the users directly interacted with the system during daily operations. This included product management, inventory updates, and transaction processing using the Point-of-Sale (POS) feature. The objective of the testing plan was to evaluate the system's functionality, usability, reliability, and performance under real-world conditions. Feedback was continuously collected from the users to identify issues, system limitations, and possible improvements. Detected errors were documented and immediately addressed to ensure continuous system enhancement.

Figure 19 illustrates the system testing process where the user interacts with the Pet Republik System to perform tasks and evaluate its functionality. During this process, the system is checked to determine if any issues are detected. If no issues are found, the process ends, indicating that the system is working correctly. However, if an issue is identified, it is documented in a report and forwarded to the project team for analysis and resolution. After fixing the problem, the project team implements a system update, which is then returned to the system for further testing. This creates a continuous cycle of testing and improvement until the system operates smoothly without errors.

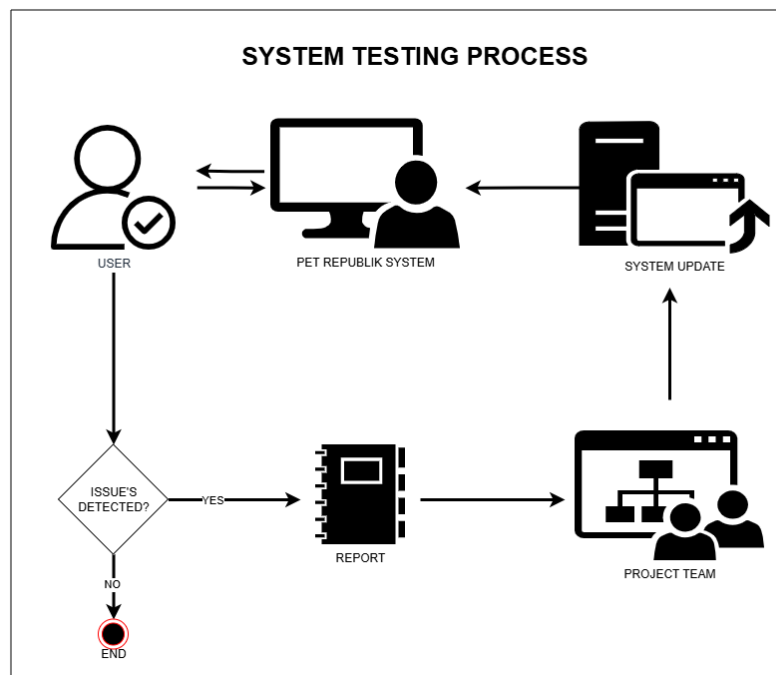


Figure 19. System Testing Process

Errors Found/Lack of Functionalities

Throughout the six-week deployment, several minor errors and system limitations were identified. These issues were mostly related to system responsiveness, product management, and transaction processing. All identified problems were addressed progressively, resulting in improved system stability over time.

Week 1 Error Found/Lack of Functionalities

As shown in Figure 20, during Week 1 of using the system, the cashier encountered an issue with the stock movement functionality. Whenever the feature was clicked, the system would generate an error, preventing access to inventory movement records. This limitation affected the cashier's ability to efficiently monitor stock changes during daily operations. The issue was immediately reported, as stock tracking is a critical component of the system. Addressing this problem was considered essential to ensure accurate inventory management and to support

smoother and more reliable system performance in the succeeding weeks.

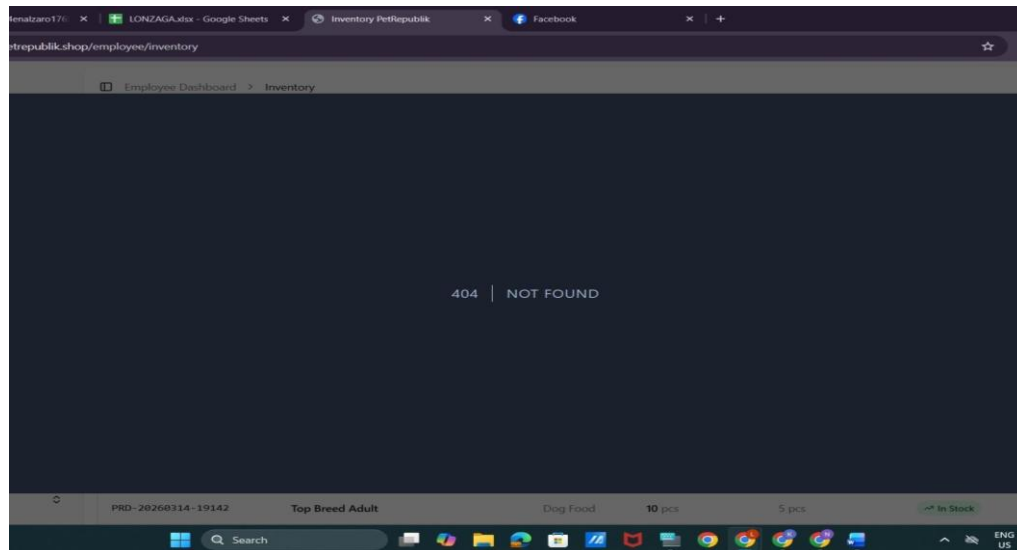


Figure 20. Week 1 Error Found/Lack of Functionalities

Week 2 Error Found/Lack of Functionalities

As shown in Figure 21, during the second week of deployment, both the admin and the cashier began to observe and discuss several minor issues while using the system in their daily operations. One of the primary concerns raised by the admin was that the total number of products displayed did not exactly match the actual number of products listed in the system. In addition, minor issues, including product upload limitations and occasional checkout errors, were identified, which slightly disrupted the smooth flow of transactions. These concerns were reported and carefully reviewed by the development team. The necessary fixes were then implemented, resulting in significant improvements in the system's accuracy and enhancing the overall reliability of the POS feature for continued use.

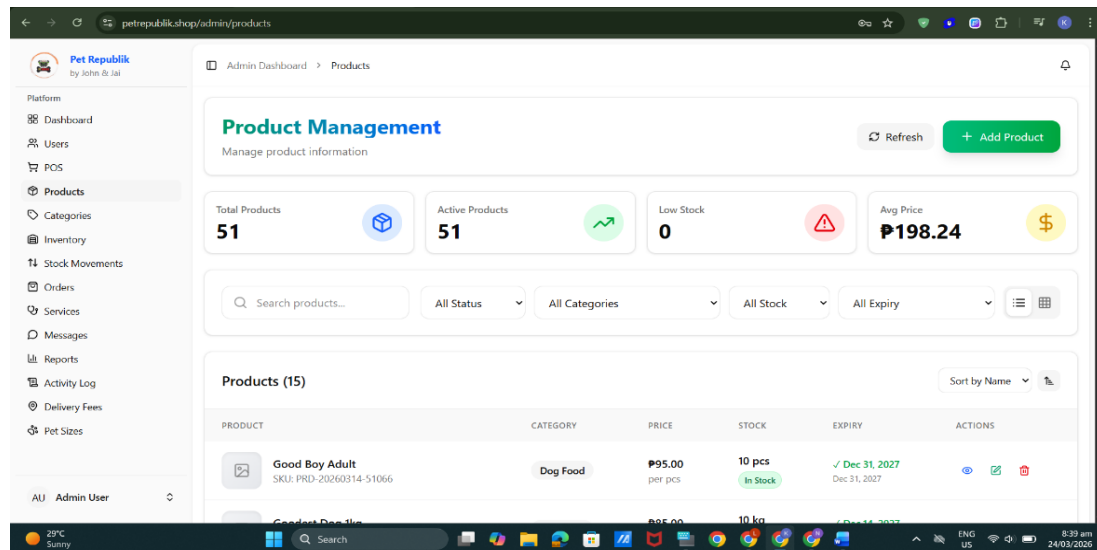


Figure 21. Week 2 Error Found/Lack of Functionalities

Week 3 Error Found/Lack of Functionalities

As shown in Figure 22, during Week 3 of using the system, the admin encountered an issue within the inventory function while performing regular tasks. Initially, the inventory feature was expected to display complete records, including the history of stock movements for monitoring and tracking purposes. However, when the admin attempted to access the history function, the system would generate an error and fail to load the necessary data. This prevented the admin from reviewing past inventory transactions and verifying stock changes. The limitation affected the overall functionality of the inventory module, as access to historical records is essential for accurate stock management and monitoring during daily operations.

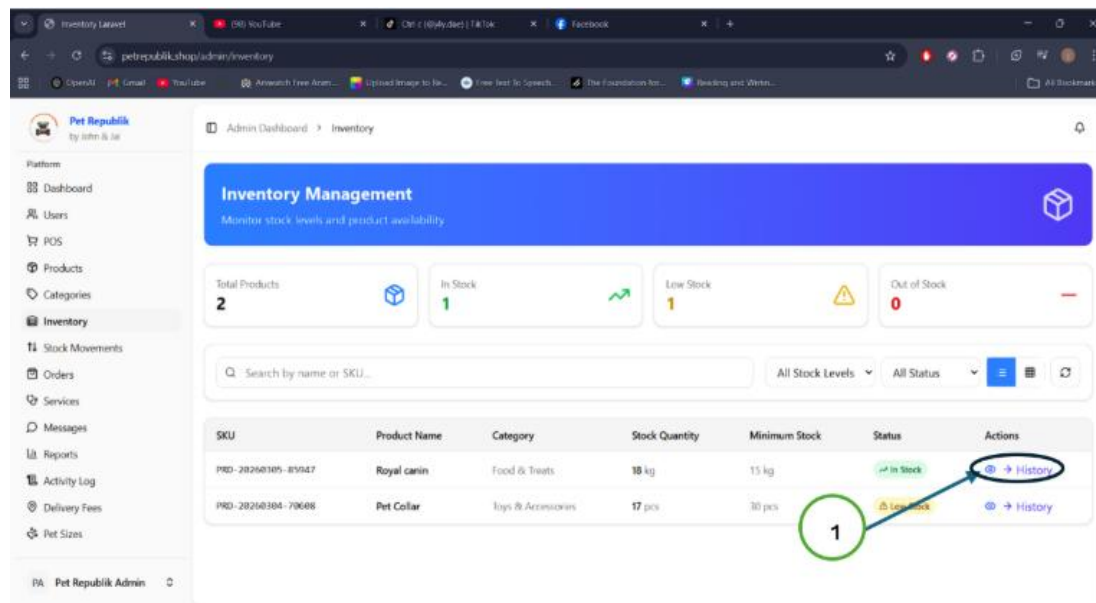


Figure 22. Week 3 Error Found/Lack of Functionality

Week 4 Error Found/Lack of Functionalities

As shown in Figure 23, during Week 4 of using the system, the customer encountered an issue related to the responsiveness of the customer-side interface while accessing the platform through a mobile phone. Although the system was intended to provide a convenient and user-friendly experience across different devices, several interface elements within the service booking function were misaligned and did not properly adapt to smaller screen sizes. Based on the figure, the filter options, search bar, and service cards appeared compressed and improperly arranged, making the interface difficult to navigate and use on mobile devices. This issue affected the customer's ability to complete service booking transactions efficiently using a smartphone.

This finding highlights a limitation in achieving the project's goal of providing a more convenient alternative to manual booking methods, as stated on Page 4, Paragraph 4. Since the system is not fully responsive on mobile devices, customers may experience difficulties when accessing the platform through smartphones, which can affect the convenience and accessibility of the service booking process. As a result, some users may still prefer desktop access or

traditional manual booking methods to complete their transactions more efficiently.

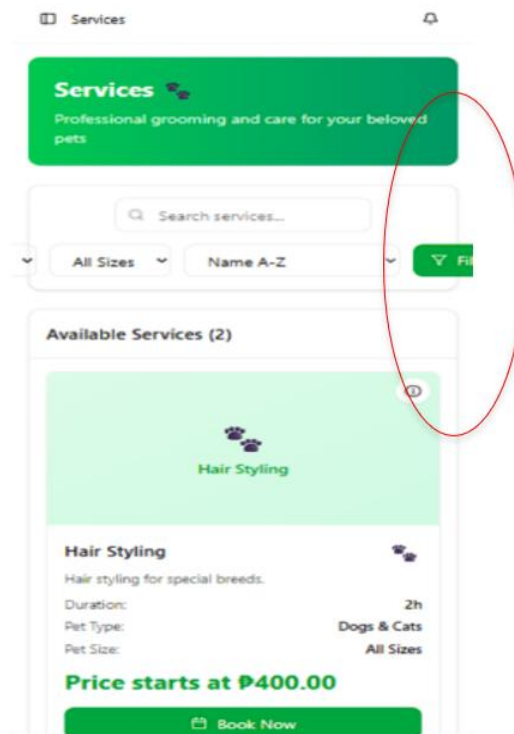


Figure 23. Week 4 Error Found/Lack of Functionalities

Week 5 Error Found/Lack of Functionalities

In week five, the team and the management staff focused on testing all the functions and features of the system to check if there were any errors unnoticed. The team carefully checked each part of the system during the testing of the system and actual usage. After testing and conducting the system, no issues and errors were found. All features worked correctly and each user role was able to access and use the system without any difficulties. The team continued to observe and check the system to make sure everything was working properly including the transaction and different processes to determine if the system had any hidden problems, but none were identified. Thus, this week five testing showed that the system is stable, functionable and free from errors during this stage of implementation.



Figure 24. Week 5 Improvements

Week 6 Error Found/Lack of Functionalities

In the sixth week of implementation, the system continued to work smoothly with no error and technical problems found. All of the functions stayed stable, and each user role was able to use the system without any difficulties. The team continued to monitor the system until the last day to make sure that everything was working properly. Each user confirmed that the system was very helpful in their daily tasks, especially in managing orders and booking service. Overall, this week was marked as successful completion of the team's system deployment.



Figure 25. Week 6 Final System Output

CHAPTER III

RESULT AND DISCUSSION

This chapter presents the results and findings from the implementation and testing of the proposed Web-Based Integrated Platform for Online Ordering and Service Appointment Management for John & Jai Pet Republik Company in Panabo City. It outlines the data collected during the six weeks of system deployment, conducted from March 9 to April 17, 2026. The results include system performance, user feedback, and the overall effectiveness of the system in supporting and improving the daily operations of the business.

During this period, the system was actively used by the Admin, Cashier, and customers. The admin and staff utilized the system to manage customer orders, handle service bookings, track transactions, and respond to customer inquiries. These functionalities contributed to more efficient operations, enabling the business to generate reports and monitor activities more easily compared to the previous manual processes. The evaluation of the system was based on both its functional and non-functional requirements, guided by the ISO/IEC 25010 software quality model. This model assesses software quality through key characteristics such as Functional Suitability, Reliability, Compatibility, Usability, Performance Efficiency, Security, and Portability.

To ensure the reliability and appropriateness of the evaluation instrument, the system questionnaire was reviewed and validated by Dr. Lanie Laureano, an external expert. Her validation confirmed that the questionnaire effectively measures the relevant aspects of system quality.

To gather meaningful data, selected end users of the system, specifically Admin, Cashier, and Customer participated in the evaluation process. They were asked to complete a structured questionnaire aligned with the identified quality characteristics. Each item in the questionnaire was designed to reflect the users' actual experiences while interacting with the system. Responses were measured using a 5-point Likert scale, allowing participants to indicate their level of agreement or satisfaction with various features of the system.

The collected feedback from the admin and cashier and customer will provide valuable insights into how the system improved order management, streamlined service booking processes, and supported overall operational efficiency of the business.

A. Results

The respondents of the study consisted of the Admin, Cashier, and Customers of John & Jai Pet Republik Company who used the system during the deployment period. A total of 18 respondents participated in the evaluation. The Admin and Cashier were selected using purposive sampling, as they are directly involved in managing system operations such as handling orders, processing transactions, and monitoring system performance. Meanwhile, customers were selected based on their actual use of the system during the deployment period, ensuring that only those who had firsthand experience in placing orders and booking services were included in the evaluation.

This approach allowed the development team to gather feedback from both internal users (Admin and Cashier) and external users (Customers), ensuring that the data collected reflects actual system usage and a range of user perspectives.

These descriptive equivalents were used to interpret the responses of the participants in evaluating the system's functionality, usability, performance, and overall effectiveness. Higher mean scores indicate a stronger level of agreement, reflecting better user satisfaction and system performance, while lower scores indicate areas that may need improvement.

Table 46 below presents the Likert scale used to interpret the responses of the participants in evaluating the proposed system. The scale ranges from 1 to 5, where each numerical value corresponds to a descriptive equivalent that reflects the level of agreement of the respondents. The use of this scale allowed the development team to clearly interpret the mean scores gathered from the evaluation. Higher values, particularly those within the range of 4.21 – 5.00,

indicate that respondents agreed that the system is very efficient and meets the expected requirements and objectives, while lower values reflect disagreement and indicate that the system does not meet expectations.

Based on this scale, the interpretation of results becomes more structured and meaningful, as each computed mean can be directly associated with a level of system effectiveness. For instance, a result interpreted as “Very Efficient” signifies that the system is performing very well in meeting its intended functions, while results falling under “Efficient” and “Moderately Efficient” indicate acceptable performance with possible areas for improvement. On the other hand, ratings under “Slightly Efficient” and “Not Efficient” highlight weaknesses in the system that may require further enhancement.

The evaluation aimed to provide a comprehensive assessment of the proposed Pet Republik: A Web-Based Integrated Platform for Online Ordering and Service Appointment Management, identify possible areas for improvement, and determine the overall impact of its implementation on the daily operations of John & Jai Pet Republik Company. The system objectives and the corresponding user assessment results are presented below:

Table 46. Likert Scale of the Pet Republik System

Numerical Description	Range	Descriptive Equivalent	Description
5	4.21 – 5.00	Very Efficient	The Web-based system meets the expected requirements and functionality of the system's objectives
4	3.41 – 4.20	Efficient	The Web-based system meets most of the expected requirements and functionality of the system's objectives
3	2.61 – 3.40	Moderately Efficient	The Web-based system meets some of the expected requirements and functionality of the system's objectives

2	1.81 – 2.60	Slightly Efficient	The Web-based system does not meet the expected requirements and functionality of the system's objectives
1	1.00 – 1.80	Not Efficient	The Web-based system fails to meet its expected requirements and functionality of the system's objectives

This section shows the user interfaces of the proposed system. The system provides a structured and user-friendly interface for both internal users and customers. The main screen displays a navigation menu that allows users to access key features such as product management, order processing, service booking, transaction tracking, and messaging. Once a user selects a specific function, the system opens the corresponding page where tasks such as managing products, processing orders, booking services, or communicating with customers can be performed efficiently.

1. Log-in to their respective accounts

1.1 Admin Log-in to the system.

The Login Page enables secure access to the system by requiring valid administrator credentials, including username and password. Upon submission, the system performs input validation and verifies the entered credentials against the administrator records stored in the database. If authentication is successful, the system initiates a secure session and grants the admin access to the administrative dashboard.

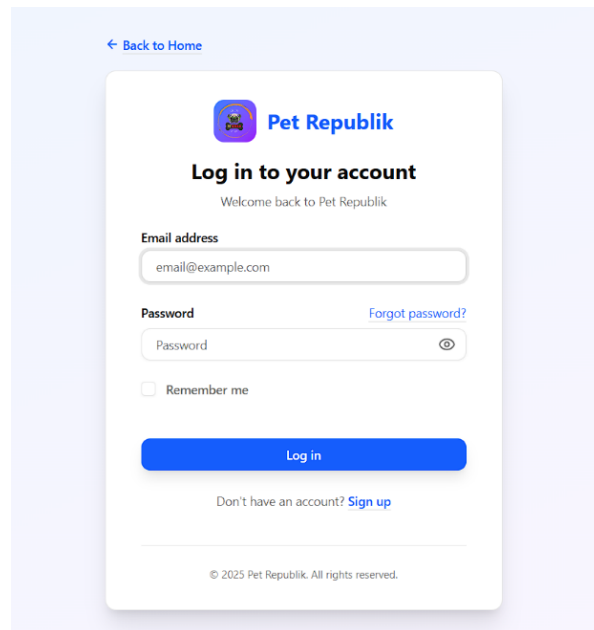
The image shows a login page for 'Pet Republik'. At the top left, there is a link '← Back to Home'. The page features the 'Pet Republik' logo, which consists of a purple square with a white dog icon and the text 'Pet Republik' in blue. Below the logo, the heading 'Log in to your account' is displayed in bold, followed by the subtext 'Welcome back to Pet Republik'. The login form includes an 'Email address' field with the placeholder 'email@example.com', a 'Password' field with a toggle icon, and a 'Remember me' checkbox. A blue 'Log in' button is positioned below the password field. To the right of the password field is a link 'Forgot password?'. At the bottom of the form, there is a link 'Don't have an account? Sign up'. The footer of the page states '© 2025 Pet Republik. All rights reserved.'

Figure 26. Log in Page

1.2 Manage user accounts.

The figure below, User Account Management is a feature of the Pet Republik System where the administrator can manage user accounts, including creating new accounts for employees. It also allows the admin to update, view, activate, or deactivate accounts to ensure proper access control and secure system usage.

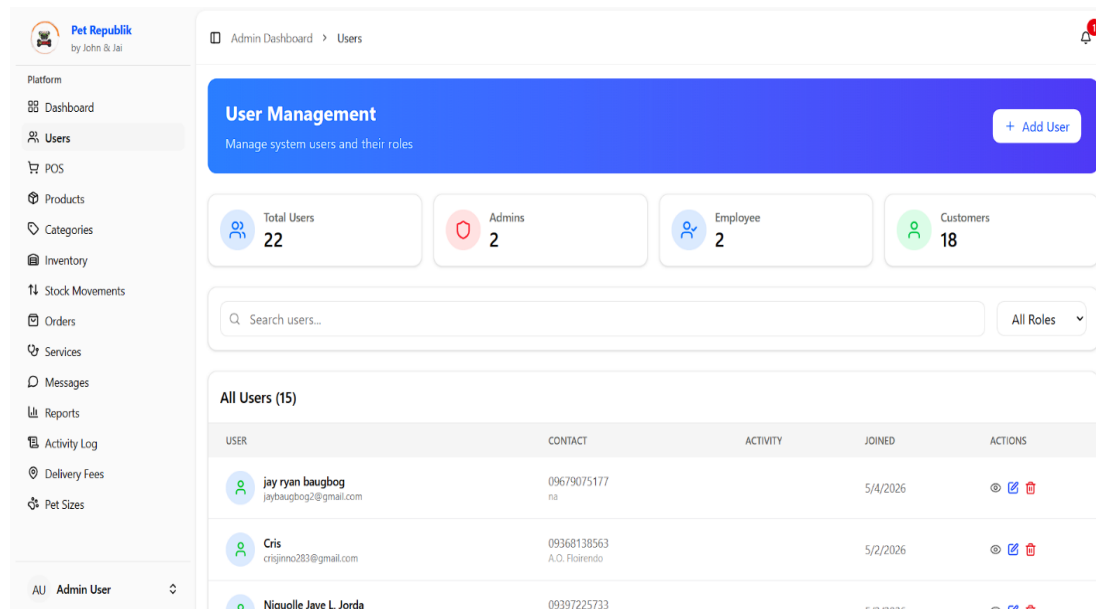


Figure 27. User Management

1.3 Manage products, categories, delivery fees and services.

Product Management is a feature of the system that allows administrators to add, view, update, and remove products. It helps organize product information such as name, description, price, and stock level, ensuring accurate and up-to-date records for smooth inventory and sales operations. This process is highly effective and well-implemented, with a mean score of 5.0 interpreted as Very Efficient, reflecting its strong efficiency, accuracy, and usability in product management.

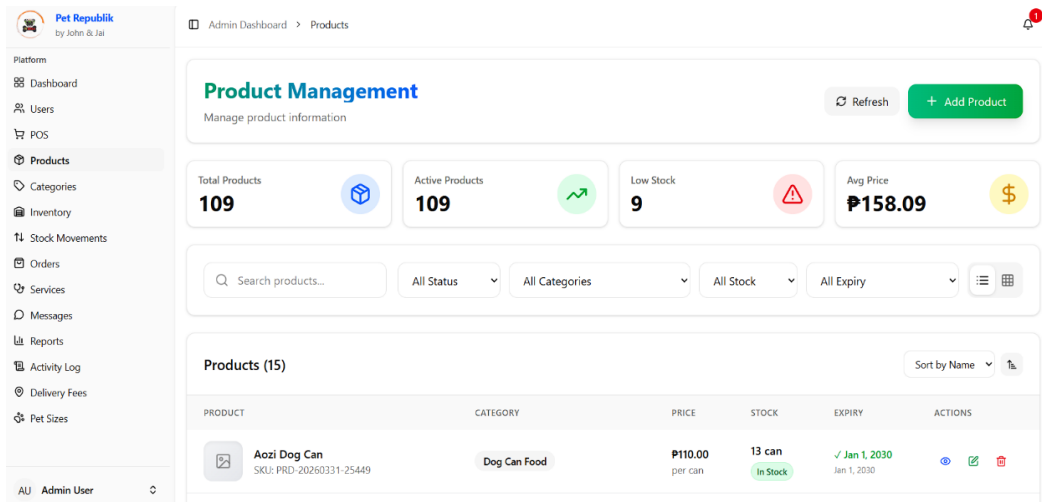


Figure 28. Product Management

Product Category Management, in figure 29, is a feature system that allows administrators to organize products into different categories. It enables the creation, updating, viewing, and deletion of categories to ensure products are properly grouped, making it easier to manage inventory and improve product navigation within the system.

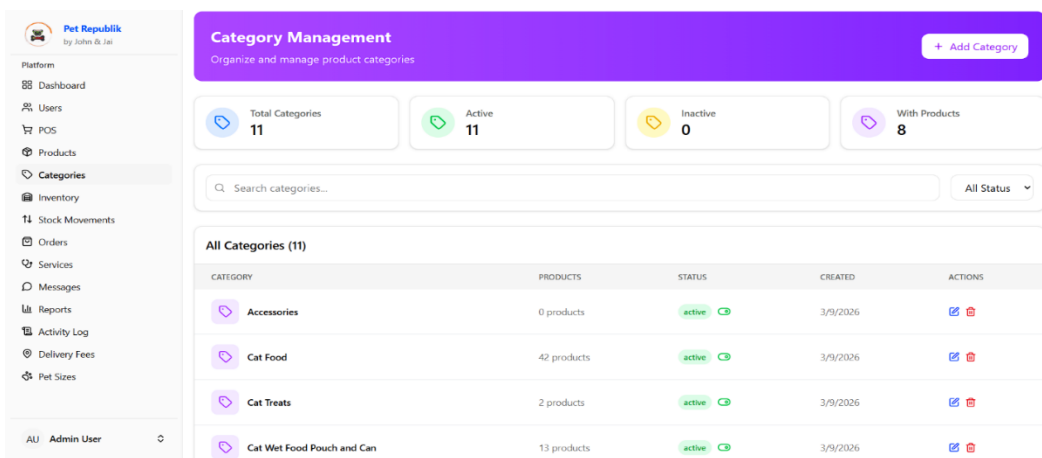


Figure 29. Category Management

Barangay Delivery Fees is a system feature that allows administrators to set and manage delivery charges based on barangay locations. It helps ensure accurate computation of delivery costs depending on the customer's address, making the checkout and order process more transparent and organized.

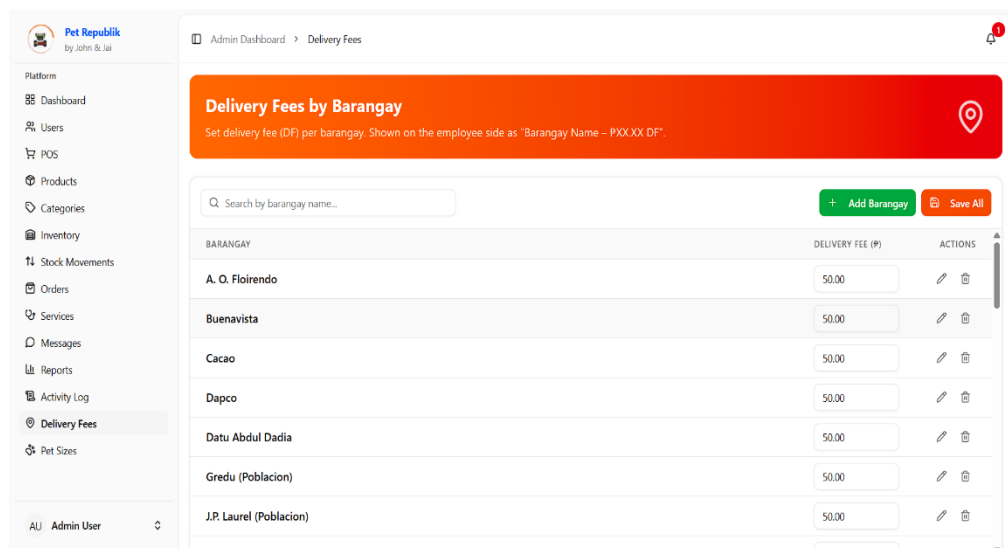


Figure 30. Barangay Delivery Fees

Service Management is a system feature that allows administrators to add, view, update, and remove services offered by the business. It helps organize service details such as descriptions, prices, and availability, ensuring accurate information and efficient management of all services in the system.

Service Management is highly effective and well-implemented, achieving a mean score of 5.0 (Very Efficient). This indicates that the system efficiently manages service details such as descriptions, prices, and availability, ensuring accurate information and smooth administration of all services offered by the business.

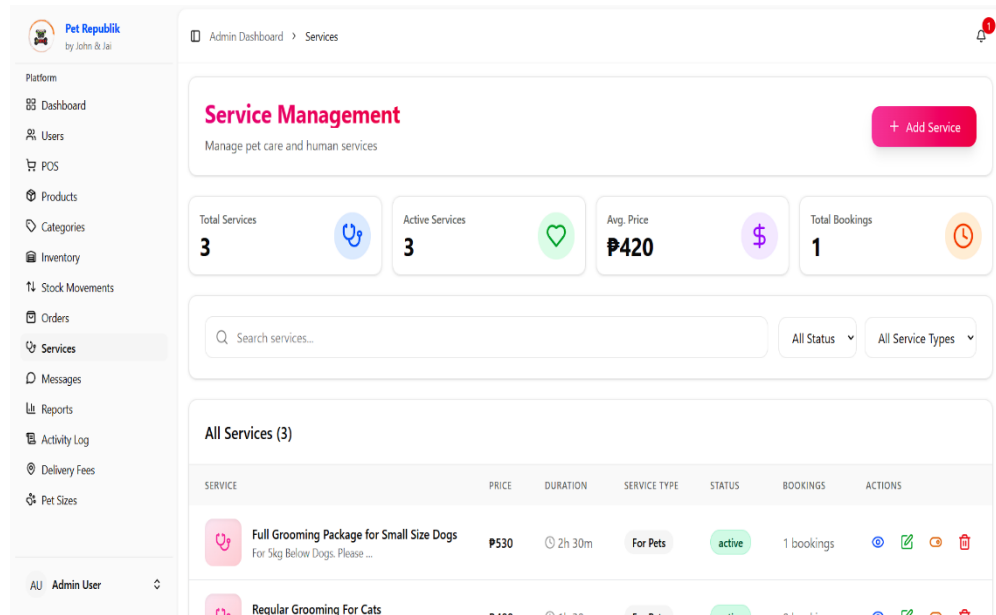


Figure 31. Service Management

1.3 Communicate with cashier

Communication with Cashier is a system feature that allows smooth coordination between the administrator and the cashier. It enables the exchange of important updates, order information, and transaction details to ensure accurate processing and efficient handling of payments and sales operations.

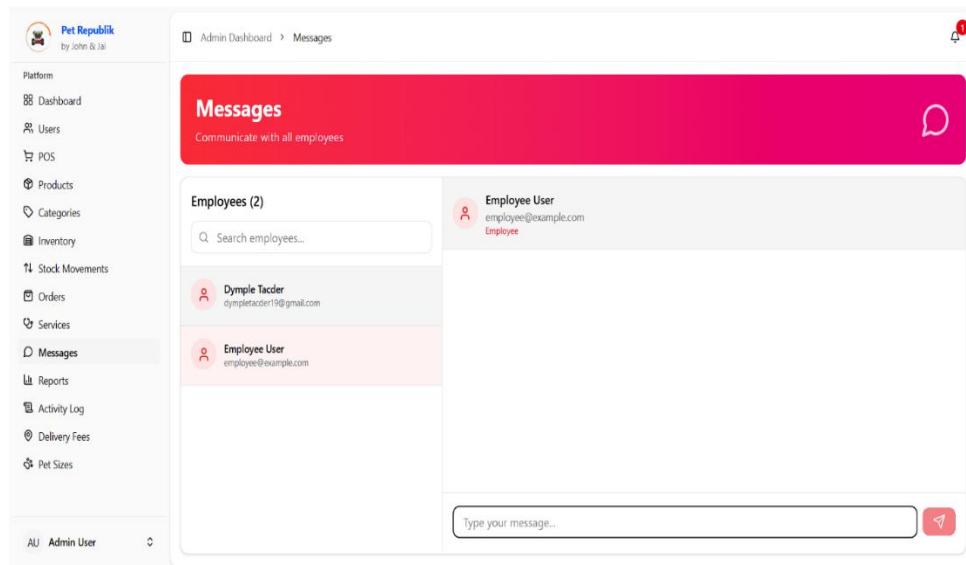


Figure 32. Messages (Admin)

1.4 Tracks inventory, stock movement, customer orders and activity logs.

Stock Movements is a system feature that tracks the changes in product inventory, including additions from restocking and reductions from sales or usage. It helps monitor stock levels accurately, ensuring proper inventory control and preventing shortages or overstocking.

Date	Product	Type	Quantity	Previous Stock	New Stock	User	Reason
5/3/2026 10:38 AM	Zoi Cat 1/2kg PRD-20260313-83376	Stock Out	-2	15	13	Dymple Tacder	POS Sale
5/3/2026 10:57 AM	Yum Yum Adult PRD-20260314-87538	Stock Out	-10	10	0	Dymple Tacder	POS Sale
5/3/2026 10:57 AM	Whiskas Jr 1kg PRD-20260313-12437	Stock Out	-1	15	14	Dymple Tacder	POS Sale
5/3/2026 10:35 AM	Top Breed Puppy PRD-20260314-59953	Stock Out	-10	10	0	Dymple Tacder	POS Sale
5/3/2026 10:35 AM	Top Breed Adult PRD-20260314-19142	Stock Out	-10	10	0	Dymple Tacder	POS Sale
5/3/2026 10:34 AM	Toei Cat 1/2kg PRD-20260313-59848	Stock Out	-1	15	14	Dymple Tacder	POS Sale
5/3/2026	Soocial Doo Adult						

Figure 33. Stock Movements (Admin)

Order Management is a system feature that allows administrators to view, process, update, and track customer orders. It helps ensure that all orders are properly handled from confirmation to completion, improving accuracy, efficiency, and overall order processing workflow.

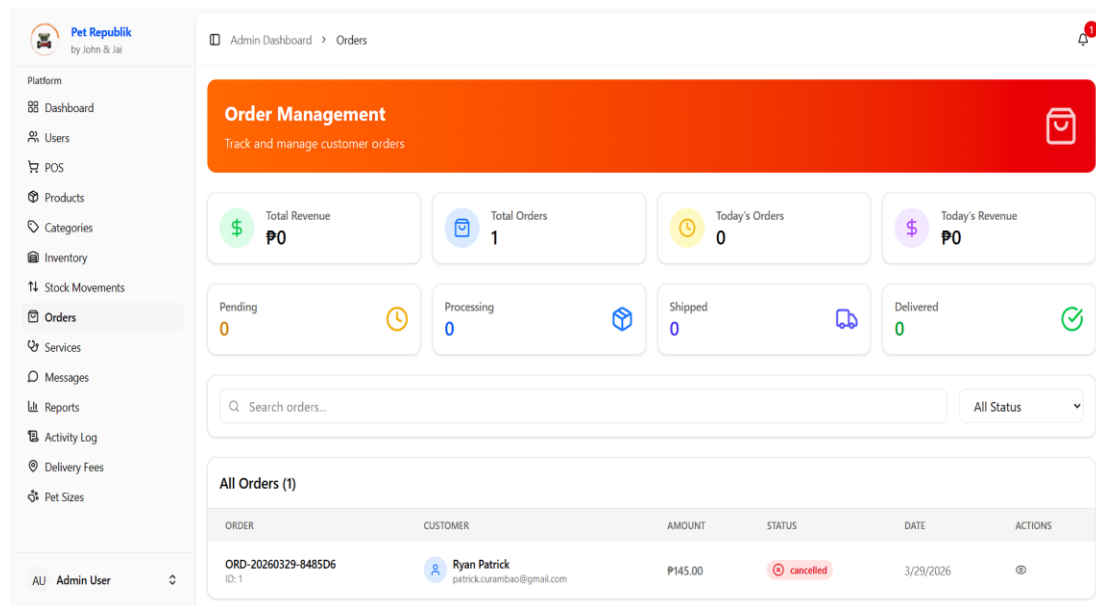


Figure 34. Order Management

Activity Log is a system feature that records all actions performed by users within the system. It tracks activities such as logins, updates, creations, and deletions to provide a clear history of system usage, helping ensure transparency, security, and accountability.

Activity Log is highly effective and well-implemented, achieving a mean score of 5.0 (Very Efficient). This indicates that the feature efficiently records and monitors all user activities such as logins, updates, creations, and deletions, ensuring transparency, security, and accountability within the system.

Activity	Performed by	Details	Date & Time
POS Sale payment processed: POS-69F6B9C0AD36C - P126.00 by Dymple Tacder	Dymple Tacder dympletacder19@gmail.com • employee	attributes: [object Object]	May 3, 2026, 10:58:08 AM
POS Sale created: POS-69F6B9C0AD36C by Dymple Tacder	Dymple Tacder dympletacder19@gmail.com • employee	attributes: [object Object]	May 3, 2026, 10:58:08 AM
POS Sale payment processed: POS-69F6B98003728 - P1050.00 by Dymple Tacder	Dymple Tacder dympletacder19@gmail.com • employee	attributes: [object Object]	May 3, 2026, 10:57:52 AM
POS Sale created: POS-69F6B98003728 by Dymple Tacder	Dymple Tacder dympletacder19@gmail.com • employee	attributes: [object Object]	May 3, 2026, 10:57:52 AM
POS Sale payment processed: POS-69F6B98EA1875 - P180.00 by Dymple Tacder	Dymple Tacder dympletacder19@gmail.com • employee	attributes: [object Object]	May 3, 2026, 10:57:18 AM

Figure 35. Activity Log

1.5 Generate detailed reports

Generate Reports is a system feature that allows administrators to create summaries of system data such as sales, inventory, orders, and user activities. It helps provide clear and organized information for monitoring performance, making decisions, and evaluating overall business operations.

Generate Reports is highly effective and well-implemented, achieving a mean score of 5.0 (Very Efficient). This indicates that the feature efficiently produces accurate and organized summaries of system data such as sales, inventory, orders, and user activities, supporting effective monitoring, decision-making, and evaluation of overall business operations.

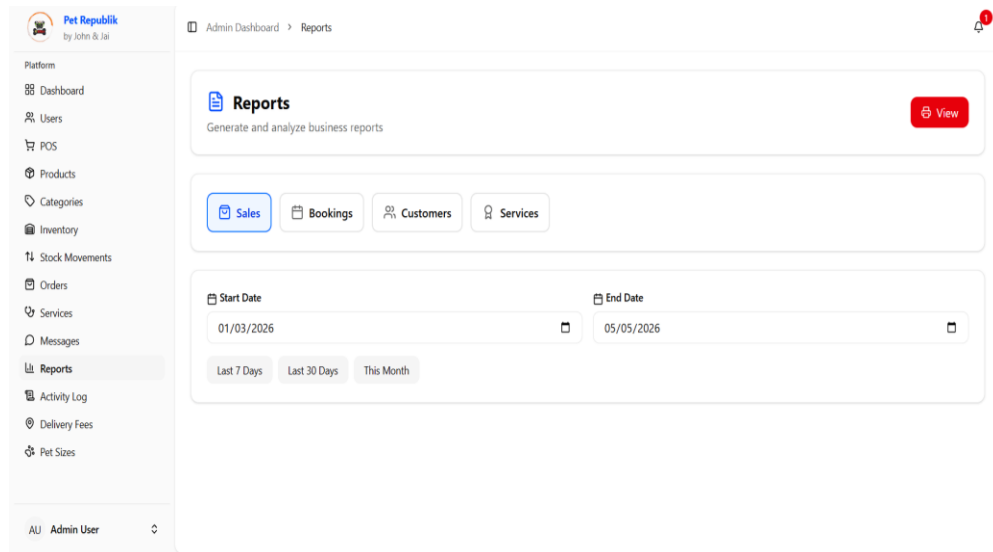


Figure 36. Generate Reports

2. Cashier functionalities

2.1 Add and update product information

Add and Update Product Information (Cashier Access) is a feature available to the cashier that allows them to view and update basic product details when needed. It ensures that product information such as price, availability, and stock levels remains accurate during transactions for smooth and efficient sales processing.

Figure 37. Add Product (Cashier)

2.2 Manage customers' orders and bookings.

Order Management (Cashier) is a system feature that allows the cashier to process customer orders by creating, viewing, and updating order transactions. It helps ensure accurate order entry, proper item listing, and correct payment processing for smooth and efficient sales operations.

Order Management (Cashier) is highly effective and well-implemented, achieving a mean score of 5.0 (Very Efficient). This indicates that the feature efficiently supports the cashier in processing customer orders by creating, viewing, and updating transactions, ensuring accurate order entry, correct item listing, and proper payment processing for smooth and efficient sales operations.

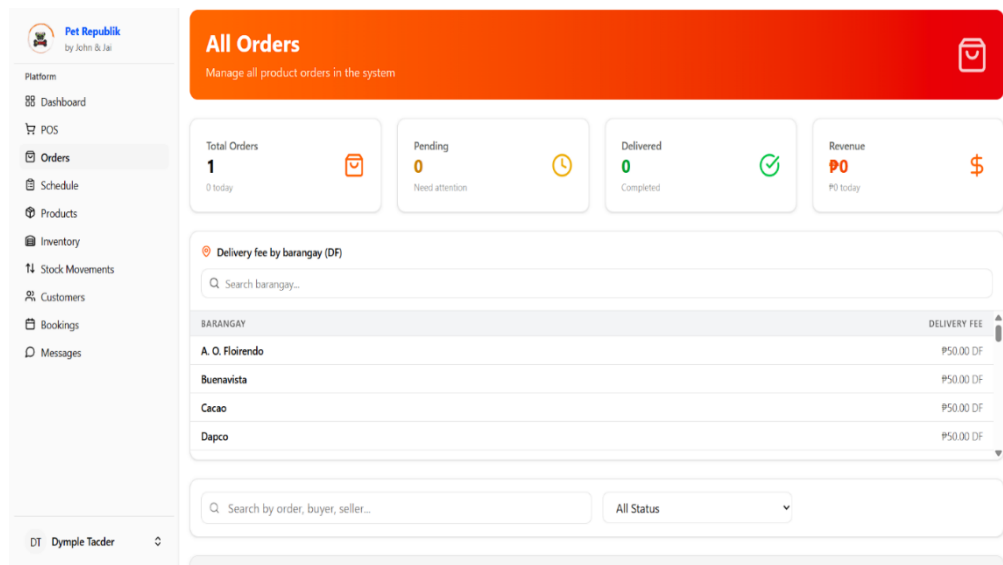


Figure 38. Order Management (Cashier)

Booking Management (Cashier) is a system feature that allows the cashier to view, confirm, and update customer bookings. It helps ensure that all reservations or service appointments are properly recorded, scheduled, and processed accurately for smooth and organized operations.

Booking Management (Cashier) is highly effective and well-implemented, achieving a mean score of 5.0 (Very Efficient). This indicates that the feature efficiently enables the cashier to view, confirm, and update customer bookings, ensuring that all reservations or service appointments are properly recorded, scheduled, and processed accurately for smooth and organized operations.

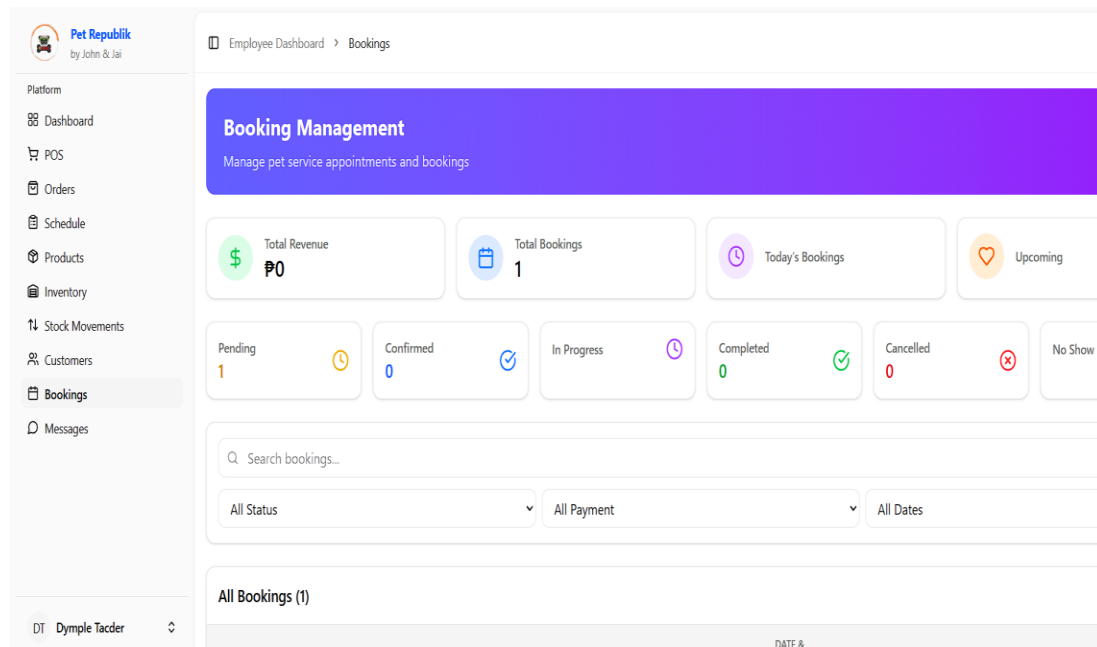


Figure 39. Booking Management

2.3 Communicate with customers and admin

Messages (Cashier) is a system feature that allows the cashier to communicate with both the administrator and customers. It enables sending and receiving important updates, inquiries, and notifications related to orders, bookings, and transactions, ensuring clear and efficient communication for smooth operations.

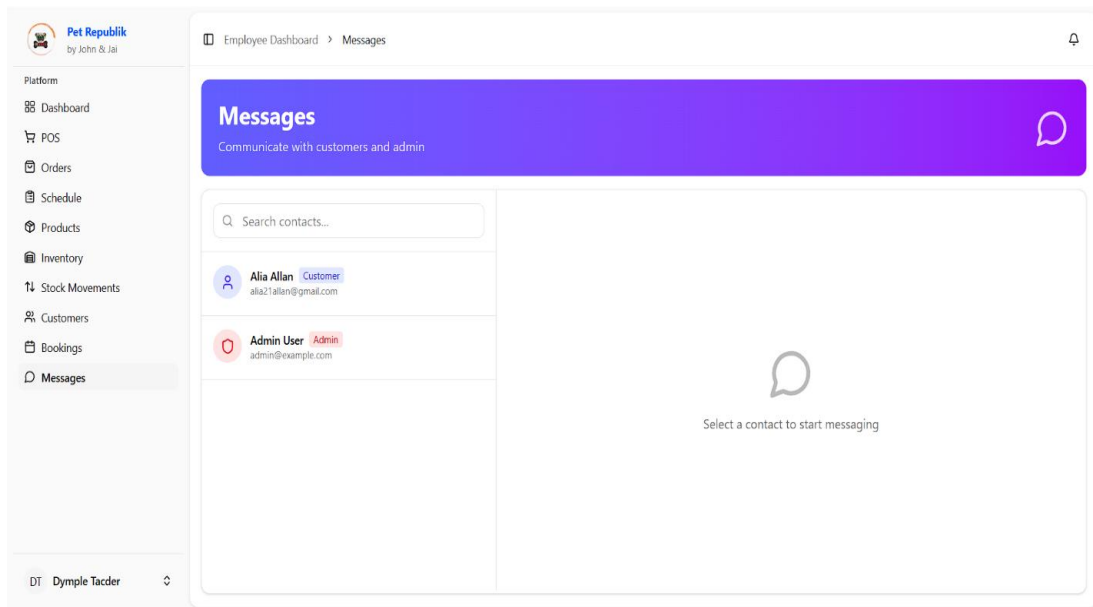
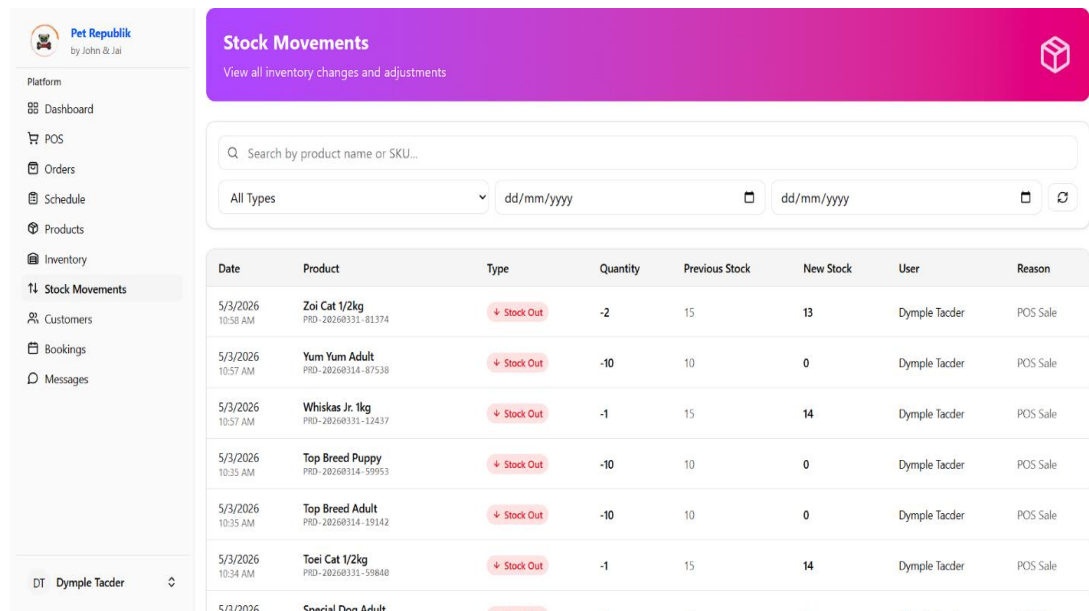


Figure 40. Messages (Cashier)

2.5 Track inventory, POS and stock movements

Stock Movements (Cashier) is a system feature that allows the cashier to record and monitor changes in inventory levels during daily transactions. It helps track stock additions and deductions caused by sales or adjustments, ensuring accurate inventory updates and smooth store operations.



Date	Product	Type	Quantity	Previous Stock	New Stock	User	Reason
5/3/2026 10:56 AM	Zoi Cat 1/2kg PRD-20260331-81374	Stock Out	-2	15	13	Dymple Tacder	POS Sale
5/3/2026 10:57 AM	Yum Yum Adult PRD-20260314-87538	Stock Out	-10	10	0	Dymple Tacder	POS Sale
5/3/2026 10:57 AM	Whiskas Jr. 1kg PRD-20260331-12437	Stock Out	-1	15	14	Dymple Tacder	POS Sale
5/3/2026 10:35 AM	Top Breed Puppy PRD-20260314-59953	Stock Out	-10	10	0	Dymple Tacder	POS Sale
5/3/2026 10:35 AM	Top Breed Adult PRD-20260314-19142	Stock Out	-10	10	0	Dymple Tacder	POS Sale
5/3/2026 10:34 AM	Toei Cat 1/2kg PRD-20260331-59848	Stock Out	-1	15	14	Dymple Tacder	POS Sale
5/3/2026	Special Dog Adult						

Figure 41. Stock Movements (Cashier)

Point of Sale (POS) for Cashiers is a system feature that allows the cashier to process customer transactions by adding items, computing total payments, and handling payments accurately. It helps ensure fast, efficient, and organized sales operations.

Point of Sale (POS) for Cashier is highly effective and well-implemented, achieving a mean score of 5.0 (Very Efficient). This indicates that the system efficiently supports the cashier in processing transactions by adding items, computing total payments, and handling payments accurately, ensuring fast, organized, and efficient sales operations.

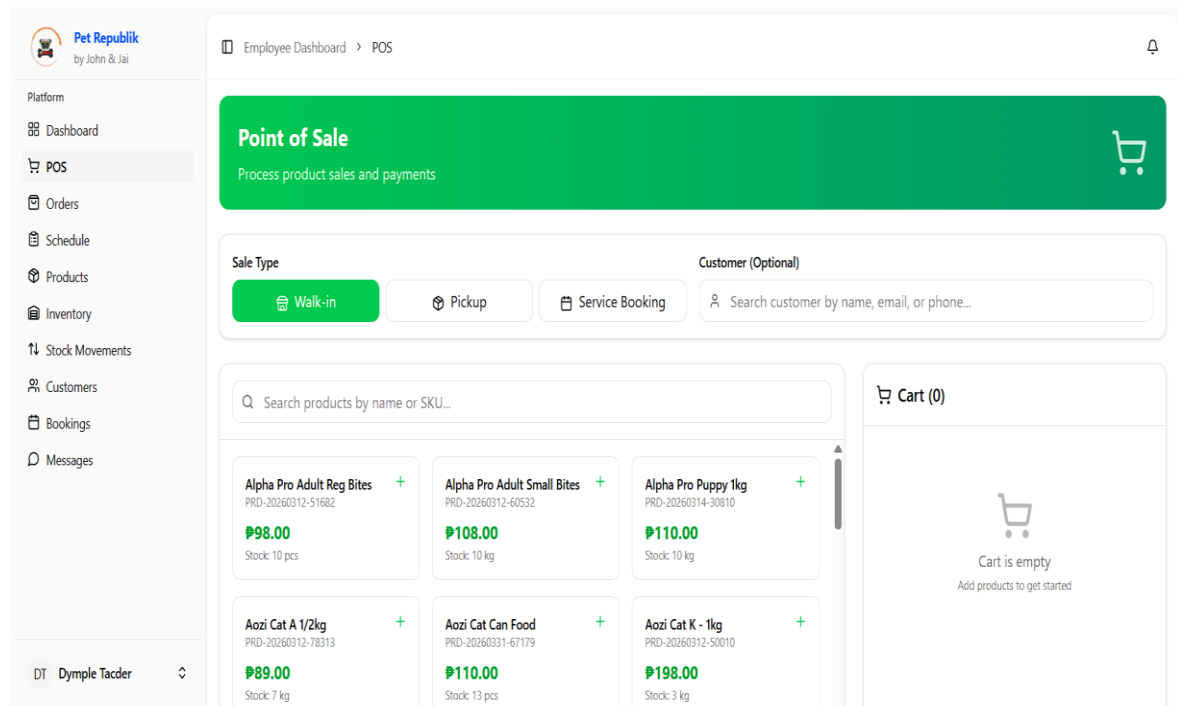


Figure 42. Point of Sale

3. Customer functionalities

3.1 Place orders and book pet services

Ordering is a system feature that allows customers or staff to place orders for products or services. It helps manage the selection, confirmation, and submission of items for purchase, ensuring accurate and efficient processing of transactions.

Ordering is highly effective and well-implemented, achieving a mean score of 4.8 (Very Efficient). This indicates that the system efficiently supports customers or staff in placing orders for products or services by managing selection, confirmation, and submission of items, ensuring accurate and smooth transaction processing.

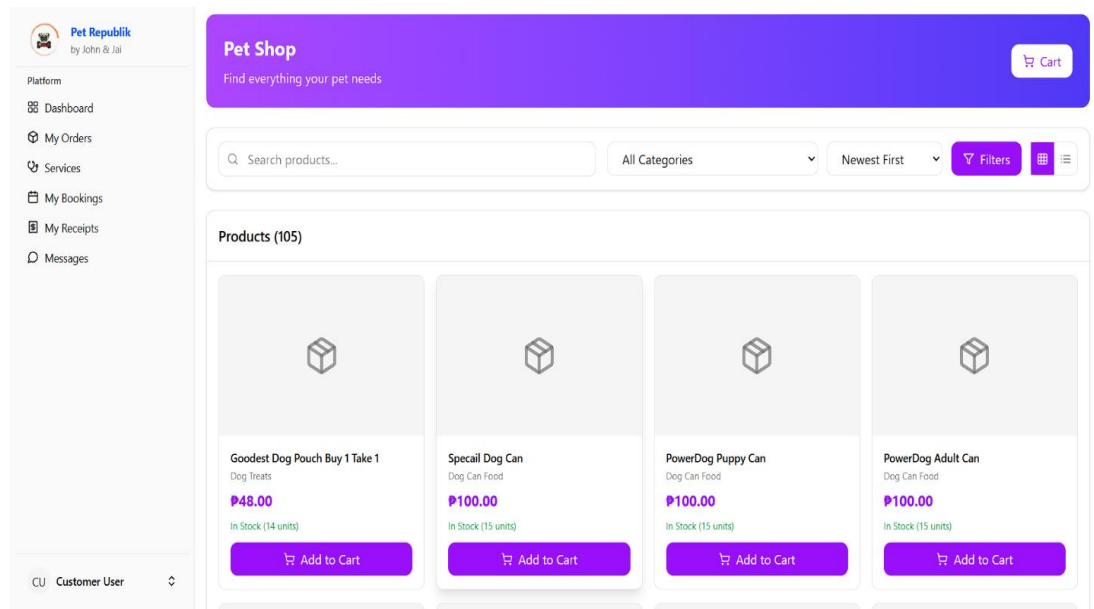


Figure 43. Ordering

Book Pet Services is a system feature that allows customers to schedule appointments for pet grooming and other related services. It helps manage the selection, confirmation, and submission of service bookings, ensuring accurate and efficient processing of reservations.

Book a Pet Services is effective and well-implemented, achieving a mean score of 4.6 (Very Efficient). This indicates that the system supports customers and staff in booking pet services by managing the selection, confirmation, and submission of service appointments, ensuring accurate and efficient processing of reservations.

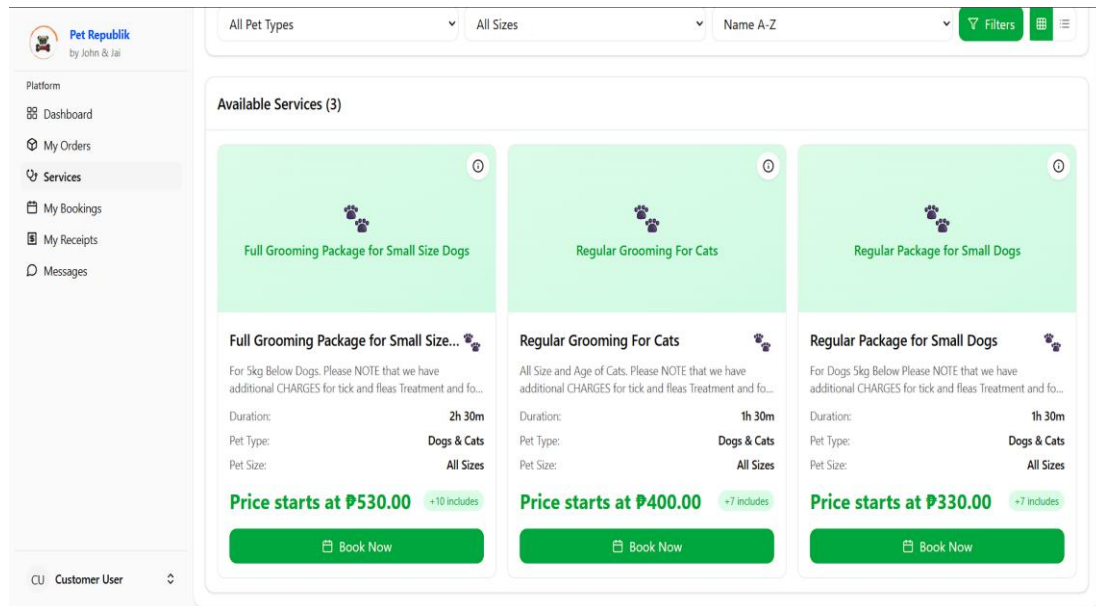


Figure 44. Book Pet Services

3.2 Track orders and bookings status

Order History is a system feature that allows customers to view and track all their placed orders. It provides details such as order status, items purchased, payment information, and delivery updates, helping users easily monitor their transactions in one place.

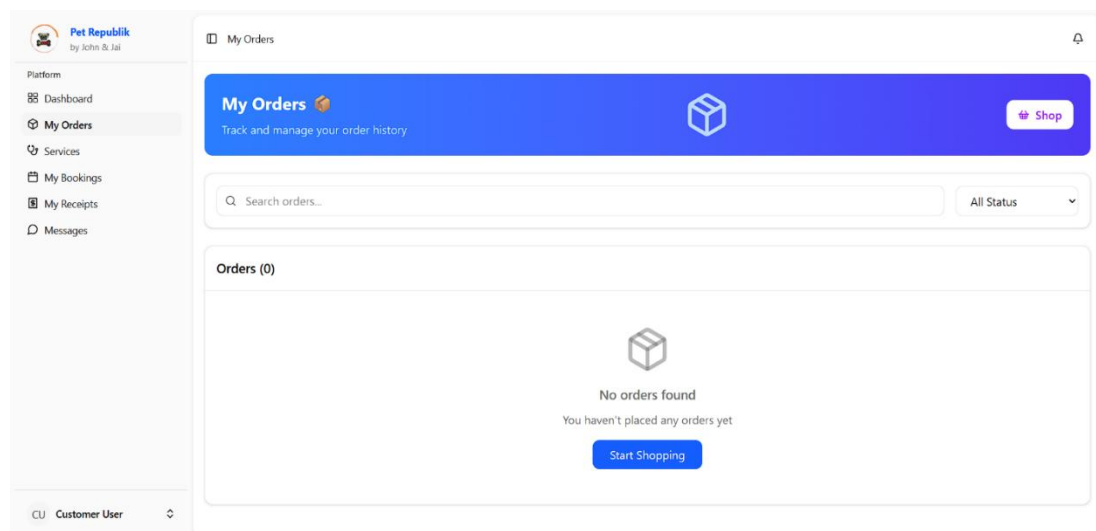


Figure 45. Order History

In figure 46, Booking History is a system feature that allows customers to view and manage all their scheduled pet service appointments. It provides details such as booking date, time, service type, and status, helping users easily track and keep updated on their reservations.

Booking History is effective and well-implemented, achieving a mean score of 4.8 (Very Efficient). This indicates that the system efficiently supports customers in managing their pet service appointments by providing clear and organized booking details for easy tracking and updates.

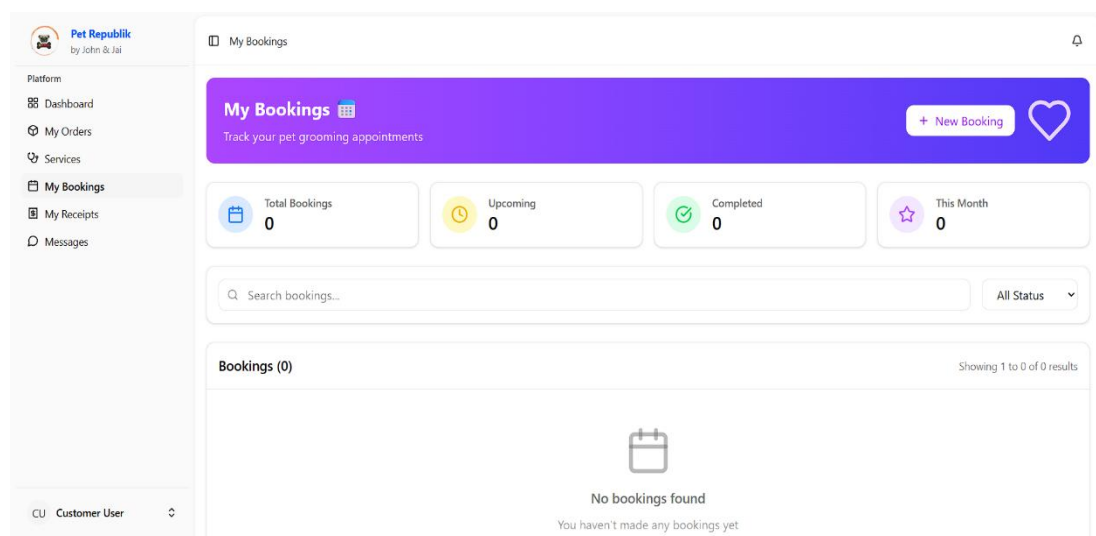


Figure 46. Booking History

3.3 Communicate with the cashier

Figure 47 shows the messaging feature of the system that allows customers to send and receive messages regarding their orders and bookings. It enables clear communication with the cashier.

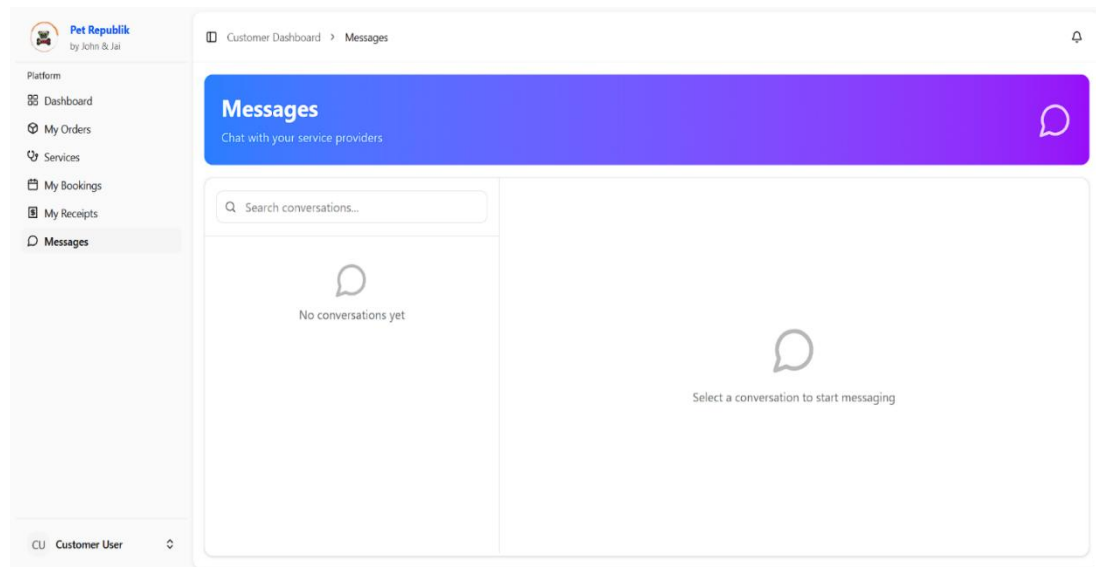


Figure 47. Messages (Customer)

The table below shows the functionality & non-functionality of the proposed Web-Based Integrated Platform for Online Ordering and Service Appointment Management based on feedback from the Admin, Cashier, and Customers of John & Jai Pet Republik Company. The evaluation used a 5-point Likert scale to measure user agreement, with scores ranging from 5 (Very Efficient) to 1 (Not Efficient).

The survey assessed how effectively users were able to manage product listings, process customer orders, handle service bookings, and track transactions within the system. It also evaluated the system's ability to facilitate communication between users, generate accurate reports, and provide a smooth and responsive user experience. Additionally, the system's capability to support daily business operations, improve efficiency, and reduce manual workload was considered as part of the overall evaluation.

Table 47 further corroborates the functional suitability of the system for John & Jai Pet Republik Company. With an overall mean score of 4.6 the system achieved a descriptive equivalent of "Very Efficient", reflecting its high level of performance and reliability. Administrative and operational functions including product

management, inventory monitoring, transaction tracking, and report generation received unanimous perfect scores (M = 5.00) from both the Admin and Cashier groups. These results suggest that the core business logic and backend features are fully optimized for the company's daily operations. Customer-facing functionalities also demonstrated strong performance, with ratings ranging from 4.3 to 4.9. While most features aligned with the "Very Efficient" threshold, specific tasks such as browsing products, placing orders, and accessing transaction history were noted for their effectiveness. The slight variance in customer feedback, particularly regarding service bookings, suggests that while the system is highly functional and meets user needs, there remains a narrow margin for enhancing the user experience for external clients. Collectively, the data confirms that the system is a robust solution that successfully facilitates pet shop management and customer engagement.

Table 47. Functional Requirements Results

FUNCTIONAL SUITABILITY	Admin	Cashier	Customer	MEAN	Descriptive Equivalent
1. The system supports management of product information.	5			5	Very Efficient
2. The system allows monitoring of inventory records.	5			5	Very Efficient
3. The system supports monitoring of transactions and activities.	5			5	Very Efficient
4. The system provides features for managing services and bookings.	4			4	Efficient
5. The system generates reports for sales, bookings, customers, and services.	4			4	Efficient
6. The system		5		5	Very Efficient

supports processing customer transactions through the POS feature.					
7. The system allows accurate management of customer orders.		5		5	Very Efficient
8. The system supports handling of service bookings.		5		5	Very Efficient
9. The system provides access to customer and transaction information.		4		4	Efficient
10. The system features support daily shop operations.		4		4	Efficient
11. The system allows customers to create and log in to their accounts.			4.9	4.9	Very Efficient
12. The system enables customers to browse available pet products.			4.8	4.8	Very Efficient
13. The system enables customers to place product orders.			4.8	4.8	Very Efficient
14. The system allows customers to book pet services.			4.6	4.6	Very Efficient
15. The system provides access to order history.			4.8	4.8	Very Efficient
16. The system provides access to booking history.			4.7	4.7	Very Efficient
17. The system supports communication between customers and cashiers.			4.3	4.3	Very Efficient

OVERALL MEAN				4.6	Very Efficient
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The results for the non-functional requirements of the system, as detailed in Table 48, the evaluation of the system based on the ISO 25010 software quality characteristics showed that the system achieved an overall descriptive rating of “Very Efficient” from the Admin, Cashier, and Customer respondents.

The system obtained a mean score of 4.6 in Performance Efficiency, indicating that it processes transactions, updates, and operations quickly while maintaining stable performance.

In Usability, the system received a mean score of 4.9, showing that users found the interface easy to understand, navigate, and use. For Reliability, the system achieved a mean score of 4.6, confirming accurate transaction processing and consistent system operation. Security also received a high mean score of 4.9, demonstrating effective protection of user accounts, transactions, and confidential data.

Meanwhile, Maintainability obtained a mean score of 4.5, indicating that the system supports efficient updates and management. Compatibility achieved a mean score of 4.6, proving that the system works properly across different devices, browsers, and operating systems.

Lastly, Portability received a mean score of 4.8, showing that the system can be accessed reliably from different devices and locations. Overall, the results indicate that the system successfully met the required non-functional requirements and provided efficient, secure, reliable, and user-friendly services.

Table 48. Non-Functional Requirements Results

NON - FUNCTIONAL SUITABILITY	Admin	Cashier	Customer	MEAN	Descriptive Equivalent
PERFORMANCE EFFICIENCY					
1. The system processes administrative tasks without delay.	5			5	Very Efficient
2. The system loads reports and dashboards promptly.	5			5	Very Efficient
3. The system maintains performance when handling multiple records.	5			5	Very Efficient
4. The system processes updates quickly.	5			5	Very Efficient
5. The system responds promptly during administrative operations.	4			4	Efficient
6. The system processes sales transactions without delay.		5		5	Very Efficient
7. The system loads order and booking records promptly.		5		5	Very Efficient
8. The system maintains stable performance during peak usage.		4		4	Efficient
9. The system updates transaction records immediately after processing.		5		5	Very Efficient
10. The system responds quickly during task execution.		4		4	Efficient

11. The system efficiently loads pages when accessing different sections.			4.5	4.5	Very Efficient
12. The system processes ordering and booking transactions efficiently.			4.7	4.7	Very Efficient
13. The system responds quickly to user inputs and selections.			4.6	4.6	Very Efficient
14. The system maintains stable performance during continuous use.			4	4	Efficient
15. The system performs efficiently under multiple user access.			4.5	4.5	Very Efficient
TOTAL				4.6	Very Efficient
USABILITY					
16. The system interface is easy to understand.	4			4	Efficient
17. The system allows clear navigation between administrative features.	5			5	Very Efficient
18. The system provides clear options and labels.	5			5	Very Efficient
19. The system enables administrators to complete data management tasks easily.	5			5	Very Efficient
20. The system prevents errors during administrative	5			5	Very Efficient

operations.					
21. The system interface is easy to understand.		4		4	Efficient
22. The system allows clear and efficient navigation between features.		4		4	Efficient
23. The system provides clear instructions and labels.		5		5	Very Efficient
24. The system enables users to complete tasks easily.		5		5	Very Efficient
25. The system prevents user errors during transactions.		4		4	Efficient
26. The system provides an interface that is easy to understand.			4.5	4.5	Very Efficient
27. The system allows users to navigate the platform efficiently.			4.6	4.6	Very Efficient
28. The system presents instructions and labels clearly.			4.4	4.4	Very Efficient
29. The system enables users to complete transactions easily.			4.6	4.6	Very Efficient
30. The system prevents errors during user transactions.			4.3	4.3	Very Efficient
TOTAL				4.9	Very Efficient
RELIABILITY					
31. The system maintains accurate inventory records.	5			5	Very Efficient

32. The system stores sales and transaction data consistently.	5			5	Very Efficient
33. The system records and stores activity logs correctly.	5			5	Very Efficient
34. The system remains available during continuous use.	5			5	Very Efficient
35. The system recovers quickly from errors or system failures.	5			5	Very Efficient
36. The system records sales and transactions accurately.		5		5	Very Efficient
37. The system updates order status correctly after processing.		5		5	Very Efficient
38. The system operates consistently without system failures.		4		4	Efficient
39. The system remains available during continuous use.		4		4	Efficient
40. The system recovers quickly from errors or interruptions.		4		4	Efficient
41. The system performs transactions reliably without errors.			4.3	4.3	Very Efficient
42. The system accurately records orders.			4.9	4.9	Very Efficient

43. The system reliably saves and tracks service bookings.			4.7	4.7	Very Efficient
44. The system remains consistently available during use.			4	4	Efficient
45. The system recovers quickly from errors or failures.			4.4	4.4	Very Efficient
TOTAL				4.6	Very Efficient
SECURITY					
46. The system restricts access to administrative functions to authorized users.	5			5	Very Efficient
47. The system manages user permission levels appropriately.	5			5	Very Efficient
48. The system protects sensitive business information.	5			5	Very Efficient
49. The system enforces secure administrative access controls.	5			5	Very Efficient
50. The system protects employee accounts through secure authentication.		5		5	Very Efficient
51. The system restricts access to authorized users only.		5		5	Very Efficient
52. The system protects sensitive customer and transaction data.		5		5	Very Efficient

53. The system maintains confidentiality of system records.		5		5	Very Efficient
54. The system effectively protects users' personal information.			4.7	4.7	Very Efficient
55. The system ensures secure login and authentication processes.			4.8	4.8	Very Efficient
56. The system maintains secure and protected transactions.			4.6	4.6	Very Efficient
57. The system preserves the confidentiality of transaction records.			4.7	4.7	Very Efficient
TOTAL				4.9	Very Efficient
MAINTAINABILITY					
58. The system allows updating of product and service information.	5			5	Very Efficient
59. The system supports maintenance of inventory records.	4			4	Efficient
60. The system allows modification of system configurations when necessary.	4			4	Efficient
61. The system supports monitoring of system performance and status.	4			4	Efficient
62. The system allows easy updating		5		5	Very Efficient

of transaction records.					
63. The system supports efficient management of bookings and orders.		5		5	Very Efficient
64. The system logs cashier activities for monitoring and tracking.		5		5	Very Efficient
65. The system supports system updates without affecting core functions.		4		4	Efficient
66. The system allows efficient updating of products, services, and user data.			4.4	4.4	Very Efficient
67. The system supports easy modification of system features.			4.4	4.4	Very Efficient
68. The system enables administrators to manage system content efficiently.			4.5	4.5	Very Efficient
69. The system organizes components and data for easy maintenance.			4.5	4.5	Very Efficient
TOTAL				4.5	Very Efficient
COMPATIBILITY					
70. The system operates properly across different web	4			4	Efficient

browsers.					
71. The system functions correctly on different operating systems.	4			4	Efficient
72. The system supports integration with existing shop workflows.	4			4	Efficient
73. The system operates reliably alongside other business applications.	5			5	Very Efficient
74. The system works properly with POS devices used in the store.		4		4	Efficient
75. The system functions correctly on different computers.		4		4	Efficient
76. The system operates effectively across various web browsers.		4		4	Efficient
77. The system works reliably alongside other shop applications.		4		4	Efficient
78. The system works properly across different web browsers.			4.7	4.7	Very Efficient
79. The system operates effectively on mobile devices.			4.6	4.6	Very Efficient
80 The system runs properly on desktop and laptop			4.6	4.6	Very Efficient

computers.					
81. Operation remains stable alongside other applications on the device.			4.4	4.4	Very Efficient
82. The system works reliably alongside other applications.			4.3	4.3	Very Efficient
TOTAL				4.6	Very Efficient
PORTABILITY					
83. The system can be accessed from different administrative devices.	5			5	Very Efficient
84. The system performs consistently across different devices.	5			5	Very Efficient
85. The system allows remote access to administrative functions.	5			5	Very Efficient
86. The system adapts to different screen sizes.	5			5	Very Efficient
87. The system can be accessed from different employee workstations.		5		5	Very Efficient
88. The system performs consistently across different devices.		5		5	Very Efficient
89. The system can be accessed from		5		5	Very Efficient

different locations.					
90. The system adapts to different screen sizes.		5		5	Very Efficient
91. The system can be accessed reliably with an internet connection.			4.3	4.3	Very Efficient
92. The system adapts effectively to different screen sizes.			4.2	4.2	Efficient
93. The system can be accessed from various locations.			4.8	4.8	Very Efficient
94. The system performs consistently across different devices.			4.7	4.7	Very Efficient
TOTAL				4.8	Very Efficient

CHAPTER IV

CONCLUSION AND RECOMMENDATION

A. Conclusion

In conclusion, the development of Pet Republik: A Web-Based Integrated Platform for Online Ordering and Service Appointment Management for John & Jai Company successfully addressed the major challenges associated with the pet shop's manual processes. The proposed system achieved its intended objectives by providing key functionalities, including managing user accounts, handling product listings and inventory, processing customer orders, scheduling service bookings, and generating necessary reports for business operations.

The system offers a practical solution to the issues of manual transaction recording, appointment scheduling, and inventory tracking, which are often prone to delays, errors, and miscommunication. By digitalizing these processes, the system ensures more accurate data handling, real-time updates, and improved coordination between the admin, cashier, and customers. This reduces operational inefficiencies and supports better decision-making for the business.

In addition, automating core processes such as order management, booking services, and report generation helps lessen the workload of staff while improving the speed and reliability of daily operations. The system also enhances customer convenience by allowing them to browse products, place orders, and book services online without the need for physical visits or manual inquiries.

The system is designed to be beneficial for different users within the organization. Administrators can efficiently manage system data and monitor performance through reports, cashiers can handle transactions and bookings more effectively, and customers experience a more accessible and organized platform for their pet care needs. These improvements contribute to better service delivery, increased accuracy, and overall operational efficiency.

Overall, the Pet Republik system provides a more organized, efficient, and reliable approach to managing pet shop operations. It supports the transition from manual to digital processes and establishes a strong foundation for improving

service quality, customer satisfaction, and business growth for John & Jai Pet Republik Pet Shop.

B. Recommendations

This section presents the recommendations of the project team based on the findings and results gathered from the development and implementation of the proposed Pet Republik Information System. The recommendations aim to further improve the functionality, efficiency, and reliability of the system in supporting the operations of John & Jai Pet Republik Pet Shop. It also provides suggestions for future enhancements and improvements that may help the business adapt to changing technological and customer needs while ensuring continuous system effectiveness and user satisfaction.

1. Enhancement of GCash Payment Integration

In today's digital era, cashless payments are widely used because of their convenience and efficiency in handling transactions. Enhancing the GCash payment integration through automated payment verification can help the system process transactions more accurately, reduce human errors, and provide customers with faster and more reliable payment experience.

The enhancement was recommended by the organization to improve the existing GCash payment feature by automating the payment verification process instead of relying on manual screenshot submissions. Integrating API-based or reference number validation will allow the system to automatically confirm payments, reduce delays, and minimize transaction errors, resulting in a more efficient and seamless booking and ordering experience.

2. Implementation of Login Location and Account Monitoring Notification Feature

As online services continue to grow, implementing a reliable account security and monitoring feature can help protect user accounts and improve overall system

safety. Integrating login location notifications will allow users to receive alerts whenever their accounts are accessed from unfamiliar locations or outside the expected service area within Panabo City.

The idea was suggested by the project team's system analyst to strengthen account security and improve user protection within the system. The feature also includes account monitoring and reporting notifications to help detect suspicious activities, bogus users, and harmful or misleading posts that may negatively affect the business reputation and customer experience. This enhancement will improve the system's security, reliability, and trustworthiness while providing a safer platform for both customers and administrators.

3. Implementation of Data Backup and Recovery System

As digital systems continue to store important customer, transaction, and business records, having a reliable data backup and recovery system is essential to prevent data loss and maintain business continuity. Implementing an automated backup and recovery feature can help secure important system information and ensure that data can still be restored during unexpected system failures, technical issues, or cyber-related incidents.

The enhancement was recommended by the project team programmer to strengthen the system's reliability through automated database backups, recovery management, and secure cloud or local storage integration. This feature can help protect important records, reduce the risk of permanent data loss, and improve system stability during emergencies. In addition, the implementation of a backup and recovery system can enhance data security, operational efficiency, and overall user trust within the organization.

4. Implementation of Barcode Scanning Feature

As businesses continue to modernize their retail operations, barcode technology has become an effective tool for improving transaction speed, product accuracy, and inventory management. The implementation of a barcode scanning feature in the Pet Republik web system was carried out to simplify product

transactions and minimize manual processing errors.

This implementation was proposed by the project team designer to improve the efficiency of product transaction processing within the system. Through barcode integration, cashiers can quickly scan products instead of manually entering item details. This feature improves transaction accuracy, reduces processing time, and provides a more efficient and convenient experience for both employees and customers. In addition, barcode integration supports better stock monitoring by automatically updating product records during sales and restocking activities.

5. Implementation of Delivery Rider Account Feature

As online ordering and delivery services continue to expand, having a dedicated delivery management feature can improve the coordination and efficiency of delivery operations. Implementing a delivery rider account within the system can help manage deliveries more effectively and provide faster order fulfillment for customers.

The idea was recommended by the project manager to implement a delivery rider account feature that allows riders to access delivery information, update order statuses, and confirm successful deliveries in real time. This implementation will improve coordination among administrators, cashiers, customers, and riders while reducing delays and miscommunication during delivery operations. Moreover, customers will be able to track their orders more effectively, resulting in a more convenient and reliable delivery experience.